Informatica® Enterprise Data Catalog
10.5.11

Upgrading from Versions 10.5.0 and Later

Informatica Enterprise Data Catalog Upgrading from Versions 10.5.0 and Later
10.5.11
August 2026

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Preface

Follow the instructions in the *Informatica® Upgrading from Versions 10.4.0 and Later* to upgrade Enterprise Data Catalog. The guide includes information about upgrading Enterprise Data Catalog on an internal and external cluster. The additional information includes the prerequisite and post-requisite tasks that you need to perform to complete the upgrade and troubleshooting tips. Unless otherwise noted, the upgrade path includes hotfixes and service packs.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the Support option.

CHAPTER 1

Enterprise Data Catalog Upgrade

This chapter includes the following topics:

- Enterprise Data Catalog Upgrade Overview, 10
- Upgrade Paths, 10
- Deployment Planning, 12

Enterprise Data Catalog Upgrade Overview

The installer provides a domain upgrade wizard to guide you through the upgrade process. The upgrade wizard installs Informatica in the directory you specify. It does not modify the files in the directory of the previous version.

You can upgrade Enterprise Data Catalog deployed on either an external or internal cluster to version 10.5.11. The upgrade process creates a non-Hadoop cluster that you will use instead of the previous internal or external Hadoop cluster. The non-Hadoop cluster can be referred to as the *catalog cluster*.

Upgrade Paths

The upgrade path that you take depends on the products and versions that you're currently using.

Consider the following rules and guidelines to applying the hotfix or upgrade:

- You can apply the hotfix if you are on any version associated with the major release. For example 10.5 is a major release, so you can apply the hotfix from any version of 10.5.x.
- If you're not currently using a version associated with the major release, you can upgrade to the current release from any supported previous release.
- If you are on a version that doesn't support a direct upgrade, you must first upgrade to a supported version.
- Any supported update also includes hotfixes, service packs, and cumulative patches.

Important: If Data Engineering is in the same domain as any other product supported by the Informatica installer, verify that all products are upgraded to the same supported upgrade version before you upgrade to 10.5.11.

The following table provides details for the rules and guidelines:

Product	Versions
PowerCenter	You can apply the 10.5.11 hotfix to version 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5.11 from the following previous versions: - 10.2 including any hotfix, service pack, or cumulative patch - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch
Informatica Data Quality	You can apply the 10.5.11 hotfix to version 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5.11 from the following previous versions: - 10.2 including any hotfix, service pack, or cumulative patch - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch
Data Engineering Integration Data Engineering Quality	You can apply the 10.5.11 hotfix to version 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5 from the following previous versions, and then you can apply the 10.5.11 hotfix: - 10.2 including any hotfix, service pack, or cumulative patch - 10.2.1 including any hotfix, service pack, or cumulative patch - 10.2.2 including any hotfix, service pack, or cumulative patch - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch **Important:** If Data Engineering is in the same domain as any other product supported by the Informatica installer, verify that all products are upgraded to the same supported upgrade version before you upgrade to 10.5.11.
Data Engineering Streaming	You can apply the 10.5.11 hotfix to version 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5 from the following previous versions, and then you can apply the 10.5.11 hotfix: - 10.2.2 including any hotfix, service pack, or cumulative patch - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch
Data Privacy Management	You can apply the 10.5.11 hotfix to versions 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5 from the following previous versions, and then you can apply the 10.5.11 hotfix: - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch
Enterprise Data Catalog	You can apply the 10.5.11 to version 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5, 10.5.1, 10.5.2, 10.5.3, or 10.5.4 from the following previous versions, and then you can apply the 10.5.11 hotfix: - 10.4 (including any service packs or cumulative patches) - 10.4.1 (including any service packs or cumulative patches) **Important:** If Enterprise Data Catalog and Data Engineering are in the same domain of a version 10.4.1, 10.4 or earlier, upgrade them both to version 10.5, 10.5.1, 10.5.2, 10.5.3, or 10.5.4 before you upgrade to 10.5.11.

Product	Versions
Enterprise Data Preparation	You can apply the 10.5.11 hotfix to version 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. **Note:** You cannot upgrade to version 10.5.11 from Enterprise Data Preparation version 10.5.7. You can upgrade to version 10.5, 10.5.1, 10.5.2, 10.5.3, or 10.5.4 from the following previous versions, and then you can apply the 10.5.11 hotfix: - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch
Test Data Management	You can apply the 10.5.11 hotfix to versions 10.5, 10.5.1, 10.5.2, 10.5.3, 10.5.4, 10.5.5, 10.5.6, 10.5.7, 10.5.8, 10.5.9, or 10.5.10 including any service pack or cumulative patch. You can upgrade to version 10.5 from the following previous versions, and then you can apply the 10.5.11 hotfix: - 10.4 including any service pack or cumulative patch - 10.4.1 including any service pack or cumulative patch

Deployment Planning

You can plan to deploy Enterprise Data Catalog on one, three, or six nodes.

Deployment of Enterprise Data Catalog involves creating and configuring the Informatica Cluster Service. The Informatica Cluster Service runs and manages your Enterprise Data Catalog deployment. If you plan to deploy Enterprise Data Catalog on more than one node, high availability is enabled by default to provide resiliency and fail over for the Informatica Cluster Service.

The Informatica Cluster Service uses a set of applications and associated services bundled with the installer to manage and run Enterprise Data Catalog.

When you configure the Informatica Cluster Service, you can specify one, three, or six nodes on which the associated applications and services can run. These nodes are referred to as the data nodes in the cluster. The nodes on which you want to run profiling jobs or metadata scan jobs are referred to as processing nodes in the cluster. There are no restrictions on the number of processing nodes that you can configure.

Important: For the associated application services, you cannot configure more than three service instances for a deployment.

Informatica Cluster Service and Associated Services

The Informatica Cluster Service uses the following services to run and manage Enterprise Data Catalog:

Service	Description
MongoDB	Service to manage the MongoDB database used as the metadata store. Enterprise Data Catalog stores the metadata extracted from external sources within the MongoDB database.
Nomad	Service to manage the Nomad orchestration service used to schedule resource scan jobs.
Apache Solr	Service to manage Apache Solr used to index and search catalog assets.

Service	Description
Apache ZooKeeper	Service to manage Apache ZooKeeper used as the service co-ordination application
PostgreSQL	Service to manage PostgreSQL database used as the store for similarity profiling data.

CHAPTER 2

Apply the hotfix

This chapter includes the following topics:

- Hotfix installation and rollback overview, 14
- Applying the services hotfix, 16
- Rolling back the services hotfix, 27
- Applying the client hotfix, 30
- Rolling back the client hotfix, 33

Hotfix installation and rollback overview

When you apply the hotfix, the installer backs up existing directories, applies the new version of the product, and restores or updates required configuration and run-time files. When you roll back a hotfix, the installer restores Informatica to the previously installed version.

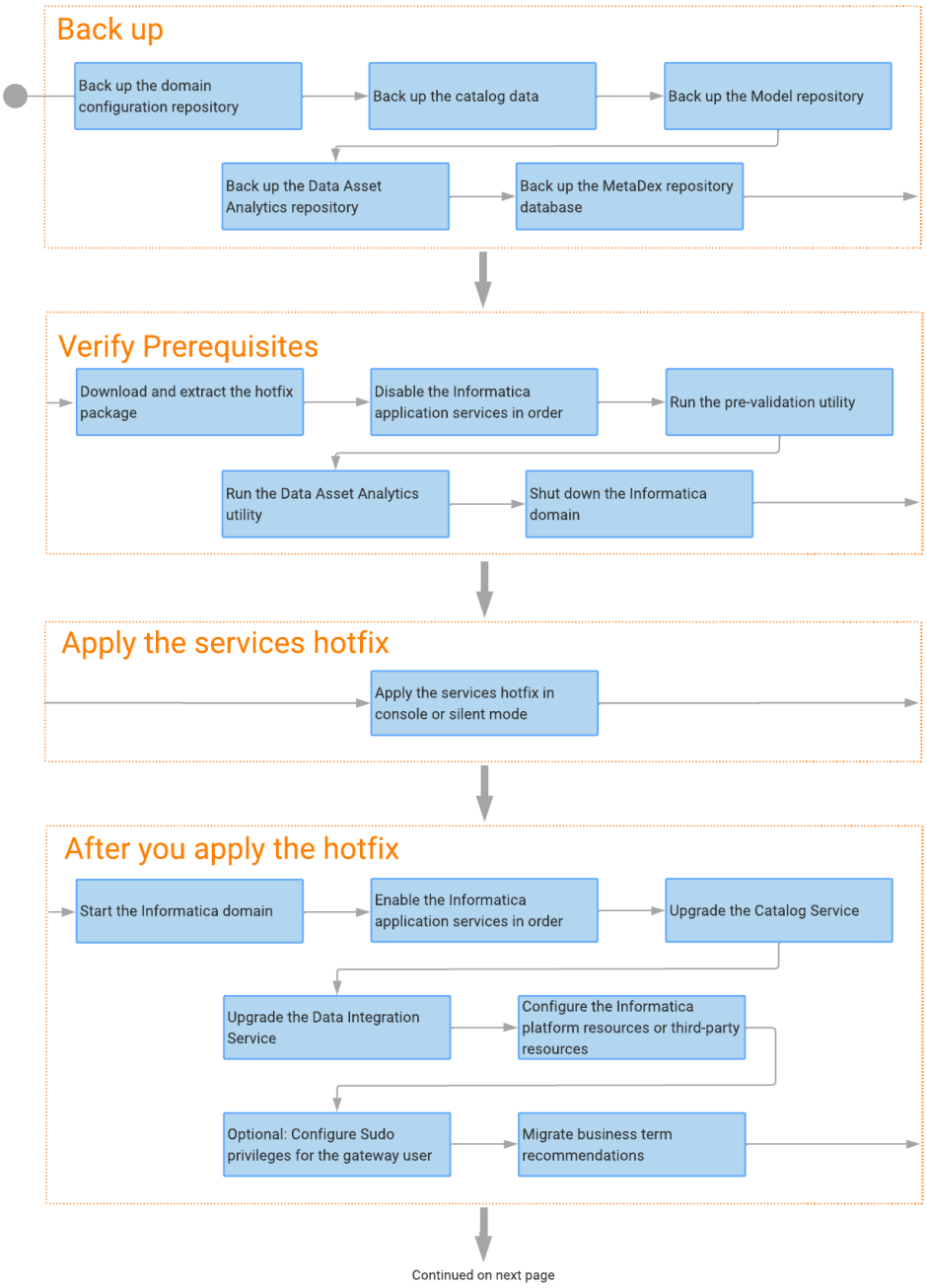
Informatica provides installers to apply hotfixes to the Informatica services.

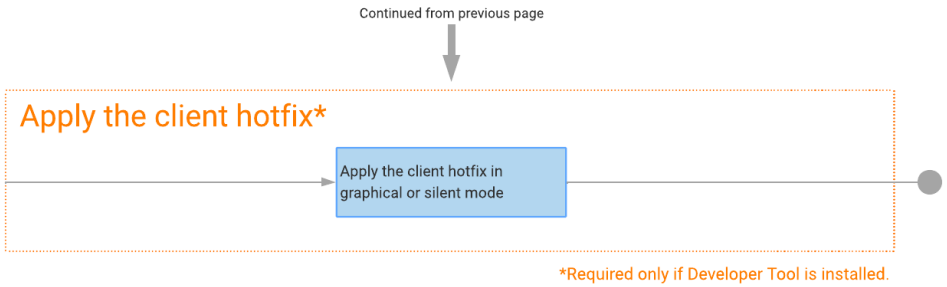
To apply the hotfix, perform the following steps:

1. Back up the catalog data, model repository, and other components.
2. Disable the Catalog Service and other Informatica application services.
3. Shut down the Informatica Cluster Service and the Informatica domain.
4. Apply the hotfix to Informatica services in console or silent mode.
5. Start the Informatica domain and complete the post-upgrade tasks.

If you want to roll back the hotfix, you can run the installer in console mode or silent mode.

The following image shows the steps that you need to perform to apply the hotfix:


Back up
Back up the domain configuration repository
Back up the catalog data
Back up the Model repository
Back up the Data Asset Analytics repository
Back up the MetaDex repository database
Verify Prerequisites
Download and extract the hotfix package
Disable the Informatica application services in order
Run the pre-validation utility
Run the Data Asset Analytics utility
Shut down the Informatica domain
Apply the services hotfix
Apply the services hotfix in console or silent mode
After you apply the hotfix
Start the Informatica domain
Enable the Informatica application services in order
Upgrade the Catalog Service
Upgrade the Data Integration Service
Configure the Informatica platform resources or third-party resources
Optional: Configure Sudo privileges for the gateway user
Migrate business term recommendations
Continued on next page



Applying the services hotfix

When you apply the hotfix, the installer backs up existing directories, applies the new version of the product, and restores or updates required configuration and run-time files.

You can run the installers to apply hotfixes to Informatica services and clients.

Ensure that you back up the catalog data and the Model repository before you apply the hotfix. Without a valid backup, you cannot perform a successful rollback of the hotfix. After you apply the hotfix, the installer creates the `BACK_FROM_HF` directory. The files within the directory are required for rollback. If you delete the directory, you cannot roll back the hotfix.

Before you apply the hotfix

Before you run the installer to apply the hotfix, perform the following prerequisite tasks.

Verify distribution package

Verify distribution package requirements

The Informatica domain and client require the distribution packages to process complex files within the Informatica domain, or to connect to Hadoop or Databricks environment when you process within the Informatica domain, or to push processing to Hadoop or Databricks environment.

If you need a distribution package, you can install it through the installer or through Integration Package Manager (the package manager) at any time.

You can use the same distribution package that you used in the previous Informatica version to process complex files within the Informatica domain or to connect to the Hadoop or Databricks environment when you process within the Informatica domain.

For the supported distribution packages when you push processing to Hadoop or Databricks environment, see the Product Availability Matrix.

Download Installer Files

Download and extract the installer files

The installer files are distributed as compressed files. You can get the Informatica installation file and distribution packages from the FTP link contained in your fulfillment email. Download the Informatica installation tar file from the Informatica Electronic Software Download site to a directory on your machine.

To download other compressed files and utilities, raise a shipping request with Informatica Global Customer Support.

Note: Make sure that you download the file to a local directory or a shared network drive that is mapped on your machine. You can then extract the installer files. However, you cannot run the installer from a mapped file. Copy the extracted files to a local drive and then run the installer.

1. Download and extract the hotfix package.
2. If you want to install distribution packages, download the ZIP files of the distribution packages.
3. Copy the ZIP files of the distribution packages to the location: `<Informatica installer files>/source`

   Note: The installer fails if the ZIP files for distribution packages aren't available in the source directory.

Verify Informatica Cluster Service Requirements

To successfully upgrade the ICS MongoDB component when updating to the latest version of the Enterprise Data Catalog, ensure the following requirements are met:

1. Install the numactl package on all Informatica Cluster Service hosts.
2. Configure sudo access for the ICS user on cluster boxes for the following commands:
   - `sudo systemctl stop mongod.service`
   - `sudo systemctl start mongod.service`
   - `sudo sed -i`
   - `sudo systemctl daemon-reload`

Pre-validation Check for Passwordless SSH Before You Install or Upgrade to the Latest Enterprise Data Catalog Version

Effective from 10.5.9, before you install or upgrade to the latest version of Enterprise Data Catalog, run a pre-validation check for passwordless SSH prior to the MongoDB upgrade process.

Note: If the Informatica Cluster Service gateway host is not included in the list of Mongo hosts, you must configure passwordless SSH on the primary Mongo host `$mongohost[0]`.

The configuration ensures secure, password-free communication between Mongo and the gateway hosts, and internally within the primary Mongo host.

Steps to Configure Passwordless SSH

1. Log in to the host. For example, to log in to`<hostname>.informatica.com` with a user credentials, use the following command:

   ```
   ssh <username>@<hostname>.informatica.com
   ```
2. Check for the existing SSH keys. If there are none, generate one with the following command:

   ```
   ssh-keygen -t rsa -b 4096 -N "" -f ~/.ssh/id_rsa
   ```

   The command generates a new SSH with no passphrase.
3. Copy the SSH public key from the source host to target host:

   ```
   ssh-copy-id <username>@<hostname>.informatica.com
   ```

Example

You have the following hosts in the `username>@<hostname>.informatica.com` format:

- Domain host: `lspe002577.informatica.com`
- Gateway host: `lspe002578.informatica.com`

- Mongo hosts: `lspe002579.informatica.com, lspe002580.informatica.com, lspe002581.informatica.com`

Perform the following configurations in the given sequence to configure passwordless SSH:

1. Configure passwordless SSH on Domain`lspe002577.informatica.com` to connect to the Mongo host `lspe002579.informatica.com`.
2. Configure passwordless SSH on `lspe002579.informatica.com` to connect to itself`lspe002579.informatica.com`.
3. Configure passwordless SSH on `lspe002579.informatica.com` to connect to the gateway host `lspe002578.informatica.com`.

Prepare for hotfix Installation

Perform the following prerequisite tasks:

1. Stop all processes that access the installation directory and subdirectories, including command prompts and command line programs.
2. Close the Informatica clients on all machines where you want to apply the hotfix.
3. Complete the following steps to back up the configuration metadata for the Informatica domain:
   a. Run the infasetup backupDomain command to back up the domain configuration database tables to a file.
   b. Back up the metadata configuration files to any directory accessible by the machines where you install Informatica.

   infasetup is located in the following directory: `<Informatica installation directory>/isp/bin`
4. Back up the binaries from the Informatica installation directory. Also, back up the following files from the Informatica domain:
   - `<Informatica installation directory>/isp/config/keys/siteKey`
   - `<Informatica installation directory>/domains.infa`
   - `<Informatica installation directory>/tomcat/conf/server.xml`
   - `<Informatica installation directory>/isp/config/nodemeta.xml`

   The nodemeta.xml file contains the metadata for a node. It is stored in the following directory on each node in the domain: `isp/config`. If you use the same backup directory name on all the nodes, rename the file before copying it to the backup location. For example, you back up nodemeta.xml to the /nodebak directory on nodeA and nodeB. Rename the configuration files so that on nodeA the file is backed up to /nodebak/nodemeta_A.xml, and on nodeB the file is backed up to /nodebak/nodemeta_B.xml.

   The domains.infa file contains connectivity information for the gateway nodes. It is stored in the Informatica installation directory or the location configured through the INFA_DOMAINS_FILE environment variable.
5. Back up the catalog data using the infacmd ldm backupContents command.

   Note: Verify that you take a full backup of the catalog data. If you deployed the Informatica Cluster Service on multiple nodes, verify that you configured the cluster shared file system path for the Informatica Cluster Service. If you do not configure the shared file system, the SEARCH store is not backed up, and a warning message appears in the console log. If the SEARCH store is not backed up, the backup validation fails when you apply or roll back the hotfix.
6. Back up the Model repository using the infacmd mrs backupContents command.
7. Back up the Data Asset Analytics repository.

8. Back up the MetaDex repository.
9. Back up the following SSL configuration files:
   - The domain truststore and keystore files.
   - The default keystore files.
   - The keystore and truststore files that the application services use.

   By default, the SSL configuration files are stored in the following directory: `<Informatica installation directory>/services/shared/security`
10. Disable the following Informatica application services:
    a. Catalog Service
    b. Content Management Service
    c. Data Integration Service
    d. Model Repository Service
    e. Analyst Service
    f. Email service
11. Shut down the Informatica Cluster Service using the infacmd ics shutdowncluster command or from the Informatica Administrator.
12. Run the pre-validation utility to validate the prerequisites to apply the hotfix. The utility is located in the following directory: `<Installer directory>/properties/utils/prevalidation/InformaticaClusterValidationUtility.jar`
    For information about how to run the pre-validation utility, see the Knowledge Base article.

    Note: If you see a permission validation error message for the cluster custom directory, you can ignore the error.
13. Run the Data Asset Analytics utility. For information on how to run the utility, see "Run the Data Asset Analytics Utility" on page 49
14. Shut down the Informatica domain using the infaservice shutdown command. By default, the infaservice executable file is located in the following directory: `<Informatica installation directory>/tomcat/bin`.
15. If the machine uses HTTPS to connect to Informatica Administrator, verify that the keystore file is in the default location and that it uses the default file name. The default location for the keystore file is `<Informatica installation directory>/tomcat/conf`. The default file name for the keystore file is Default.keystore.
16. Set the SKIP_VENDOR_CHECK environment variable to true to remove the sudo prompt from the hotfix installation on UNIX.

    Note: If you don't have sudo privileges, set the environment variable to true before you apply or roll back a hotfix. If you have sudo privileges, you don't need to set the environment variable.

Apply the services hotfix in console mode

Run the 10.5.11 installer to apply the hotfix to version 10.5 or 10.5.x.

When you apply the hotfix in console mode, the words Help, Quit, and Back are reserved words. Do not use them as input text.

Important: You cannot revert to the previous version of MetaDex after you apply the hotfix.

You cannot roll back the hotfix automatically using the installer. You must perform the steps listed in the "Roll back the services hotfix in console mode" on page 28 section to roll back the hotfix.

1. Run the installer using the ./install.sh command.
2. Press **Y** to confirm that you want to proceed with the installation.
3. Press **5** to select the option to apply the hotfix.
4. Press **1** to apply the hotfix.
5. Press **2** to agree to the terms and conditions and apply the license.
6. Verify that all installation requirements are met before you continue to apply the hotfix.
7. Press **Enter**.
8. Enter the directory where you installed Informatica 10.5 or 10.5.x.
   The directory names in the path must not contain spaces or the following special characters: @|* $ # ! % ( ){ } [ ] , ; '
9. Press **2** to confirm that you configured the Catalog Service on the node.

   Note: If you confirm that you configured the Catalog Service on the node, you must validate the backup.
10. Press **2** to confirm that you backed up the catalog data using the infacmd ldm backupContents command.
    Press **1** to skip the backup validation and proceed.

    You must skip the validation in the following scenarios:

    - You configured multiple nodes for the Catalog Service and you do not want to validate the backup on each node.
    - You did not configure the cluster shared file system path for the Informatica Cluster Service on a multi-node setup.
    - You enabled the Model Repository Service and the Data Integration Service, but you did not create the Catalog Service and the Informatica Cluster Service.
11. Enter the location of the backup file. For example, `/data/Informatica/LDM_BACKUP1050SAML12.zip`.

    Note: If the SEARCH store is not included in the backup, the validation fails with an error. You can skip the backup validation if the backup does not include the SEARCH store.
12. Select one of the following options based on the type of installation:

    - Press **1** to set Sandbox environment for a basic environment used for proof of concept with minimal users.
    - Press **2** to set Development environment for the design environment.
    - Press **3** to set Test environment for high volume processing that is closest to a production environment.
    - Press **4** to set Production environment for high volume processing with high levels of concurrency meant for end users. Advanced production environments are typically multi-node setups.

    Default is **1** for Sandbox.
13. Choose whether you want to install distribution packages through the Informatica installer.

    - Press **1** if you don't need distribution packages or if you want to install them later.
    - Press **2** if you want to install distribution packages through the installer.

    Default is 1.
14. If you choose to install distribution packages, select one or more packages from the list that you want to install. Separate multiple packages with a comma.
    Default is 1.

15. Press **Enter**.
16. At the Pre-Installation Summary, review the installation information and press **Enter** to apply the hotfix. The installer copies the Informatica files to the installation directory.
17. At the Post-Installation Summary, verify that the hotfix application completed successfully and press **Enter** to exit the installer.

Apply the services hotfix in silent mode

To apply the hotfix in silent mode, create the SilentInput.properties file and then run the installer.

Informatica provides a sample properties file, called SilentInput_hotFix.properties, that includes the parameters that the installer requires.

Important: You cannot revert to the previous version of MetaDex after you apply the hotfix.

1. Go to the root of the directory that contains the installation files.
2. Locate the SilentInput_hotFix.properties file.
3. Back up the file before you modify it.
4. Use a text editor to open the file and modify the values. The following table describes the installation properties that you can modify:

Property Name	Description
INSTALL_TYPE	Indicates whether to apply the hotfix or roll back the hotfix. If the value is 0, the installer applies the hotfix to the current Informatica installation. If the value is 1, the installer rolls back the current hotfix. Set the value to 0 to apply the hotfix.
USER_INSTALL_DIR	Directory for the Informatica installation where you want to apply or roll back the hotfix. To apply the hotfix to Informatica services, enter the absolute path for the directory in which the current version of Informatica services is installed.
ENABLE_USAGE_COLLECTION	Enables the software to automatically transmit to Informatica in the USA information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at https://www.informatica.com/in/privacy-policy.html. You may disable usage collection in Administrator tool. For more information about how to disable sending usage statistics, see the *Informatica Administrator Guide*. You must set the value to 1 to apply the hotfix.

Property Name	Description
INSTALLATION_ENVIRONMENT	If you apply the hotfix, specify the environment type associated with the Informatica services installation. • Press **1** to set Sandbox environment for a basic environment used for proof of concept with minimal users. • Press **2** to set Development environment for the design environment. • Press **3** to set Test environment for high volume processing that is closest to a production environment. • Press **4** to set Production environment for high volume processing with high levels of concurrency meant for end users. Advanced production environments are typically multi-node setups. Default is 1 for Sandbox.
EDC_CATALOG_INST	Required for Enterprise Data Catalog, Data Privacy Management, and Enterprise Data Preparation installations. Set the property to true if the Catalog Service runs on the node on which you want to apply the hotfix.
EDC_HF_BACKUP	Required for Enterprise Data Catalog, Data Privacy Management, and Enterprise Data Preparation installations, if you have set EDC_CATALOG_INST=true. Set the property to true if you have taken a backup of the catalog.
EDC_BACK_UP_DIR	Required for Enterprise Data Catalog, Data Privacy Management, and Enterprise Data Preparation installations, if you set the EDC_CATALOG_INST and EDC_HF_BACKUP properties to true. Enter the location of the catalog data backup file.
INSTALL_HADOOP_LIBRARIES	Determines whether to install distribution packages through the installer. Set the value to true if you want to install distribution packages through the installer. Set the value to false if you don't need distribution packages or if you want to install them later.
SELECTED_HADOOP_LIBRARIES	Determines the distribution packages that you want to install from the supported packages list. Enter the distribution packages that you want to install, separating multiple packages with a comma.

5. Save the properties file with the name SilentInput.properties in the same folder.
6. Go to the root of the services installer directory.
7. Verify that the directory contains the SilentInput.properties file that you edited and saved.
8. To apply the hotfix, on a shell command line, enter the following command to run the silent installer: ./silentInstallHotFix.sh.

   The installer copies the Informatica files to the installation directory.
9. Complete the post-installation tasks. Verify that the hotfix completed successfully by reviewing the log file or the **Post-Installation Summary** page.

Sample silent input properties

The following example shows the contents of the SilentInput.properties file when you apply or roll back the hotfix in silent mode.

```
# Use the following guidelines when editing this file:
# * Use this file to install without user interaction. Save the file
#   with the following name:  SilentInput.properties
# * Any error condition that leads to failure, such as an invalid
```

```
#   installation directory, generates a log file in the user home
#   directory. For example: /home/<user name>/silentErrorLog.log

############################################################
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www.informatica.com/trademarks.html
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############################################################


#######
# Use this file (SilentInput.properties) to install Informatica domain services without
user interaction.
#     Use this sample properties file to define the parameters for the silent
installation.
#     To upgrade Informatica, use the SilentInput_upgrade.properties or
SilentInput_upgrade_NewConfig.properties file.
# Use the following guidelines when you edit this file:
#     Back up the file before you modify it.
#     Any error condition that causes the installation to fail, such as an installation
directory that is not valid,
#     generates a log file in /home/<user name>. For example: /home/informatica/
silentErrorLog.log
#######

# Set RESUME_INSTALLATION to true if you want to resume installation
# Set RESUME_INSTALLATION to false and INFA_SERVICES_INSTALLED to true in case of
standalone installation
RESUME_INSTALLATION=false

# Set ENABLE_USAGE_COLLECTION  to 1 to accept the product usage toolkit end user license
agreement.
# You must set the value as 1 to install the Informatica platform.
# The product usage toolkit end user license agreement is available at: http://
www.informatica.com/us/eula/en-support-eula.aspx.
# As further described in the EULA, your use of the Informatica platform will enable the
product usage toolkit
# to collect certain product usage and failure information. You may disable this feature
at any time.
# For more information on how to disable this feature refer to the Informatica
Administrator Guide.

ENABLE_USAGE_COLLECTION=0

# The LICENSE_KEY_LOC property represents the absolute path and file name of the license
key file.
#     Set the property if you are installing Informatica domain services.

LICENSE_KEY_LOC=/home/license.key


# The USER_INSTALL_DIR property represents the directory in which to install the new
version of Informatica.
#     Set the property if you are installing or upgrading Informatica.
#     The property must point to a valid directory with write permissions enabled.

USER_INSTALL_DIR=/home/Informatica/10.5.11


#The INSTALLATION_ENVIRONMENT property represents the installation environment
# Set the property to one of the following installation environment types: Sandbox,
Development, Test, or Production. Values are case-sensitive.
# Set to Sandbox for a basic environment used for proof of concept with minimal
users.
# Set to Development for the design environment.
```

```
# Set to Test for high volume processing that is closest to a production environment.
# Set to Production for high volume processing with high levels of concurrency meant for
end users. Advanced production environments are typically multi-node setups.

INSTALLATION_ENVIRONMENT=


# The INSTALL_TYPE property determines whether to install or upgrade Informatica.
#     Set INSTALL_TYPE=0 to perform a new installation of Informatica.
#     To upgrade Informatica, use the SilentInput_upgrade.properties file.
#     To upgrade Informatica to a different node configuration, use the
SilentInput_upgrade_NewConfig.properties file.

INSTALL_TYPE=0


# The INSTALL_LDM property determines whether to install or upgrade Enterprise Data
Catalog along with Informatica domain services.
#     Set INSTALL_LDM=0 to perform a new installation of Informatica without Enterprise
Data Catalog.
#     Set INSTALL_LDM=1 to perform a new installation of Informatica with Enterprise
Data Catalog.

INSTALL_LDM=0

# The INSTALL_EDP property determines whether to install or upgrade Enterprise Data
Preparation along with Enterprise Data Catalog.
#     Set INSTALL_EDP=0 to perform a new installation of Enterprise Data Catalog without
Enterprise Data Preparation.
#     Set INSTALL_EDP=1 to perform a new installation of Enterprise Data Catalog with
Enterprise Data Preparation.

INSTALL_EDP=0


# This property determines whether to install standalone Enterprise Data Catalog.
# Set INFA_SERVICES_INSTALLED=true if the current version of Informatica is already
installed and you want to install only Enterprise Data Catalog on top of it.
# Set INFA_SERVICES_INSTALLED=false if the current version of Informatica is not
installed and you want to install Enterprise Data Catalog.
# If you want to set INFA_SERVICES_INSTALLED to true set RESUME_INSTALLATION to false
INFA_SERVICES_INSTALLED=false

# This property determines whether to install standalone Enterprise Data Preparation.
# Set EIC_SERVICES_INSTALLED=true if the current version of Enterprise Data Catalog is
already installed and you want to install only Enterprise Data Preparation on top of it.
# Set EIC_SERVICES_INSTALLED=false if the current version of Enterprise Data Catalog is
not installed and you want to install Enterprise Data Preparation.
EIC_SERVICES_INSTALLED=false

# The INSTALL_HADOOP_LIBRARIES property determines whether to install packages to
integrate the domain with Hadoop or Databricks.
# Set INSTALL_HADOOP_LIBRARIES=true if you want to install the integration packages to
push processing to Hadoop or Databricks, process complex files within the domain, or
connect to Hadoop but process within the domain.
# If you don't install the integration packages now, you can install them later through
Integration Package Manager.
INSTALL_HADOOP_LIBRARIES=false

# The SELECTED_HADOOP_LIBRARIES property determines the integration packages that you
want to install.
# Lists the supported integration packages and their versions.
# Enter the integration package that you want to install. Separate multiple packages
with a comma.
# You can choose from the following options:
# 1->Azure HDInsight 5.1
# 2->Cloudera 7.1
# 3->Cloudera 7.218
# 4->Databricks 14.3
# 5->Elastic MapReduce 7.1
```

```
# 6->Elastic MapReduce 7.9
# 7->MapR 7.9
# 8->Select all
SELECTED_HADOOP_LIBRARIES=
```

After you apply the hotfix

After you run the installer to apply the hotfix, start the domain and perform the post requisite tasks to complete the hotfix application.

Enable and Upgrade the Informatica Application Services

The post-upgrade tasks to complete the hotfix application involve starting the Informatica domain, enabling the Informatica application services, and upgrading the services.

1. Run the ./infaservice.sh startup command to start the Informatica domain. By default, infaservice.sh is located in the following directory: `<Informatica installation directory>/tomcat/bin`
2. Enable the following Informatica application services from the Administrator tool:
   a. Model Repository Service
   b. Data Integration Service
   c. Content Management Service
   d. Analyst Service
   e. Informatica Cluster Service
3. Enable and upgrade the Catalog Service.

   Note:

   - If you enabled Data Asset Analytics, it takes longer to start the Catalog Service. The service will not be available until the Data Asset Analytics repository upgrade is complete.
   - To know the status of the Data Asset Analytics repository upgrade, review the daa_upgrade.log file from the installation directory.

   This completes the upgrade to Enterprise Data Catalog 10.5.11.
4. Upgrade the Data Integration Service.

Remove the distribution packages from the backup location

After you apply the hotfix, remove the distribution packages from the backup folder to mitigate the security vulnerabilities.

When you apply the hotfix, all the existing files in the Informatica installation directory are backed up. You need to remove the distribution packages backup from the following location: `<Informatica installation directory>\BACK_FROM_<hotfix version number>\services\shared\hadoop\<distribution package_version>`

The following example shows the backup location for a distribution package when you apply the hotfix: `C:\Informatica\BACK_FROM_3\services\shared\hadoop\EMR_6.4`

Note: The deletion process doesn't remove all files of the distribution packages from the backup folder. Some files might still exist in the following location: `<Informatica installation directory>\BACK_FROM_<hotfix version number>\services\shared\` You can delete these files manually.

Important: When you roll back the hotfix, distribution packages are not restored. You can't enable functionality associated with the Hadoop or Databricks distribution packages in the rolled-back version. To install the distribution packages, contact Informatica Global Customer Support.

Configure Informatica Platform resources or third-party resources

If you did not configure Informatica Platform resources or third-party resources, download the scanner binaries and resource drivers from the Informatica Electronic Software Download (ESD) site mentioned in the fulfillment email.

Copy the scanner binaries to the following directory: `<Informatica installation directory>/services/CatalogService/ScannerBinaries`

Note: The ScannerBinaries.zip file includes the Informatica Platform resource binary files for 10.5.11 and the ExtendedScannerBinaries.zip file includes the Informatica Platform resource binary files from previous versions.

Configure Sudo Privileges for the Gateway User (Optional)

If you plan to change the properties for the Informatica Cluster Service or replace the SSL certificates configured for the Informatica Cluster Service, assign sudo privileges to the gateway user.

The following commands require sudo privileges for the gateway user when you enable the Informatica Cluster Service for the first time:

- mkdir
- chown
- chmod
- echo
- systemctl
- cp
- mv
- sysctl
- rm
- tee
- sed
- sshd

To configure sudo privileges for the commands, you must add the commands to the `/etc/sudoers` file as shown in the following sample: `/bin/mkdir, /bin/chown, /bin/chmod, /bin/echo, /bin/systemctl, /bin/cp, /bin/mv, /usr/sbin/sysctl, /bin/rm, /usr/bin/tee, /usr/bin/sed, /usr/sbin/sshd`.

Note: The `tee` command does not require sudo privileges in 10.5.1.1 and later versions.

You can determine the directory where each command is located using the `which <command name>` command.

After you enable the Informatica Cluster Service for the first time, you can choose to disable the validation for sudo permissions for the Informatica Cluster Service. To disable the validation, configure the following custom property for the Informatica Cluster Service: `IcsCustomOptions.IcsGatewayUserSudoEnabled` and set the value to false. After you disable the validation, sudo privileges are not required when you restart the Informatica Cluster Service. However, to shut down the Informatica Cluster Service, sudo permission is still required for the `systemctl` command.

To run the `infacmd ics cleanCluster` command to clean the Informatica Cluster Service, sudo permission must be configured for the `systemctl` and `rm` commands.

Note: The services associated with the Informatica Cluster Service such as Apache Solr, PostgreSQL, and Nomad require a non-root user with sudo privileges to run the services. Sudo allows a user to run programs or commands with elevated privileges for a specific time frame. Enterprise Data Catalog requires a non-root user with sudo permissions to run certain commands when performing administrative tasks such as installation, upgrade, and service status monitoring. For more information on the commands used and the role of each command during the upgrade, see the
Enable sudo privileges for a non-root user in Enterprise Data Catalog KB article.

Migrate business term recommendations

To migrate business term recommendations, run the infacmd ldm upgradePropagationStageFrom105 command from the following directory: `<Informatica installation directory>/isp/bin`

After you run the command, the Business Glossary recommendations appear in the Enterprise Data Catalog tool. For more information about the command, see *Informatica® Command Reference*.

Rolling back the services hotfix

You can roll back the hotfix if required. To roll back the hotfix, run the installer.

Before you roll back the hotfix

Before you roll back the hotfix, perform the following prerequisite tasks:

1. Back up the catalog data using the infacmd ldm backupContents command. For more information about the command, see *Informatica Command Reference*.
2. Back up the Model repository using the infacmd mrs BackupContents command. For more information about the command, see *Informatica Command Reference*.
3. Disable the following Informatica application services:
   a. Catalog Service
   b. Analyst Service
   c. Content Management Service
   d. Data Integration Service
   e. Model Repository Service
   f. Informatica Cluster Service
   g. Email Service

   Note: To disable a service, in Informatica Administrator, select the service and then click **Actions** > **Disable Service**.
4. Shut down the Informatica Cluster Service using the infacmd ics shutdowncluster command or from the Informatica Administrator.
5. Clean the Informatica Cluster Service using the infacmd ics cleancluster command.
6. For the Model Repository Service associated with the Catalog Service, you must delete the content of the service.
7. Shut down the Informatica domain using the infaservice shutdown command. The command is located in the following directory by default: `<Informatica installation directory>/tomcat/bin`.

Roll back the services hotfix in console mode

You can roll back the hotfix that you applied. To roll back the hotfix in console mode, run the installer.

1. Run the installer using the ./install.sh command.
2. Press **Y** to confirm that you want to proceed with the installation.
3. Press **5** to roll back the hotfix.
4. Press **2** to roll back the hotfix.
5. Press **2** to accept the license.
6. Verify the prerequisites and press **Enter**.
7. Enter the directory where you installed Informatica and press **Enter**.
8. Press **2** to confirm that you configured the Catalog Service on the node.

   Note: If you confirm that you configured the Catalog Service on the node where you plan to roll back the hotfix, you must validate the backup on the node. Press **1** to skip the backup validation and proceed to step 11.

   You must skip the validation in the following scenarios:

   - You did not configure the cluster shared file system path for the Informatica Cluster Service on a multi-node setup.
   - You configured multiple nodes for the Catalog Service and you do not want to validate the backup on each node.
   - You enabled the Model Repository Service and the Data Integration Service, but you did not create the Catalog Service and the Informatica Cluster Service.
9. Press **2** to confirm that you backed up the catalog data using the infacmd ldm backupContents command.
10. Enter the location of the backup file. For example, `/data/Informatica/LDM_BACKUP1050SAML12.zip`.

    Note: If the SEARCH store is not included in the backup, the validation fails with an error. You can skip the backup validation if the backup does not include the SEARCH store.
11. Press **Enter** to start the rollback.
12. At the Pre-Installation Summary, press **Enter** to continue.
13. Press **Enter** to exit the installer when the rollback completes successfully.

    Important: When you roll back the hotfix, distribution packages are not restored. You can't enable functionality associated with the Hadoop or Databricks distribution packages in the rolled-back version. To install the distribution packages, contact Informatica Global Customer Support.

Roll back the services hotfix in silent mode

To roll back the hotfix in silent mode, update the SilentInput.properties file and then run the silent installer.

Informatica provides a sample properties file, called SilentInput_hotFix.properties, that includes the parameters that are required by the installer.

1. Go to the root of the directory that contains the installation files.
2. Locate the SilentInput_hotFix.properties file.
3. Back up the file before you modify it.

4. Use a text editor to open the file and modify the values. The following table describes the installation properties that you can modify:

Property Name	Description
INSTALL_TYPE	Indicates whether to apply the hotfix or roll back the hotfix. Set the value to 1 to roll back the current hotfix.
USER_INSTALL_DIR	Directory for the Informatica installation where you want to roll back the hotfix.
ENABLE_USAGE_COLLECTION	Enables the software to automatically transmit to Informatica in the USA information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at https://www.informatica.com/in/privacy-policy.html. You may disable usage collection in Administrator tool. For more information about how to disable sending usage statistics, see the *Informatica Administrator Guide*. You must set the value to 1 to roll back the hotfix.
INSTALLATION_ENVIRONMENT	You can ignore the property during rollback.
EDC_CATALOG_INST	Required for Enterprise Data Catalog, Data Privacy Management, and Enterprise Data Preparation installations. Set the property to true if the Catalog Service runs on the node on which you want to apply the hotfix.
EDC_HF_BACKUP	Required for Enterprise Data Catalog, Data Privacy Management, and Enterprise Data Preparation installations, if you have set EDC_CATALOG_INST=true. Set the property to true if you have taken a backup of the catalog.
EDC_BACK_UP_DIR	Required for Enterprise Data Catalog, Data Privacy Management, and Enterprise Data Preparation installations, if you set the EDC_CATALOG_INST and EDC_HF_BACKUP properties to true. Enter the location of the catalog data backup file.

5. Save the properties file with the name SilentInput.properties.
6. Go to root of the Informatica services installer directory.
7. Verify that the directory contains the SilentInput.properties file that you edited and saved.
8. On a shell command line, enter the following command to run the silent installer:

   ./silentInstallHotFix.sh.

   The installer removes the last hotfix and restores the previous version of Informatica.
9. Complete the post rollback tasks. Verify that the rollback completed successfully by reviewing the log file or the **Post-Installation Summary** page.

   Important: When you roll back the hotfix, distribution packages are not restored. You can't enable functionality associated with the Hadoop or Databricks distribution packages in the rolled-back version. To install the distribution packages, contact Informatica Global Customer Support.

After you roll back the hotfix

After you run the installer to roll back the hotfix, start the Informatica domain and perform the post requisite tasks to complete the rollback.

1. Start the Informatica domain.
2. Use the infacmd mrs deleteContents command to remove the Model repository content for version 10.5.11.
3. Enable the Model Repository Service.
4. Use the infacmd mrs RestoreContents command to restore the Model repository content.
5. Enable the Data Integration Service.
6. Enable the Content Management Service.
7. Enable the Analyst Service.
8. Enable the Email service.
9. Enable the Informatica Cluster Service.
10. Run the infacmd ldm restoreContents command to restore the catalog data.
11. From the Informatica Administrator, remove any custom properties that you had configured for the Catalog Service after you upgraded to version 10.5.11.
12. Enable the Catalog Service.

Note: The installer retains the Data Asset Analytics configuration properties on the Catalog Service page when you view the Catalog Service in Informatica Administrator.

Applying the client hotfix

When you apply the hotfix, the installer backs up existing directories, applies the new version of the client, and restores or updates required configuration and run-time files.

You can run the client installer to apply a hotfix to Informatica clients that you use. After you apply the hotfix, you can remove the backup files of the distribution packages from the backup folder.

Apply the client hotfix in graphical mode

When you run the installer in graphical mode, a console window opens and runs in the background. If you close this window, the installation stops.

1. Log in to the machine with a system user account. The user account must be the same account used to install the previous version of Informatica services.
2. Close all other applications.
3. To begin the installation, run install.bat from the root directory where you unzipped or untarred the client hotfix installation package.
4. On the Informatica 10.5.11 page, select **Apply hotfix to a version from 10.5 through 10.5.10, or roll back the hotfix**, and then click **Start**.

   The **Installation Type** page appears.
5. Select **Apply the hotfix** to choose to apply the hotfix.
6. Verify that all requirements are met before you continue the hotfix, and then click **Next**.

The **Installation Directory** page appears.

7. Enter the absolute path for the directory in which the current version of the Informatica client is installed and specify the installation environment.
8. If you want to install distribution packages through the Informatica installer, select the check box.
9. If you choose to install distribution packages, select one or more packages from the list that you want to install.
10. Click **Next**.

    The **Pre-Installation Summary** page appears.
11. Review the installation information, and then click **Install** to apply the hotfix.

    The installer copies the Informatica files to the installation directory.
12. Complete the post-installation tasks. Verify that the hotfix completed successfully by reviewing the log file or the **Post-Installation Summary** page.

Apply the client hotfix in silent mode

To apply the hotfix in silent mode, create the SilentInput.properties file and then run the installer.

Informatica provides a sample properties file, called SilentInput_hotFix.properties, that includes the parameters that the installer requires.

1. Go to the root of the directory that contains the client installation files.
2. Locate the SilentInput_hotFix.properties file.
3. Back up the file before you modify it.
4. Use a text editor to open the file and modify the values. The following table describes the installation properties that you can modify:

Property Name	Description
INSTALL_TYPE	Indicates whether to apply the hotfix or roll back the hotfix. If the value is 0, the installer applies the hotfix to the current Informatica client installation. If the value is 1, the installer rolls back the current hotfix. Set the value to 0 to apply the hotfix.
USER_INSTALL_DIR	Directory for the Informatica client installation where you want to apply or roll back the hotfix. To apply the hotfix, enter the absolute path for the directory in which the current version of the Informatica clients is installed.
INSTALL_HADOOP_LIBRARIES	Determines whether to install distribution packages through the installer. Set the value to true if you want to install distribution packages through the installer. Set the value to false if you don't need distribution packages or if you want to install them later.
SELECTED_HADOOP_LIBRARIES	Determines the distribution packages that you want to install from the supported packages list. Enter the distribution packages that you want to install, separating multiple packages with a comma.

5. Save the properties file with the name SilentInput.properties in the same folder.
6. Double-click the silentInstallHotFix.bat to apply the hotfix.

   The installer copies the Informatica client files to the installation directory.

7. Complete the post-installation tasks. Verify that the hotfix completed successfully by reviewing the log file or the **Post-Installation Summary** page.

Sample silent input properties

The following example shows the contents of the `SilentInput.properties` file when you apply or roll back the hotfix client in silent mode.

```
############################################################
# Informatica Installer Build Details
# Copyright (c) 1993-2026 Informatica LLC
# This software contains confidential and proprietary
# information of Informatica LLC.
# All Rights Reserved.
############################################################

# Use the following guidelines when editing this file
# * Use this file to install without user interaction.
# * After you create the properties file, save the file with the name
SilentInput.properties
# * Any error condition that leads to failure, such as an invalid
# installation directory, generates a log file in the user home directory.
# For example:  c:\silentErrorLog<time_stamp>.log

############################################################


#######
## *INSTALL_TYPE - Set this value to '0' to apply the hotfix. Set this value to '1' to
roll back the hotfix.

INSTALL_TYPE=0


# * The USER_INSTALL_DIR variable must point to a valid directory with write
#     permissions enabled.

USER_INSTALL_DIR=c:\Informatica\10.5.11

#######
# After you create the properties file, save the file with the name
SilentInput.properties and
#     run the silent installer to perform the Informatica client installation.
#######
```

Remove the distribution packages from the backup location

After you apply the hotfix, remove the distribution packages from the backup folder to mitigate the security vulnerabilities.

When you apply the hotfix, all the existing files in the Informatica installation directory are backed up. You need to remove the distribution packages backup from the following location: `<Informatica installation directory>\BACK_FROM_<hotfix version number>\clients\DeveloperClient\hadoop\<distribution package_version>`

The following example shows the backup location for a distribution package when you apply the hotfix: `C:\Informatica\BACK_FROM_3\clients\DeveloperClient\hadoop\EMR_6.4`

Note: The deletion process doesn't remove all files of the distribution packages from the backup folder. Some files might still exist in the following location: `<Informatica installation directory>\BACK_FROM_<hotfix version number>\clients\DeveloperClient\` You can delete these files manually.

Important: When you roll back the hotfix, distribution packages are not restored. You can't enable functionality associated with the Hadoop or Databricks distribution packages in the rolled-back version. To install the distribution packages, contact Informatica Global Customer Support.

Rolling back the client hotfix

You can roll back the hotfix if required. To roll back the hotfix, perform the required prerequisite tasks and then run the installer.

Rolling back the client hotfix in graphical mode

When you run the installer in graphical mode, a console window opens and runs in the background. If you close this window, the rollback stops.

1. Log in to the machine with a system user account. The user account must be the same account used to install the previous version of Informatica services.
2. Close all other applications.
3. To begin the installation, run install.bat from the root directory where you unzipped or untarred the client hotfix installation package.
4. On the Informatica 10.5.11 page, select **Apply hotfix to a version from 10.5 through 10.5.10, or roll back the hotfix**, and then click **Start**.

   The **Installation Type** page appears.
5. Select **Roll back the hotfix** to choose to roll back the hotfix.
6. Verify that all requirements are met before you continue the rollback, and then click **Next**.

   The **Installation Directory** page appears.
7. Enter the absolute path for the directory in which the current version of Informatica clients is installed, and then click **Next**.

   The **Pre-Installation Summary** page appears.
8. Review the installation information, and then click **Install** to roll back the hotfix.

   The installer removes the last hotfix and restores Informatica clients to the previous version.
9. Complete the post-installation tasks. Verify that the rollback completed successfully by reviewing the log file or the **Post-Installation Summary** page.

   Important: When you roll back the hotfix, distribution packages are not restored. You can't enable functionality associated with the Hadoop or Databricks distribution packages in the rolled-back version. To install the distribution packages, contact Informatica Global Customer Support.

Rolling back the client hotfix in silent mode

To roll back the hotfix in silent mode, create the SilentInput.properties file and then run the installer with the file.
Informatica provides a sample properties file, called SilentInput_hotFix.properties, that includes the parameters that the installer requires.

1. Go to the root of the directory that contains the client installation files.
2. Locate the SilentInput_hotFix.properties file.

3. Back up the file before you modify it.
4. Use a text editor to open the file and modify the values. The following table describes the installation properties that you can modify:

Property Name	Description
INSTALL_TYPE	Indicates whether to apply the hotfix or roll back the hotfix. If the value is 0, the installer applies the hotfix to the current Informatica client installation. If the value is 1, the installer rolls back the current hotfix. Set the value to 1 to roll back the hotfix.
USER_INSTALL_DIR	Directory for the Informatica client installation where you want to roll back the hotfix. Enter the absolute path for the directory in which the current version of the Informatica clients is installed.

5. Save the properties file with the name SilentInput.properties in the same folder.
6. To roll back the hotfix on Windows, double-click the silentInstallHotFix.bat file.

   The installer removes the last hotfix and restores the previous version of Informatica clients.
7. Complete the post-installation tasks. Verify that the rollback completed successfully by reviewing the log file or the **Post-Installation Summary** page.

   Important: When you roll back the hotfix, distribution packages are not restored. You can't enable functionality associated with the Hadoop or Databricks distribution packages in the rolled-back version. To install the distribution packages, contact Informatica Global Customer Support.

CHAPTER 3

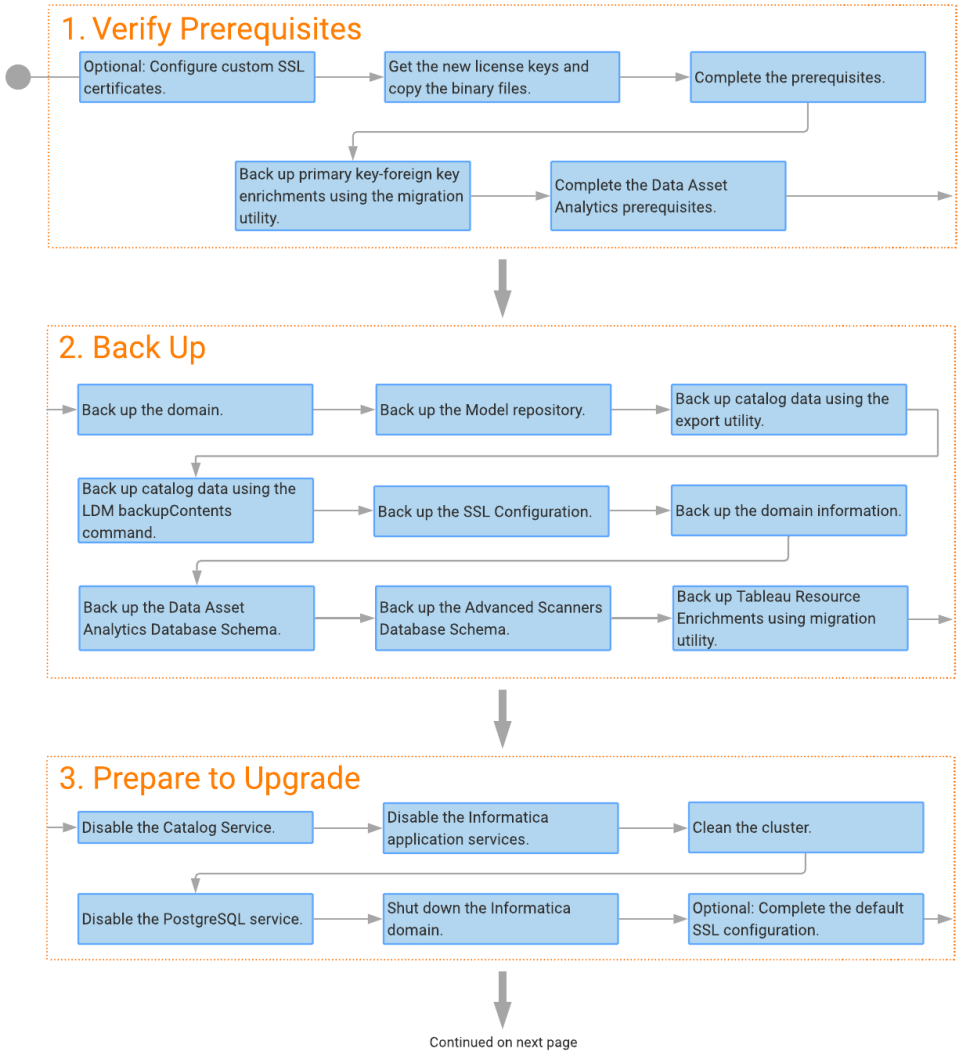
Upgrade Enterprise Data Catalog in an Internal Cluster

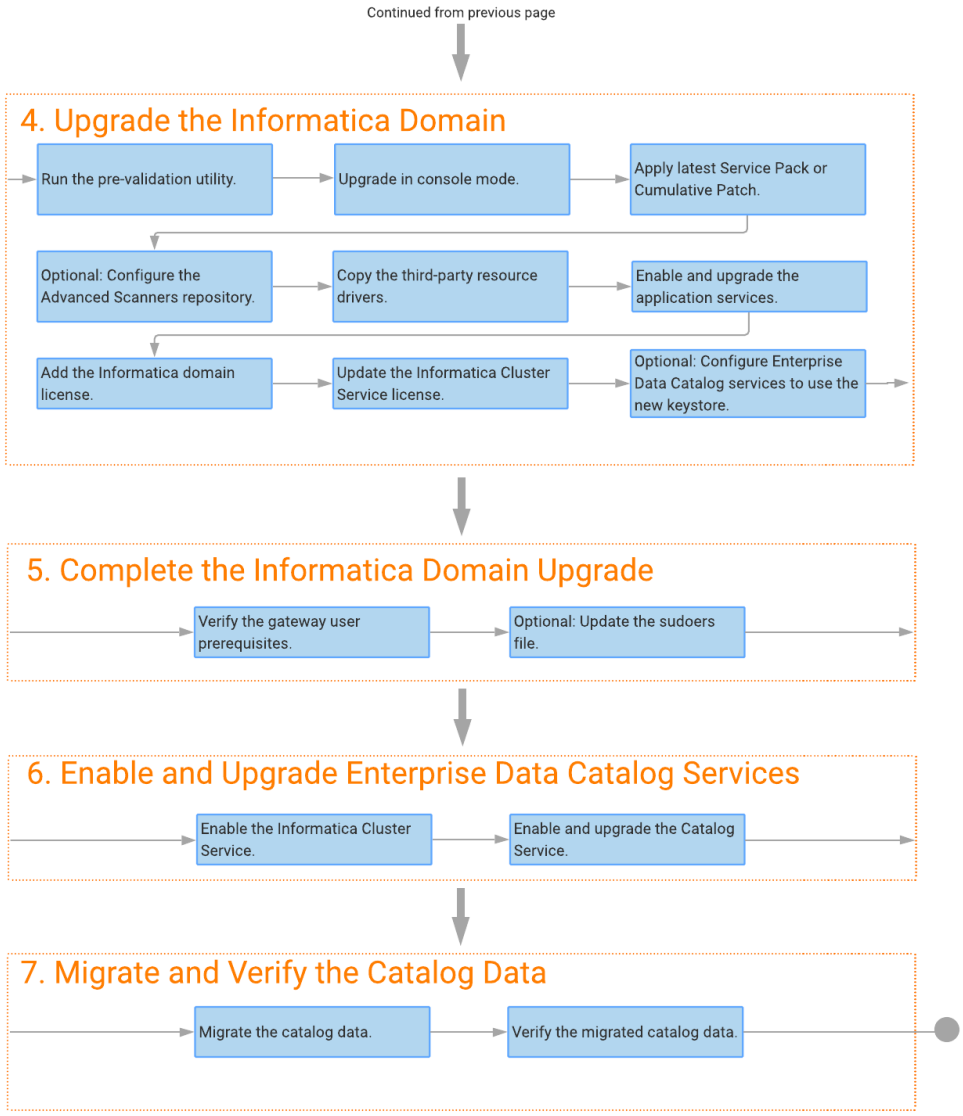
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Upgrade Enterprise Data Catalog in an Internal Cluster Overview

The following image shows the steps that you need to perform to upgrade Enterprise Data Catalog:





Step 1. Verify Prerequisites to Upgrade

Before you start the upgrade as a non-root user with sudo privileges, validate the following prerequisites:

- Verify that at least 100 GB of free disk space is available on the machine where the Informatica domain runs.
- Verify that Java Development Kit (JDK) 17 is installed.
- If the cluster is enabled for Kerberos and SSL, back up the Informatica domain truststore and default keystore files.
- Verify if the domain configuration repository is connected during the upgrade.
- Verify that either ntpd or chronyd is synchronized between the Informatica domain node and the cluster nodes.

- Install the following applications and packages on all nodes before you upgrade Enterprise Data Catalog:
  - Bash shell
  - rsync
  - libcurl
  - xz-libs
  - systemctl
  - zip
  - unzip
  - tar
  - wget
  - scp
  - rpm
  - curl
  - nslookup
  - md5sum
  - netstat
  - ping
  - ifconfig
  - cksum
  - dnsdomainname
  - libncurses5
  - OpenSSL version 1.0.1e-30.el6_6.5.x86_64 or later.

  OpenSSL 3.0 is not supported.

  If the machine on which the PostgreSQL service runs is FIPS enabled, verify that compat-openssl10 is installed.

  Verify that the $PATH variable points to the `/usr/bin` directory to use the correct version of Linux OpenSSL.

  In 10.5.1.0.1 and later versions, rsync is used instead of scp for file transfers.

  In 10.5.1.0.2 and later versions, the MaxStartups parameter in the SSH server configuration file is validated. The maximum number of unauthenticated concurrent SSH connections must be set to a value greater than or equal to 30. To set the value of this parameter to 30, open the `/etc/ssh/sshd_config` file, change the value of the MaxStartups parameter to 30:30:100, and then restart the sshd daemon.

  Note: Unauthenticated SSH connections are socket connections to the SSH daemon that are established without user authentication.
- The following table lists the default ports that must be available for the Informatica Cluster Service:

Service	Default Port
HTTP/HTTPS	9075
Nomad Serf	4648

Service	Default Port
Nomad HTTP	4646
Nomad RPC	4647
ZooKeeper	2181
ZooKeeper peer	2888
ZooKeeper leader	3888
Solr	8983
MongoDB	27017
PostgreSQL	5432

- If you configured the repository for MetaDex, back up the MetaDex repository database.

  Note: The default MetaDex server memory is 4 GB.
- Remove the values configured for the following environment variables using the `unset` command:

  - INFA_HOME

  - INFA_DOMAINS_FILE

  - DISPLAY

  - JRE_HOME

  - INFA_TRUSTSTORE

  - INFA_TRUSTSTORE_PASSWORD
- Verify that the LD_LIBRARY_PATH and the PATH environment variables do not contain earlier versions of Informatica.
- Delete the following JAR file from the nodes on which the Catalog Service runs: `<Informatica Installation Directory>/logs/<NodeName> /services/CatalogService/<CatalogServiceName>/migrate/restore.jar`
- Effective in version 10.5, Universal Connectivity Framework (UCF) is deprecated. To use UCF resources that you had configured in earlier versions, copy the UCF jar files from the existing installation location to the new upgrade location:`<$INFA HOME>/services/CatalogService/ScannerJars/bin`.

  Note: The UCF jar files have the `com.infa.ldm.extended. *.jar` file pattern.

  After you upgrade the Catalog Service, the existing UCF resources are listed in the **Select Resource Type** dialog box in Catalog Administrator.

Note: Keep a record of the asset count from the catalog, Catalog Administrator, and the Data Asset Analytics dashboard for reference. You can use the `/data/search` REST API to view the asset count in JSON format as shown in the following sample: `http://<CatalogAdmin>:<port>/access/2/catalog/data/search?defaultFacets=false&disableSemanticSearch=true&enableLegacySearch=false&facet=true&facetId=core.classType&facetId=core.resourceName&facetId=core.resourceType&highlight=false&includeRefObjects=false&offset=0&pageSize=1&q=*&tabId=all`.

For more information, see the *Enterprise Data Catalog REST API Reference*.

If you use Axon Data Governance with Enterprise Data Catalog, make sure that the Axon SuperAdmin has disabled the asset onboarding job and the reference link retrieval job before you upgrade. For more information, see the Informatica Knowledge Base article.

Configure Custom SSL Certificates Overview

During an install or an upgrade, you can use SSL certificates of your choice, referred to as custom SSL certificates, to secure components associated with the Informatica Cluster Service. You can choose to use CA-signed custom SSL certificates or self-signed custom SSL certificates as the client and cluster certificates.

If you choose to use CA-signed custom SSL certificates, use the generate_csr.sh and generate_certs.sh scripts to generate the required certificates.

If you choose to use self-signed custom SSL certificates, use the custom SSL utility bundled with the installer to generate the required certificates.

Requirements to Enable Custom SSL Certificates

Before you enable custom SSL for Enterprise Data Catalog, secure the Informatica domain with custom SSL certificates.

Verify the following prerequisites for the Informatica domain:

- The custom truststore and keystore files are placed in a custom directory. The truststore file name must be infa_truststore.jks.
- The directory that stores the keystore and truststore files are accessible to the command line programs.

Important: The installer places the default infa_truststore.jks and keystore files in the `<Informatica installation directory>/services/shared/security` directory on each node. Do not overwrite, delete, or move the default truststore and keystore files. Do not place the custom truststore and keystore files in the directory.

To know about how to convert a non-SSL Informatica domain to an SSL-enabled Informatica domain, see the Knowledge Article Convert a non-SSL Informatica domain to an SSL-enabled Informatica domain.

Generate CA-signed Custom SSL Certificates

You have a custom SSL certificate for the Informatica domain, and you want to use CA-signed SSL certificates as the client and cluster certificates for the Informatica Cluster Service.

Use the `generate_csr.sh` and `generate_certs.sh` scripts to generate the Certificate Signing Request (CSR) to send it to a CA and generate the required SSL certificates. You can download the scripts from the Informatica Electronic Software Download site.

Perform the following steps to use the scripts to generate the certificates:

1. Extract the generate_csr.sh and generate_certs.sh scripts from the following location: `<Location of installer files>/properties/utils/CustomSSLScriptsUtil_ExternalCA`.
2. Set the JAVA_HOME environment variable to point to JDK 17.
3. In the gen_csr.properties file, provide the values for the following parameters:

Parameter	Description
InfaDomainKeystorePassword	The Informatica domain keystore password in plain text.
ServerHosts	The Informatica Cluster Service hosts that include the data nodes, processing nodes, and gateway node. Enter a comma-separated list of FQDNs of cluster nodes.

Parameter	Description
ClientHosts	Comma-separated list of unique host names of domain nodes and cluster nodes.
InfaDomainName	The Informatica domain name.
ICSServiceName	The name of the Informatica Cluster Service.
KeysOutputDir	The directory to store the generated keys. Specify the $CUSTOM KEYSTORE LOC directory to avoid the additional steps to copy the generated keys. The $ICS SERVICENAME/client_certs and the $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM KEYSTORE LOC directory. $CUSTOM KEYSTORE LOC is the directory where the custom keystore for the Informatica domain (infa_keystore.jks) is located. $ICS SERVICENAME is the name of the Informatica Cluster Service.
CertsOutputDir	The $CUSTOM TRUSTSTORE LOC directory to store the generated truststore files. The $ICS SERVICENAME/client_certs and the $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM TRUSTSTORE LOC directory. The $CUSTOM TRUSTSTORE LOC is the directory where the custom truststore for the Informatica domain (infa_truststore.jks) is located.
DNSDomainName	The DNS domain name for the cluster nodes.
ClusterCert_OrganizationUnit	Optional. The value for the OrganizationUnit for the cluster nodes certificate.
ClusterCert_Organization	The value for the Organization for the cluster nodes certificate. **Note:** Verify that the combination of the Organization (O) and the Organizational Unit (OU) parameters in the certificate subject is distinct for the cluster and client certificates
ClusterCert_Location	The value for the Location for the cluster nodes certificate.
ClusterCert_State	The value for the State for the cluster nodes certificate.
ClusterCert_CountryCode	The value for the Country Code for the cluster nodes certificate.
DomainCert_OrganizationUnit	Optional. The value for the Organization Unit for the domain nodes certificate.
DomainCert_Organization	The value for the Organization for the domain nodes certificate. **Note:** Verify that the combination of the Organization (O) and the Organizational Unit (OU) parameters in the certificate subject is distinct for the cluster and client certificates
DomainCert_Location	Optional. The value for the Location for the domain nodes certificate. Default is the ClusterCert_Location parameter.
DomainCert_State	Optional. The value for the State for the domain nodes certificate. Default is the ClusterCert_State parameter.

Parameter	Description
DomainCert_CountryCode	Optional. The value for the Country Code for the domain nodes certificate. Default is the ClusterCert_CountryCode parameter.
Custom_Server_Certificate_CN	Optional. The value for the Common Name in the cluster nodes certificate that can be used instead of the default $InfaDomainName-$ICSServiceName.$DNSDomainName value. **Note:** You must enter RFC2253 compliant values. The following special characters are supported: `, + " \ < > ;`. Double quotes (") must be used in pairs. The characters \ and " must not be used together. The value cannot contain a space.
Custom_Client_Certificate_CN	Optional. The value for the Common Name in the domain nodes certificate that can be used instead of the default $InfaDomainName-$ICSServiceName.$DNSDomainName value. Default is the Custom_Server_Certificate_CN parameter. **Note:** You must enter RFC2253 compliant values. The following special characters are supported: `, + " \ < > ;`. Double quotes (") must be used in pairs. The characters \ and " must not be used together. The value cannot contain a space.

Note: If the values contain spaces or special characters, you must enclose the values within double-quotes.

4. Run the generate_csr.sh script using the following command to generate the .csr files to send to an external CA: `./generate_csr.sh gen_csr.properties`
   The following files are generated for the cluster:

   infa_nodecert.csr

   infa_privkey.key

   infa_privkey.pem

   keystore.jks

   The following files are generated for the client:

   infa_nodecert.csr

   infa_privkey.key

   infa_privkey.pem

   keystore.jks

   browser_cert.csr

   browser_keystore.jks

   browser_privkey.key

   browser_privkey.pem
5. Validate the contents of the .csr files. Run the following command to view the contents: `keytool -printcertreq -file $PATH TO CSR`
6. Send the following .csr files to an external CA for signing:
   - <CertsOutputDir>/<ICSServiceName>/client_certs/infa_nodecert.csr
   - <CertsOutputDir>/<ICSServiceName>/cluster_certs/infa_nodecert.csr

- <CertsOutputDir>/<ICSServiceName>/client_certs/browser_cert.csr

Note: The browser_cert.csr file is required if you want to create the browser certificates to view the scan job logs on Nomad.

7. After you receive the certificates or certificate chains from the CA in .pem format, "Validate the CA-signed Certificates" on page 44 and store the certificates to a location under the $INFA HOME directory on your machine.

   Note: If you receive the certificates or certificate chains from the CA in .cer format, run the following command to convert the files to pem format: `openssl x509 -inform der -in <certificate file name>.cer -outform pem -out <certificate file name>.pem`.

   If you receive a certificate chain from the CA, you must extract the root certificate, intermediate certificates, and the end user certificate.

8. In the gen_certs.properties file, provide the values for the following parameters:

Parameter	Description
InfaDomainKeystorePassword	The Informatica domain keystore password in plain text.
InfaDomainTruststorePassword	The Informatica domain truststore password in plain text.
ClusterCertificate	The path to the cluster certificate signed by the CA in .pem format. This is an end user certificate.
ClientCertificate	The path to the client certificate signed by the CA in .pem format. This is an end user certificate.
BrowserCertificate	The path to the browser certificate signed by the CA in .pem format.
ICSServiceName	The name of the Informatica Cluster Service.
IsCACertificateChainAvailable	Specify if the CA certificate chain is available as a single .pem file. Enter true or false. **Note:** The certificate chain must contain only the root and intermediate certificates.
SingleCACertificateChain	The path to the CA certificate chain in .pem format.
IndividualCertificatesFromCAChain	Optional. Only required if the IsCACertificateChainAvailable parameter is set to false. Comma-separated paths to the public certificates in the CA certificate chain in .pem format if the complete CA certificate chain is available as individual .pem files.

Parameter	Description
KeysOutputDir	The $CUSTOM KEYSTORE LOC directory store the generated keys. The $ICS SERVICENAME/client_certs and $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM KEYSTORE LOC directory. $CUSTOM KEYSTORE LOC is the directory where the custom keystore for the Informatica domain (infa_keystore.jks) is located. $ICS SERVICENAME is the name of the Informatica Cluster Service
CertsOutputDir	The $CUSTOM TRUSTSTORE LOC directory to store the generated truststore files. The $ICS SERVICENAME/client_certs and the $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM TRUSTSTORE LOC directory. The $CUSTOM TRUSTSTORE LOC is the directory where the custom truststore for the Informatica domain (infa_truststore.jks) is located.

9. Run the generate_certs.sh script using the following command to generate the certificates: `./generate_certs.sh gen_certs.properties`
   The keystore.jks keystore and the infa_privkey.pem private keys are stored at $CUSTOM KEYSTORE LOC/$ICSServiceName/client_certs and $CUSTOM KEYSTORE LOC/$ICSServiceName/cluster_certs directories.

   The truststore.jks truststore and the infa_nodecert.pem, infa_nodecertkey.pem, and infa_pubcert.pem public keys are stored at $CUSTOM TRUSTSTORE LOC/$ICSServiceName/client_certs and $CUSTOM TRUSTSTORE LOC/$ICSServiceName/cluster_certs directories.
10. Optional. The directories $CUSTOM KEYSTORE LOC and $CUSTOM TRUSTSTORE LOC are generally the same. If the <KeysOutputDir> location is not the same as $CUSTOM KEYSTORE LOC and <CertsOutputDir> location is not the same as $CUSTOM TRUSTSTORE LOC, move the keys and certificates to the respective directories.

    Note: Verify that the $CUSTOM KEYSTORE LOC and the $CUSTOM TRUSTSTORE LOC directories have the required user privileges. Also, validate that the user has minimum chmod 700 permissions configured for the directories and chmod 600 permissions configured for the files that are copied to the directories.

To access the Nomad Web UI and Solr Admin UI when the Informatica Cluster Service is SSL enabled, you must import the browser certificates. To know more about how to import the browser certificates, see the Knowledge Article
Access Nomad Web UI and Solr Admin UI when Informatica Cluster Service is SSL enabled in Enterprise Data Catalog.

Validate the CA-signed Certificates

You must use a single CA for the cluster and client certificates. After you receive the signed certificates from the CA, you must verify that each certificate is an X.509 certificate in .pem format.

Run the following command to view the contents of the signed cluster and client certificates: `keytool -printcert -file $PATH TO CERTIFICATE PEM FILE`.

Validate the following requirements for the cluster and client certificates:

Prerequisite	Certificate Requirement
Mandatory fields	For the cluster certificate, consider the following key usage requirements: keyUsage = digitalSignature,keyEncipherment extendedKeyUsage = serverAuth,clientAuth For the client certificate, consider the following key usage requirements: keyUsage = digitalSignature extendedKeyUsage = clientAuth
Subject Alternate Name (SAN)	For the cluster certificate, the SAN must include the list of cluster nodes in the following format: SAN=DNS:$CLUSTER HOST1 FQDN,DNS:$CLUSTER HOST2 FQDN,DNS:$CLUSTER HOST3 FQDN The client certificate must contain the FQDNs for the cluster nodes. For the client certificate, the SAN must include the list of all Informatica nodes in the following format: SAN=DNS:$INFA DOMAIN HOST1 FQDN,DNS:$INFA DOMAIN HOST2 FQDN The client certificate must contain the FQDNs for both the domain and cluster nodes.

CLUSTER HOST FQDN represents the fully qualified domain name for the cluster gateway host, processing nodes, and data nodes in the cluster.

INFA DOMAIN HOST FQDN represents the fully qualified domain name of the Informatica domain gateway host, domain nodes, cluster gateway host, processing nodes, and data nodes in the cluster.

Note: Verify that the custom certificate location for the domain nodes contains the infa_truststore.pem file. Also, verify that all the certificates in the CA certificate chain are present in the truststore.jks and the infa_pubcert.pem files.

Generate Self-signed Custom SSL Certificates

You have a custom SSL certificate for the Informatica domain, and you want to use self-signed custom SSL certificates as the client and cluster certificates for the Informatica Cluster Service.

Use the custom SSL utility bundled with the installer to generate the required SSL certificates.

Perform the following steps to use the custom SSL utility to generate the certificates:

1. Extract `GenerateCustomSslUtility.zip` from the following location: `<Location of installer files>/properties/utils/CustomSslCertsUtility/`.
2. Set the JAVA_HOME environment variable to point to JDK 17.
3. Configure the following parameters in the `input.properties` file that you extracted from the `GenerateCustomSslUtility.zip` file:

Parameter	Description
KeystoreFile	Path to a custom keystore file along with file name. The keystore type must be in JKS. X509 format. The file must contain a single private key entry with the complete certificate chain. Verify that the file has signing capability.
KeystorePassword	Password of the custom keystore file in plain text format.

Parameter	Description
TruststoreFile	Path to a custom truststore file along with file name. the truststore type must be in JKS. X509 format. The file must contain the public certificates corresponding to the private key entry in the keystore file.
TruststorePassword	Password of the custom truststore file in plain text format.
ISPDomainKeystorePassword	Password of Informatica domain keystore in plain text format. The utility uses the password for the cluster and client keystore.jks file.
ISPDomainTruststorePassword	Password of the Informatica domain truststore in plain text format. The utility uses the password for the cluster and client truststore.jks file
KeystoreOutputDir	Represents the location of the Informatica domain custom keystore that you provided when you installed Enterprise Data Catalog.
TruststoreOutputDir	Represents the location of the Informatica domain custom truststore that you provided when you installed Enterprise Data Catalog. **Note:** Verify that the KeystoreOutputDir and TruststoreOutputDir parameters point to the same directory.
ServerNodes	Comma-separated list of fully qualified domain names of nodes that you plan to configure as data nodes, processing nodes, service hosts, and gateway node when you configure the Informatica Cluster Service.
IcsServiceName	Name of the Informatica Cluster Service.
ClientNodes	Comma-separated list of fully qualified domain names of nodes that you plan to configure as Informatica domain hosts, data nodes, processing nodes, service hosts, and gateway node.
IspDomainName	The Informatica domain name.
ClusterNodeDNSDomain	Domain name of the gateway host that you plan to configure for the Informatica Cluster Service.

4. Run the utility using the following command: `java -jar GenerateCustomSslUtility.jar -in input.properties`. The utility generates the following keys and client and cluster certificates:
   - Keys:
     - keystore.jks
     - infa_privkey.pem

     Note: The cluster keys are generated in the following directory: `<Keystore output directory>/<IcsServiceName>/cluster_certs`. The client keys are generated in the following directory: `<Keystore output directory>/<IcsServiceName>/client_certs`
   - Certificates:
     - truststore.jks
     - infa_pubcert.pem
     - infa_nodecert.pem

- infa_nodecertkey.pem

  Note: The cluster certificates are generated in the following directory: `<Truststore output directory>/<IcsServiceName>/cluster_certs`. The client certificates are generated in the following directory: `<Truststore output directory>/<IcsServiceName>/client_certs`

5. Copy the generated certificates to the required folders.
6. Assign the ownership of the following directories to the Informatica domain user:
   - <Keystore output directory>/<IcsServiceName>/cluster_certs
   - <Truststore output directory>/<IcsServiceName>/cluster_certs
   - <Keystore output directory>/<IcsServiceName>/client_certs
   - <Truststore output directory>/<IcsServiceName/client_certs

Note: Verify that all the client certificates are included in a single directory. Similarly, verify that all the cluster certificates are included in a single directory. You must also verify that the directories have the `chmod 700` permission configured and the files under the directories have the `chmod 600` permission configured.

Download and Extract the Installer Files

The installer files are distributed as compressed files. You can get the Informatica installation file and distribution packages from the FTP link contained in your fulfillment email.

Download the Informatica installation tar file and the required distribution package ZIP files from the Informatica Electronic Software Download site. You can download them to a local directory or a shared network drive that is mapped on your machine.

To download other compressed files and utilities, raise a shipping request with Informatica Global Customer Support.

Extract the Informatica installer files to a directory on your machine. The user that runs the installer must have read and write permissions on the installer files directory and execute permissions on the executable file.

Note: Ensure that you extract the installer files to a local directory as you can't run the installer from a mapped file.

Copy the ZIP files of the distribution packages to the following location: `<Informatica installer files>/source`

Note: The installer fails if the ZIP files for distribution packages aren't available in the source directory.

Verify Installer Package Checksum on UNIX

Before you run the services installer, verify the install package integrity through the cksum command. The cksum command calculates the checksum value for the installers.

Verify the checksum for the specific installer files against the checksum of the installation files downloaded from the Informatica Electronic Software Download site.

The following table lists the checksum and file size for the Informatica services for UNIX:

File	Checksum Value	File Size
informatica_10510_server_linux-x64.tar	1346986189	17774100480 bytes

A checksum mismatch can occur when there are data errors during download due to network issues or when data corruption occurs in the file on disk. For more information about the checksum errors, see HOW TO: Identify file errors after downloading Informatica installation files.

Verify the License Keys and Copy the Binary Files

Before you start the upgrade, perform the following steps:

- Verify the new license keys for the Informatica domain and Informatica Cluster Service.
- Extract the installer binary files to a directory on the machine where you plan to upgrade Enterprise Data Catalog. You must have read and write permissions on the installer files directory and execute permissions on the executable file.
- Copy the scanner binaries from the Informatica Electronic Software Download site and place it in the following directory: `<Location where you extracted the installer binary files>/source`

Verify Distribution Package Requirements (Linux and UNIX)

You can use third-party distribution packages to integrate the Informatica domain with the Hadoop or Databricks environment.

The Informatica domain and client require the distribution packages to process complex files within the Informatica domain, or to connect to Hadoop or Databricks environment when you process within the Informatica domain, or to push processing to Hadoop or Databricks environment.

If you need a distribution package, you can install it through the installer or through Integration Package Manager (the package manager) at any time.

Process within the Informatica domain

You can use the same distribution package that you used in the previous Informatica version to process complex files within the Informatica domain or to connect to the Hadoop or Databricks environment when you process within the Informatica domain.

The following adapters require distribution packages for processing within the Informatica domain:

- PowerExchange for Amazon S3
- PowerExchange for Google Cloud Storage
- PowerExchange for Google Cloud Storage for PowerCenter
- PowerExchange for Hadoop for PowerCenter
- PowerExchange for HBase
- PowerExchange for HDFS
- PowerExchange for Hive
- PowerExchange for JDBC V2
- PowerExchange for Kafka for PowerCenter
- PowerExchange for MapR-DB
- PowerExchange for Microsoft Azure Blob Storage
- PowerExchange for Microsoft Azure Data Lake Storage Gen1
- PowerExchange for Microsoft Azure Data Lake Storage Gen2

Process with Hadoop or Databricks environment

When you push processing to the Hadoop or Databricks environment, the Informatica domain and client require distribution packages. For more information about the supported distribution packages, see the Product Availability Matrix.

Complete the Prerequisites

Complete the following prerequisites before you upgrade the Informatica domain:

- Clear the configured values for the INFA_TRUSTSTORE and INFA_TRUSTSTORE_PASSWORD environment variables if the domain is enabled for Secure Sockets Layer (SSL).
- Download and install the libidn.so.11 library if you use RHEL 8.3.

Back Up Primary Key-Foreign key Enrichments Using the Migration Utility

To use the enhancements added to the primary key-foreign key enrichments in Enterprise Data Catalog, back up the enrichments using the migration utility before you upgrade to Enterprise Data Catalog version 10.5.11.

Perform the following steps to run the utility:

1. From the Informatica Electronic Software Download (ESD) site, download and extract the `Informatica_<version>_PkFkEnrichmentMigrationUtility.zip` file on the domain machine.
2. Run the following command:

```
java -jar PKFKEnrichmentsRestoreUtility.jar export <Catalog Service hostname>
<Catalog Service admin port number> <username> <password> <location to download
enrichments json> <comma separated resource type IDs> <SSL Enabled true false>
```

Note: Verify that the resource name length does not exceed the maximum file name length supported by the operating system.

Complete the Data Asset Analytics Prerequisites

If you configured Data Asset Analytics for Enterprise Data Catalog, you must configure the process termination timeout before you shut down the Informatica domain for an upgrade. You must run the Data Asset Analytics utility before you upgrade to Enterprise Data Catalog 10.5.11.

Configure the Process Termination Timeout

1. In the Administrator tool, go to the custom properties section for the domain.
2. Add the following custom property:
   ProcessTerminationTimeout=900
3. Stop the Informatica Services using the `infaservice.sh shutdown` command.
4. Start the Informatica Services using the `infaservice.sh startup` command.

Verify that the file size of the transaction or redo log files for the Data Asset Analytics repository database is large enough. For an Oracle database, ensure that the database has a minimum of three redo log files of 2 GB each.

Run the Data Asset Analytics Utility

Data Asset Analytics displays key metrics and key charts on the dashboard tabs based on pre-computed analytical data. Before you perform the upgrade, you must populate the pre-computed database tables with analytical data.

Perform the following steps to run the Data Asset Analytics utility:

1. Back up the Data Asset Analytics database schema.
2. Verify the daa.log log file and make sure that there are no unprocessed data asset events to avoid losing event data such as user logins, searches, and asset usage.
3. From the Informatica Electronic Software Download (ESD) site, download and extract the informatica_<version>_daa_utility.zip file on the Informatica domain.

   Note: If you applied a service pack, you must download the utility corresponding to the service pack version.
4. Go to the directory where you extracted the files and navigate to the `config` directory.
5. Open the input.properties file and configure the following connection parameters for the Data Asset Analytics repository.

Property	Description
databaseType	The Data Asset Analytics database type. Enter ORACLE, POSTGRESQL, or SQLSERVER.
databaseHostName	The fully qualified host name of the machine where the database is hosted. **Note:** If the value contains a special character, you must enclose the value within single quotes (').
databasePort	The port number for the database engine service.
databaseServiceName	The service name for the Oracle database.
databaseName	The database name for the PostgreSQL or SQL Server database.
databaseUserName	The name of the user account to connect to the database.
databaseAdvanceParameter	Optional. Enter the advanced JDBC connection parameters used to connect to the Data Asset Analytics repository. **Note:** You must enclose the parameters within single quotes (').

6. On a shell command line, enter the following command to run the utility from the directory where you extracted the files: `./daautil.sh`.
7. Press **1** to select the upgrade mode.
8. Enter the password to connect to the database.

Note: To know the execution status, see the daautil.log file from the directory from which you run the utility.

Step 2. Back Up the Repositories and Configuration Details

Back Up the Informatica Domain

Before you upgrade the domain, back up the configuration metadata for the Informatica domain.

Complete the following steps to back up the Informatica domain:

- Run the infasetup BackupDomain command to back up the domain configuration database tables to a file.
- Back up the metadata configuration files to any directory accessible by the machines where you install Informatica.

Informatica includes command line programs to back up and restore the domain. infasetup is located in the following directory:

```
<Informatica installation directory>/isp/bin
```

To back up the domain with infasetup, use the following syntax:

```
BackupDomain
<<-DatabaseAddress|-da> database_hostname:database_port|

<-DatabaseConnectionString|-cs> database_connection_string>

<-DatabaseUserName|-du> database_user_name

<-DatabasePassword|-dp> database_password

<-DatabaseType|-dt> database_type

[<-DatabaseServiceName|-ds> database_service_name]

<-BackupFile|-bf> backup_file_name

[<-Force|-f>]

<-DomainName|-dn> domain_name

[<-Tablespace|-ts> tablespace_name (used for IBM DB2 only)]

[<-SchemaName|-sc> schema_name (used for Microsoft SQL Server only)]

[<-DatabaseTlsEnabled|-dbtls> database_tls_enabled]

[<-DatabaseTruststorePassword|-dbtp> database_truststore_password]

[<-TrustedConnection|-tc> trusted_connection (used for Microsoft SQL Server only)]

[<-EncryptionKeyLocation|-kl> encryption_key_location]
```

The following table describes the metadata files and the locations of the files:

Metadata File	Description	Location
nodemeta.xml	Contains metadata for a node.	Stored in the `isp/config` directory on each node in the domain. If you use the same backup directory name on all the nodes, rename nodemeta.xml before copying it to the backup location. For example, you back up nodemeta.xml to the /nodebak directory on nodeA and nodeB. Rename the configuration files so that on nodeA the file is backed up to /nodebak/nodemeta_A.xml, and on nodeB the file is backed up to /nodebak/nodemeta_B.xml.
domains.infa	Contains connectivity information for the gateway nodes.	Stored in one of the following locations: - The Informatica installation directory. - The location configured through the INFA_DOMAINS_FILE environment variable.

Back Up the Model Repository

Back up the contents of the Model repository using the Administrator tool.

To back up the Model repository, select the Model Repository Service in the Administrator tool. Then, on the **Domain Actions** menu, click **Repository Contents** > **Backup**.

Back Up Catalog Data Using the Export Utility

Back up the catalog data using the export.jar utility so that you can restore the catalog data after you upgrade.

Use the export utility if you upgraded from a Hadoop cluster version to a non-Hadoop cluster version, and you want to migrate the existing catalog data to version 10.5.11. Contact Global Customer Support to download the utility.
Before you run the utility, verify the following prerequisites:

- The Informatica Cluster Service is running.
- No jobs are running in Enterprise Data Catalog. To verify, see the **Monitoring** tab in Catalog Administrator.
- The Catalog Service is running.
- If backup contains large similarity profile data, make sure that infacmd process includes 4 GB heap size. You can set the JAVA heap size value with the following property in the Infacmd.sh file: `export ICMD_JAVA_OPTS=-Xmx4096m`
- Enable the read-only mode for the Catalog Service. See the following sample curl commands to enable the service in read-only mode:
  - If the Catalog Service is SSL-enabled, use the `curl -k -u <DomainUserName>:<DomainPassword> -X POST "https://<host name>:<Catalog Service port>/access/1/catalog/catalogMode/READONLYMODE" -H "accept: */*"` command.
  - If the Catalog Service is not SSL-enabled, use the `curl -k -u <DomainUserName>:<DomainPassword> -X POST "http://<host name>:<Catalog Service port>/access/1/catalog/catalogMode/READONLYMODE" -H "accept: */*"` command.
- From your existing installation, copy the `application.properties` file from `<$INFA_HOME>/logs/<NODE_NAME>/services/CatalogService/<SERVICE_NAME>/<SCN_NAME>/spark/` to the directory where you copied the `export.jar` file.

- If the Catalog Service is SSL enabled, set the following environment variables:
  - INFA_TRUSTSTORE. Set the variable to the location of the Informatica truststore file. Default location is `$INFA_HOME/services/shared/security`.
  - INFA_TRUSTSTORE_PASSWORD. Encrypt the password that you set. See the sample command to set the encrypted password: `export INFA_TRUSTSTORE_PASSWORD="84Ve/soUbpQ/Aae5uGKXQA=="`

    Note: See the following sample command to encrypt the password: `$INFA_HOME/server/bin/pmpasswd <password>`
- If catalog data is large, run the export.jar utility with 20 GB heap size. For example, `java -Xmx20960m -jar export.jar export` For more information about performance sizing, see the *Enterprise Data Catalog Performance Tuning Guide*.
- Set the commit interval option to 48 hours when you run the export utility. For example, `java -Dexport.commit.interval.ms=172800000 -jar export.jar export`

Run the utility using the following command: - `java -jar export.jar export`

You can verify the `export.log` file generated at the location from which you ran the utility.

Back Up the Catalog Data Using the ldm backupContents Command

Back up the catalog data before you upgrade Enterprise Data Catalog. Backing up the catalog ensures the availability of the existing metadata, attributes, associations, values, schedules, and tasks in the event of an upgrade failure. Back up the catalog data to retain the catalog data from the Hadoop environment in the event of a roll back or upgrade failure. You must have the Informatica Administrator role to perform the backup.

Before you run the infacmd ldm backupContents command, verify the following prerequisites:

- Set the following environment variables if the Catalog Service is enabled for SSL:
  - INFA_TRUSTSTORE. See the following sample command to set the variable: `export INFA_TRUSTSTORE=<Location of the Informatica truststore file>` . Default location is `<Informatica installation directory>/services/shared/security`.
  - INFA_KEYSTORE. See the following sample command to set the variable: `export INFA_KEYSTORE=<Location of the keystore file>`. Default location is `<Informatica installation directory>/services/shared/security`. This variable is required only if you used custom SSL configuration. For default SSL and non-SSL configurations, you must unset the variable.
  - INFA_TRUSTSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_TRUSTSTORE_PASSWORD="84Ve/soUbpQ/Aae5uGKXQA==".`
  - INFA_KEYSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_KEYSTORE_PASSWORD="6cDE/ItyUL/Rtui9nhVRI==".` This variable is required only if you used custom SSL configuration. For default SSL and non-SSL configurations, you must unset the variable.

  You can contact Informatica Global Customer Support for the default truststore and keystore passwords.

  Note: See the following sample command to encrypt the password: `<Informatica installation directory>/server/bin/pmpasswd <password>`
- Estimate the HDFS disk space consumption of the catalog data. Run the following command to view the directory size: `hdfs -du /Informatica/LDM/<SCN>`.
- Make sure that sufficient disk space is available in a domain host so that backup does not run out of disk space.

The infacmd ldm BackupContents command as shown in the following sample, creates a .zip file of the directory in the local machine: `./infacmd.sh LDM backupContents -dn <name of the Informatica domain> -un <username to connect to the Informatica domain> -pd <password for the username> -sn <name of the Catalog Service> -of <name of the backup ZIP file>`
For more information about the syntax, parameters, and description of the command, see *Informatica Command Reference*.

Note: If backup fails with an out of memory issue, you must increase the JAVA heap size for **ICMD_JAVA_OPTS** property in the following `infacmd.sh` file. For example, ICMD_JAVA_OPTS=-Xmx20g.

Back Up the SSL Configuration

Back up the following SSL configuration files from the default location: `<Informatica installation directory>/services/shared/security`.

- Back up the domain truststore and keystore files.
- Back up the default keystore files.
- Back up all the keystore and truststore files that the application services use.

Back Up the Domain Information

Back up the following files:

- `<Informatica installation directory>/isp/config/keys/siteKey`
- `<Informatica installation directory>/domains.infa`
- `<Informatica installation directory>/tomcat/conf/server.xml`
- `<Informatica installation directory>/isp/config/nodemeta.xml`

Note: You must back up the binaries from the Informatica installation directory.

Back Up the Data Asset Analytics Database Schema

If you have configured the repository for Data Asset Analytics, perform a backup of the Data Asset Analytics database schema.

Note: Use a database tool to take backup.

Back Up MetaDex

If you have configured the repository for MetaDex, back up the MetaDex repository.

Note: In version 10.4, the MetaDex job scheduler was disabled by default. Effective in version 10.5, by default, the job scheduler removes jobs every six hours, leaving only the last three jobs. You can change that behavior by editing parameters in the repository.properties file. For more information, see "Removing jobs" in the *MetaDex Administrator Guide*.

1. Use a database tool to perform a backup of the MetaDex database schema.

   Note: Ensure that you create a backup of the entire database, not just the data content.
2. Create a backup of the following configuration files:
   - `<workspace>/etc`
   - `<workspace>/setup.env`
3. Create a backup of the connection registry.

4. In the MetaDex tool, export your configurations and projects.
   See "Projects page" and "Configuration page" in the *MetaDex Administrator Guide*.

Back Up Tableau Resource Enrichments Using the Migration Utility

If the catalog contains Tableau resources, you can back up the Tableau resource enrichments using the Tableau migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. From the Informatica Electronic Software Download (ESD) site, download and extract the `Informatica_<version>_TableauEnrichmentMigrationUtility.zip` file on the domain machine.
2. Run the following command to back up the Tableau resource enrichments:

```
java -jar EnrichmentUtility.jar export <Catalog Service hostname> <Catalog Service admin port number> <username> <password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
```

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Back Up Qlik Sense Resource Enrichments Using the Migration Utility

If the catalog contains Qlik Sense resources and you use MITI version 11.x, you must either back up the Qlik Sense resource enrichments using the Qlik Sense migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. Download and extract the `QliksenseEnrichmentsCLIMigrator.jar` file on the domain machine, from the following location: `<INFA_HOME>/services/CatalogService/ScannerBinaries/qliksense-utility/lib/`
2. Run the following command to back up the Qlik Sense resource enrichments:

```
java -jar QliksenseEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service admin port number> <username> <password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
```

The utility run time depends on the number of assets in the resource. If there are many assets in the resource, you can configure the Java memory to back up the enrichments.

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Back Up Salesforce Resource Enrichments Using the Migration Utility

If the catalog contains Salesforce resources, you must either back up the Salesforce resource enrichments using the Salesforce migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. Download and extract the `SalesforceEnrichmentcliMigrationUtility.zip` file on the domain machine from the following location: `<INFA_HOME>/services/CatalogService/ScannerBinaries/`

2. Run the following command to back up the Salesforce resource enrichments:

```
java -jar EnrichmentUtility.jar export <Catalog Service hostname> <Catalog Service admin port number> <username> <password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
```

The utility run time depends on the number of assets in the resource. If there are many assets in the resource, you can configure the Java memory to back up the enrichments.

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Back Up Erwin Resource Enrichments Using the Migration Utility

If the catalog contains Erwin resources, you must either back up the Erwin resource enrichments using the migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. Copy and extract the `ErwinEnrichmentsCLIMigrator.jar` file from the following location on the Informatica domain machine:
   `<Informatica installation directory>/services/CatalogService/ScannerBinaries/erwin-utility/lib`
2. Run the following command to back up the Erwin resource enrichments:

```
java -jar ErwinEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service port number> <Catalog administrator user name> <Catalog administrator password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
```

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Back Up IBM Cognos Resource Enrichments Using the Migration Utility

If the catalog contains IBM Cognos resources, you must either back up the resource enrichments using the Cognos migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. Copy and extract the `CognosEnrichmentsCLIMigrator.jar` file from the following location on the Informatica domain machine:
   `<Informatica installation directory>/services/CatalogService/ScannerBinaries/cognos-utility/lib`
2. Run the following command to back up the IBM Cognos resource enrichments:

```
java -jar CognosEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service port number> <Catalog administrator user name> <Catalog administrator password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
```

The utility run time depends on the number of assets in the resource.

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Back Up Relational Resource Enrichments Using the Migration Utility

If the catalog contains IBM Netezza and Oracle resources, you must either back up the resource enrichments using the Relational migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. Copy and extract the `RelationalEnrichmentsCLIMigrator.jar` file from the following location on the Informatica domain machine:
   `<Informatica installation directory>/services/CatalogService/ScannerBinaries/relational-utility/lib`
2. Run the following command to back up the IBM Netezza and Oracle resource enrichments:

```
java -jar RelationalEnrichmentsCLIMigrator.jar export <Catalog Service hostname>
<Catalog Service port number> <Catalog administrator user name> <Catalog
administrator password> <location to download enrichments json> <resourceName> <SSL
Enabled true/false>
```

The utility run time depends on the number of assets in the resource.

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Back Up Cloudera Distributed Hadoop (CDH) 6.3 Version Binaries

Informatica dropped support for Cloudera Distributed Hadoop (CDH) 6.3 version in Enterprise Data Catalog. You cannot run scanners that are compatible with Cloudera CDH or Hortonworks CDH. To run these scanners post-upgrade in Enterprise Data Catalog, you need to back up and restore Cloudera Distributed Hadoop (CDH) version 6.3 binaries.

Before you upgrade, back up Cloudera Distributed Hadoop (CDH) version 6.3 binaries from `<infa_home>/services/shared/hadoop/`.

Step 3. Prepare to Upgrade

Disable the Catalog Service

To disable the service, select the Catalog Service in Informatica Administrator and click **Actions** > **Disable Service**.

Disable the Informatica Application Services

Disable the following Informatica application services:

1. Informatica Cluster Service
2. Content Management Service
3. Data Integration Service
4. Model Repository Service

To disable an application service, select the service in Informatica Administrator and click **Actions** > **Disable Service**.

Clean the Cluster

Clean the cluster using the infacmd ihs cleanCluster command. After you run the infacmd ihs cleanCluster command, log in to each cluster node as a non-root user with sudo privileges and delete the contents of the `/tmp` directory that the yarn user owns. To delete the contents, run the following command as a non-root user with sudo privileges: `find /tmp/ -user yarn -exec rm -fr {} \;`.

ihs cleanCluster Command Syntax and Description

The infacmd ihs cleanCluster command uses the following syntax:

```
cleanCluster
<-DomainName|-dn> domain_name
<-UserName|-un> user_name
<-Password|-pd> password
<-ServiceName|-sn> service_name
[<-SecurityDomain|-sdn> security_domain]
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

The following table describes infacmd ihs cleanCluster options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-ServiceName -sn	service_name	Required. Informatica Cluster Service name.

Option	Argument	Description
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation. The name of the security domain is the same as the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or reestablish a connection to the domain. You can set the resilience timeout period with the -re option or the environment variable INFA_CLIENT_RESILIENCE_TIMEOUT. If you set the resilience timeout period with both methods, the -re option takes precedence.

Disable and Uninstall the PostgreSQL Service

You must disable PostgreSQL before you upgrade.

Disable PostgreSQL Service

Use the following command on the machine where Informatica Administrator runs to disable the PostgreSQL Service: `service postgresql-<PostgreSQL version> stop`. For example, if you are using PostgreSQL version 9.6, use the following command: `service postgresql-9.6 stop` or `systemctl disable postgresql-9.6 or systemctl disable postgresql-x.x.`

Uninstall PostgreSQL Service

Use the following command on the machine where Informatica Administrator runs to uninstall PostgreSQL Service: `sudo yum remove postgre*` and `sudo rm -rf /var/lib/pgsql/*`.

Note: To verify that the postgresql service is stopped, run the following command: `systemctl postgresql-<PostgreSQL version> status`.

Shut Down the Informatica Domain

From Informatica Administrator, shut down the Informatica domain.

Step 4. Domain Upgrade Overview

The upgrade wizard reads the domain information from files in the previous version and uses the same settings to configure the files for the upgrade. It upgrades the tables of the domain configuration repository in the same database as the previous version. You can upgrade the Informatica domain in console mode or silent mode.

Complete the pre-upgrade tasks before you start the upgrade. Run the installer on all machines that host previous versions of Informatica that you want to upgrade.

Note: In a multi-node domain, upgrade the gateway node before you upgrade other nodes and upgrade all worker nodes in parallel.

You can perform the upgrade from the directory where you downloaded the installation files.

After you upgrade the domain, upgrade the Informatica client tool to the same Informatica version.

Before you perform the upgrade, verify that the Informatica domain truststore file contains only the TrustedCertEntry. You must delete all the PrivateKeyEntry entries from the file.

Run the Cluster Validation Utility

Run the Cluster Validation utility to validate the prerequisites to upgrade Enterprise Data Catalog.

Perform the following steps to run the utility:

1. Verify that each cluster node has a minimum of four CPU cores, 16 GB of memory, and 60 GB of free disk space.

   Note: The hardware requirements listed here are the minimum recommended values to install the product and start the services. The requirements vary based on the size of deployment. See the Enterprise Data Catalog Performance Tuning Guide for additional information about hardware requirements based on deployment sizes.
2. Verify that the root directory (`/`) has a minimum of 10 GB of free disk space.
3. Download the Informatica Cluster Validation Utility for version 10.5.11 from the Informatica Electronic Software Download (ESD) site.

   Note: If you applied a service pack, you must download the utility corresponding to the service pack version.
4. Extract the ZIP file on the Informatica domain that you plan to upgrade.
5. Go to the directory where you extracted the files and open the input.properties file.

   Note: You must have read and write permissions on the directory and execute permissions on the executable files.
6. In the input.properties file, configure the required properties.
7. Disable the firewall on each host in the cluster.
8. Enable passwordless SSH between the following nodes:
   - The node that hosts the Informatica domain and the node that hosts the gateway.
   - The gateway host and the data nodes.
   - The gateway host and the processing nodes.
   - The backup nodes and the gateway node.
   - All nodes in the cluster and the data nodes.

   Note: If you enabled advanced configuration for the Informatica Cluster Service, and the MongoDB, PostgreSQL, Solr, and Nomad services run on different hosts, enable passwordless access from the gateway host to the hosts that run these services.
9. On a shell command line, go to the directory where you extracted the utility files and run the following command: `java -jar InformaticaClusterValidationUtility.jar -in <Location of the input.properties file>`.

The utility provides the following options:

Option	Description
-hp	Display the help contents.
-in	Specify the location of the input.properties file.
-lf	Optional. Specify the directory to store the log files generated by the utility.
-v	Optional. Enable the debug log.
-vf	Optional. If you moved the validation scripts to a different directory, specify the location where the validation scripts are stored.

Note: If you see a permission validation error message for the cluster custom directory, you can ignore the error.

10. Verify the validation status from the following log files located in the directory where you ran the utility:
    - clusterPrevalidation.log
    - clusterPrevalidation.err

Upgrade in Console Mode

You can upgrade in console mode on UNIX or Linux.

When you run the installer in console mode, the words quit, back, and help are reserved words. Do not use them as input text.

1. Log in to the machine with the same user account that you used to install the previous version.
2. Stop all processes that access the directory and subdirectories of the Informatica product to upgrade, including command prompts and tail logs.
3. On a shell command line, run the install file.

   The installer displays the message to verify that the locale environment variables are set.
4. If the environment variables are not set, press **n** to exit the installer and set them as required.

   If the environment variables are set, press **y** to continue.
5. Press **2** to upgrade Informatica.
6. Press **2** to skip running the Informatica Upgrade Advisor.

   Informatica provides utilities to facilitate the Informatica services installation process. You can run the utility before you upgrade Informatica services. The Informatica Upgrade Advisor helps to validate the services and checks for obsolete services in the domain before you perform an upgrade.

   The installer displays a warning to shut down the Informatica domain that you want to upgrade before you continue the upgrade.
7. Read the terms and conditions of Informatica product usage toolkit and press **2** to continue the upgrade.

   Subject to your opt-out rights described below, the software will automatically transmit to Informatica in the USA the information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at

https://www.informatica.com/in/privacy-policy.html. You can disable usage collection in the Administrator tool.

8. The **Upgrade Prerequisites** page displays the upgrade system requirements.

   Verify the requirements before you continue the upgrade.

9. Review the upgrade information and press **Enter** to continue.

   The **Upgrade Directory** panel appears.

10. At the prompt, enter the directory of the Informatica version you want to upgrade and the directory in which you want to upgrade to Informatica 10.5.11.

    The following table describes the directories you must specify:

Directory	Description
Directory of the Informatica product to upgrade	Directory that contains the version of Informatica services that you want to upgrade.
Directory for Informatica 10.5.11	Directory in which to install Informatica 10.5.11. Enter the absolute path for the installation directory. The directory cannot be the same as the directory that contains the previous version of Informatica services.The directory names in the path must not contain spaces or the following special characters: @|* $ # ! % ( ) { } [ ] , ; ' **Note:** If you use a special character such as á or €, unexpected results might occur at run time.

11. From the following installation options, select the upgrade environment based on your requirement:
    - Sandbox. Used for proof of concepts or as a sandbox with minimal users.
    - Development. Used for a design environment.
    - Test. Used for high-volume processing environment that resembles a production environment.
    - Production. Used for production environments with high-volume processing and high levels of concurrency meant for end users.
12. Select if you want to change the node host name, port numbers, or domain configuration repository. Press **1** to use the same node configuration as the previous version. Press **2** to change the node configuration.

    The **Component Selection** panel appears.

13. Press **1** to upgrade Informatica services.
14. If you created a Catalog Service on the node, you must validate the Catalog backup before you upgrade. Press **1** if the node does not have a Catalog Service or press **2** if the node has a Catalog Service and press **Enter** to continue.
15. Press **2** to confirm that you backed up the catalog using both the export utility and the infacmd LDM backupContents command.
16. Specify the location of the directory containing the catalog backup.

    The catalog backup is validated.

17. Press **1** if you want to configure the MetaDex repository and provide the following details:

Property	Description
Database type	Specify the database that you want to configure as the repository for MetaDex. You can select from the following options: - Oracle - SQLServer
Database user name	User name configured to access the database before upgrade.
Database password	Password configured for the user name.
Schema name	Not applicable if you selected Oracle as the database. Specify the schema that you want to use.
Is the database secure?	Specify if the database is enabled for SSL.
Path to the database truststore file	Applies if you specified that the database is enabled for SSL. Provide the fully qualified path to the database truststore file.
Truststore password	Applies if you specified that the database is enabled for SSL Password to access the truststore file.

Secure JDBC parameters	Applies if you specified that the database is enabled for SSL. Specify the secure database parameters as shown in the following sample:
	`EncryptionMethod=SSL;HostNameInCertificate=ORATLS.informatica.com;`
	`ValidateServerCertificate=true;CryptoProtocolVersion=TLSv1.2;`
	Enterprise Data Catalog appends the secure JDBC parameters to the JDBC connect string. If you include the secure JDBC parameters directly in the connection string, do not enter any parameter in the Secure JDBC Parameters field.

JDBC connect string	The JDBC connection string to connect to the database. Use the following formats to specify the connection strings: - Oracle. jdbc:Informatica:oracle://<host name> :<port number>;ServiceName= - SQLServer. jdbc:Informatica: sqlserver://<host name>: <portnumber>;SelectMethod=cursor;DatabaseName=
MetaDex repository server port	A unique port for the MetaDex repository server. Default is 48090.

Property	Description
Enable secure communication for the MetaDex repository ?	Specify if TLS is enabled for communication with the MetaDex repository.
SSL certificates to secure the MetaDex repository?	Specify if you want to use the default SSL certificates or custom SSL certificates to secure the MetaDex repository. Specify the following properties if you want to use custom SSL certificates: - Keystore type. Specify the type of keystore. - Keystore alias. Unique alias to access the keystore entry. - Keystore file. Specify the keystore file that contains the required keys and certificates. - Keystore password. Password to access the keystore file.

If you did not opt to configure the MetaDex repository during installation, you can configure the repository after you complete the installation. Perform the steps listed in "Configure the MetaDex Repository (Optional)" on page 65.

Note: Ensure that the user has write permissions to the nodemeta.xml node configuration file located in `<Informatica installation directory>/isp/config`.

18. Choose whether you want to install distribution packages through the Informatica installer.
    - Press **1** if you don't need distribution packages or if you want to install them later.
    - Press **2** if you want to install distribution packages through the installer.

    Default is 1.
19. If you choose to install distribution packages, select one or more packages from the list that you want to install. Separate multiple packages with a comma.

    Default is 1.
20. Press **Enter**.

    The **Domain Security Encryption Key** panel appears.
21. Enter the user name and password for the Informatica domain.

    The **Pre-installation Summary** page appears.
22. Review the upgrade information and press **Enter** to continue.

    After the domain upgrade completes, the **Domain Configuration Repository** panel appears.
23. Press **Enter** to continue the upgrade.

    The installer copies the server files to the Informatica 10.5.11 installation directory.

    The installer displays the database and user account information for the domain configuration repository to upgrade. It displays the database connection string for the domain configuration repository based on how the connection string of the previous version was created at installation:
    - If the previous version used a JDBC URL at installation, the installer displays the JDBC connection properties, including the database address and service name.
    - If the previous version used a custom JDBC connection string at installation, the installer displays the custom connection string.
24. If you use a JDBC URL, you can specify additional parameters to include in the connection string.

    If you use a custom connection string, you cannot specify additional parameters.
25. Review the information and press **Enter**.

    The upgrade completes and the **Post-installation Summary** panel appears.

26. Press **Enter** to complete the installation procedure and exit the installer.

Review the upgrade.log file to get more information about the tasks performed by the upgrade wizard and to view the configuration of installed components.

Apply Latest Service Pack or Cumulative Patch

Apply the latest available service pack or cumulative patch after you upgrade the domain.

Configure the MetaDex Repository (Optional)

If you did not configure the MetaDex repository during upgrade, you can configure the repository after you upgrade Enterprise Data Catalog.

Perform the following steps to configure the repository:

1. Go to the directory where you extracted the installer files and navigate to the following sub directory: `\properties\MetaDex\`.
2. Open the `MetaDex.properties` file and configure the following properties:

Property	Description
USER_INSTALL_DIR	Directory in which Informatica is installed.
MDX_DB_UNAME	User name configured to access the database before upgrade.
MDX_DB_PASSWD	Password to access the database.
MDX_DB_TYPE	The type of database that you want to configure. Specify either `Oracle` or `MSSQLServer`.
MDX_SQLSERVER_SCHEMA_NAME	Applies if MDX_DB_TYPE=MSSQLServer. Specify the name of the schema that you want to use. If you do not specify a schema, Enterprise Data Catalog uses the default schema.
MDX_DB_CUSTOM_STRING	The JDBC connection string to connect to the database. Use the following formats to specify the connection strings: - Oracle. jdbc:informatica:oracle: //<host name>:<port number>;ServiceName= - SQLServer. jdbc:informatica:sqlserver: //<host name>:<portnumber>;SelectMethod=cursor;DatabaseName=
MDX_SERVER_PORT	A unique port for the MetaDex repository server. Default is 48090.
MDX_SSL_ENABLED	Specify `true` if the database is enabled for SSL, else specify `false`. Default is false.
MDX_DB_TRUSTSTORE_FILE	Applies if you specified that the database is enabled for SSL. Provide the fully qualified path to the database truststore file.
MDX_DB_TRUSTSTORE_PASSWD	Applies if you specified that the database is enabled for SSL Password to access the truststore file.

Property	Description

MDX_SSL_DEFAULT_STRING	Applies if you specified that the database is enabled for SSL. Specify the secure database parameters as shown in the following sample:
	`EncryptionMethod=SSL;HostNameInCertificate=`
	`ORATLS.informatica.com;`
	`ValidateServerCertificate=true;`
	`CryptoProtocolVersion=TLSv1.2;`
	Enterprise Data Catalog appends the secure JDBC parameters to the JDBC connect string. If you include the secure JDBC parameters directly in the connection string, do not enter any parameter in the Secure JDBC Parameters field.

MDX_CUSTOM_SELECTION	Applies if you specified that the database is enabled for SSL. Specify if you want to use the default SSL certificates or custom SSL certificates to secure the repository. Set the property to `true` if you want to use custom SSL certificates. Set the property to `false` if you want to use the default SSL certificate.
MDX_KEYSTORE_FILE	Applies if you set MDX_CUSTOM_SELECTION as `true`. Specify the keystore file that contains the required keys and certificates.
MDX_KEYSTORE_PASSWD	Applies if you set MDX_CUSTOM_SELECTION as `true`. Password to access the keystore file.
MDX_KEYSTORE_TYPE	Applies if you set MDX_CUSTOM_SELECTION as `true`. Specify the type of keystore.
MDX_KEYSTORE_ALIAS	Applies if you set MDX_CUSTOM_SELECTION as `true`. Unique alias to access the keystore entry.

3. Enter the following command: `./configureMetaDex.sh`

   Note: Install MetaDex on the same node where Catalog Service resides.
4. If Catalog Service is running, recycle the service.

Copy the Third-party Resource Drivers

Complete the following post requisite after you upgrade the Informatica domain:

- Copy the third-party JAR or ZIP files that you had configured for resources such as Teradata, JDBC, and IBM Netezza from the following location `<Informatica installation directory>services/CatalogService/ScannerBinaries` to the same location in the machine that hosts the upgraded Informatica domain.

Enable and Upgrade the Application Services

To enable an application service, select the service in Informatica Administrator and click **Actions** > **Enable Service**. Upgrade the application services after you upgrade all nodes.

Enable the following Informatica application services:

1. Model Repository Service
2. Data Integration Service
3. Content Management Service

After you enable the Informatica application services, run the service upgrade wizard to upgrade the services.

Note: You must disable Data Asset Analytics before you upgrade the Informatica Cluster Service. To disable Data Asset Analytics, select the Catalog Service in Informatica Administrator. Edit the configuration properties for Data Asset Analytics and clear the **Enable Data Asset Analytics** check box.

Run the Upgrade Wizard

Use the service upgrade wizard to upgrade application services and the database contents of the databases associated with the services. The service upgrade wizard displays upgraded services in a list along with services and associated databases that require an upgrade. You can also save the current or previous upgrade report.

1. In the Informatica Administrator header area click **Manage** > **Upgrade**.
2. Select the application services and associated databases to upgrade.
3. Optionally, specify if you want to **Automatically recycle services after upgrade**.

   If you choose to automatically recycle application services after the upgrade, the upgrade wizard restarts the services after they have been upgraded.
4. Click **Next**.
5. If dependency errors exist, the **Dependency Errors** dialog box appears. Review the dependency errors and click **OK**. Then, resolve dependency errors and click **Next**.
6. Enter the repository login information.
7. Click **Next**.

   The service upgrade wizard upgrades each application service and associated database and displays the status and processing details.
8. When the upgrade completes, the **Summary** section displays the list of application services and their upgrade status. Click each service to view the upgrade details in the **Service Details** section.
9. Optionally, click **Save Report** to save the upgrade details to a file.

   If you choose not to save the report, you can click **Save Previous Report** the next time you launch the service upgrade wizard.
10. Click **Close**.
11. If you did not choose to automatically recycle application services after the upgrade, restart the upgraded services.

You can view the upgrade report and save the upgrade report. The second time you run the service upgrade wizard, the Save Previous Report option appears in the service upgrade wizard. If you did not save the upgrade report after upgrading services, you can select this option to view or save the previous upgrade report.

Add the Informatica Domain License

Perform the following steps to add the new license for the Informatica domain:

1. Log in to Informatica Administrator.
2. Select the Informatica domain.
3. Click **New** > **License**.
4. Provide a name for the new license.
5. Select the new license key from the **License** drop-down list.

Update the Informatica Cluster Service License

Perform the following steps to update the license for the Informatica Cluster Service:

1. Log in to Informatica Administrator.
2. Select the domain.
3. Select the existing license and click **Assigned Services**.
4. Click **Edit**.
5. Select the Informatica Cluster Service and click **Remove**.
6. Click **OK** to confirm.
7. Select the new license and click **Assigned Services**.
8. Click **Edit**.
9. Select the Informatica Cluster Service and click **Add**.
10. Click **OK** to confirm.

Step 5. Complete the Informatica Domain Upgrade

Verify the Gateway User Prerequisites

The gateway user must be a non-root user with sudo access. You must enable a passwordless SSH connection between the Informatica domain and the gateway host for the gateway user. The gateway user must have the required privileges to run the ping command.

Configure Sudo Privileges for the Gateway User (Optional)

If you plan to change the properties for the Informatica Cluster Service or replace the SSL certificates configured for the Informatica Cluster Service, assign sudo privileges to the gateway user.

The following commands require sudo privileges for the gateway user when you enable the Informatica Cluster Service for the first time:

- mkdir
- chown
- chmod
- echo
- systemctl
- cp
- mv
- sysctl
- rm
- tee
- sed
- sshd

To configure sudo privileges for the commands, you must add the commands to the `/etc/sudoers` file as shown in the following sample: `/bin/mkdir, /bin/chown, /bin/chmod, /bin/echo, /bin/systemctl, /bin/cp, /bin/mv, /usr/sbin/sysctl, /bin/rm, /usr/bin/tee, /usr/bin/sed, /usr/sbin/sshd`.

Note: The `tee` command does not require sudo privileges in 10.5.1.1 and later versions.

You can determine the directory where each command is located using the `which <command name>` command.

After you enable the Informatica Cluster Service for the first time, you can choose to disable the validation for sudo permissions for the Informatica Cluster Service. To disable the validation, configure the following custom property for the Informatica Cluster Service: `IcsCustomOptions.IcsGatewayUserSudoEnabled` and set the value to false. After you disable the validation, sudo privileges are not required when you restart the Informatica Cluster Service. However, to shut down the Informatica Cluster Service, sudo permission is still required for the `systemctl` command.

To run the `infacmd ics cleanCluster` command to clean the Informatica Cluster Service, sudo permission must be configured for the `systemctl` and `rm` commands.

Note: The services associated with the Informatica Cluster Service such as Apache Solr, PostgreSQL, and Nomad require a non-root user with sudo privileges to run the services. Sudo allows a user to run programs or commands with elevated privileges for a specific time frame. Enterprise Data Catalog requires a non-root user with sudo permissions to run certain commands when performing administrative tasks such as installation, upgrade, and service status monitoring. For more information on the commands used and the role of each command during the upgrade, see the Enable sudo privileges for a non-root user in Enterprise Data Catalog KB article.

Complete the Default SSL Configuration

If the Informatica domain is enabled for SSL using the default SSL certificates, you can configure the Informatica Cluster Service and Catalog Service to use the new keystore.
To configure the services to use the 10.5.11 keystore, perform the following steps:

1. Disable the Informatica Cluster Service and the Catalog Service.
2. Edit the process options for the Informatica Cluster Service and Catalog Service and provide the location of the 10.5.11 `Default.keystore` file. Default path to the file is `<Informatica installation directory>/tomcat/conf/Default.keystore`.
3. From the 10.5.11 `<Informatica installation directory>/services/shared/security/infa_truststore.jks` file, remove the existing value configured for the *infa_dflt* alias and import the `<Informatica installation directory>/tomcat/conf/Default.keystore` file.
4. Enable the Informatica Cluster Service and Catalog Service.

Step 6. Enable and Upgrade Enterprise Data Catalog Services

Enable the Informatica Cluster Service

To enable the application service, select the service in Informatica Administrator and click **Actions** > **Enable Service**.

Enable and Upgrade the Catalog Service

To enable and upgrade the Catalog Service, select the service and then click **Enable Service** from the **Actions** menu in Informatica Administrator.
After you start the Catalog Service, Informatica Administrator prompts you to upgrade the Catalog Service.

Perform the following steps:

1. Upgrade the Catalog Service.
2. Enable the email service if you had configured the service for the Catalog Service.

Note:

- The installer includes Azul JDK. After upgrade, Enterprise Data Catalog uses Azul JDK on all cluster nodes.
- If you enabled Data Asset Analytics, it takes longer to start the Catalog Service. The service will not be available until the Data Asset Analytics repository upgrade is complete.
- To know the status of the Data Asset Analytics repository upgrade, review the daa_upgrade.log file from the installation directory.
- If you enabled Data Asset Analytics, run the "Complete the Data Asset Analytics Prerequisites" on page 49.

Step 7. Migrate and Verify the Catalog Data

Migrate the Catalog Data

Use the infacmd.sh LDM migrateContents command to migrate the data that you backed up using the export utility.
Perform the following steps to migrate catalog data:

1. Set the following environment variables:
   - INFA_TRUSTSTORE. See the following sample command to set the variable: `export INFA_TRUSTSTORE= <Location of the Informatica truststore file>`. Default location is `<Informatica installation directory>/services/shared/security`.
   - INFA_KEYSTORE. See the following sample command to set the variable: `export INFA_KEYSTORE=<Location of the keystore file>`. Default location is `<Informatica installation directory>/services/shared/security`. This variable is required only if you used custom SSL configuration for the Informatica domain. For default SSL and non-SSL configurations, you must unset the variable.
   - INFA_TRUSTSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_TRUSTSTORE_PASSWORD="84Ve/soUbpQ/Aae5uGKXQA=="`.
   - INFA_KEYSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_KEYSTORE_PASSWORD="6cDE/ItyUL/Rtui9nhVRI=="`. This variable is required only if you used custom SSL configuration for the Informatica domain. For default SSL and non-SSL configurations, you must unset the variable.

You can contact Informatica Global Customer Support for the default truststore and keystore passwords.

Note: See the following sample command to encrypt the password: `<Informatica installation directory>/server/bin/pmpasswd <password>`

2. If exported catalog data is large, update the `xdocs.content.systemProperties=-Xmx20G` property in the `<Informatica installation directory>/services/CatalogService/Binaries/MigrationModuleConfigurations.properties` file, and then reset `ICMD_JAVA_OPTS=-Xmx20g` property in the `<Informatica installation directory>/isp/bin/infacmd.sh` file. For more information about performance sizing, see *Enterprise Data Catalog Performance Tuning Guide*.
3. Run the infacmd ldm migrateContents command as shown: ./infacmd.sh ldm migrateContents -un <Informatica Domain User Name> -pd <Informatica Domain Password> -dn <Informatica Domain Name> -sn <Name of the Catalog Service> -id <Export Directory>

Note: The <Export Directory> signifies the directory where the backed up content is available.

If you want to tune the performance parameters for migrating data or skipping failed resources, modify the parameters specified in the `MigrationModuleConfigurations.properties` file available at the following location: `<Informatica installation directory>/services/CatalogService/Binaries`

See the following table for more information about the performance parameters:

Property	Description
enableModule=	Set the value to true if you want to enable a specific module, else, set the property to false. For example, similarity.content.enableModule=true enables the migration of similarity profiling data.
Dblacklist=	List the resources that you want to skip in the following format: -Dblacklist=<resource name1>,<resource name2>
DmaxParallelRestores=	Specify the number of threads that you want to run in parallel when you migrate data.
systemProperties=	Use the property to specify the system resources that you want to use. For example, xdocs.content.systemProperties=-Xmx6G -XX:+HeapDumpOnOutOfMemoryError

Verify the Migrated Catalog Data

You can use the `infacmd migrateContents -verify` command as shown in the following sample to verify the migrated content: `./infacmd.sh LDM migrateContents -un Administrator -pd Administrator -dn Domain -sn CS -id /data/Installer1050/properties/utils/upgrade/EDC/export -verify`.

For more information about the syntax, parameters, and description of the command, see the *Informatica Command Reference Guide*.

After you run the `infacmd migrateContents -verify` command, the command produces a file named `results.txt`.

The following table describes the messages that appear in the `results.txt` file:

Error Message	Description
`missing from backup`	Indicates that some resources were scanned before verifying the migrated catalog data.
`missing from warehouse`	Indicates that assets were not included in version 10.5.11.

Note: Make sure that the resource is not purged or reloaded before you to verify the migrated catalog data.

CHAPTER 4

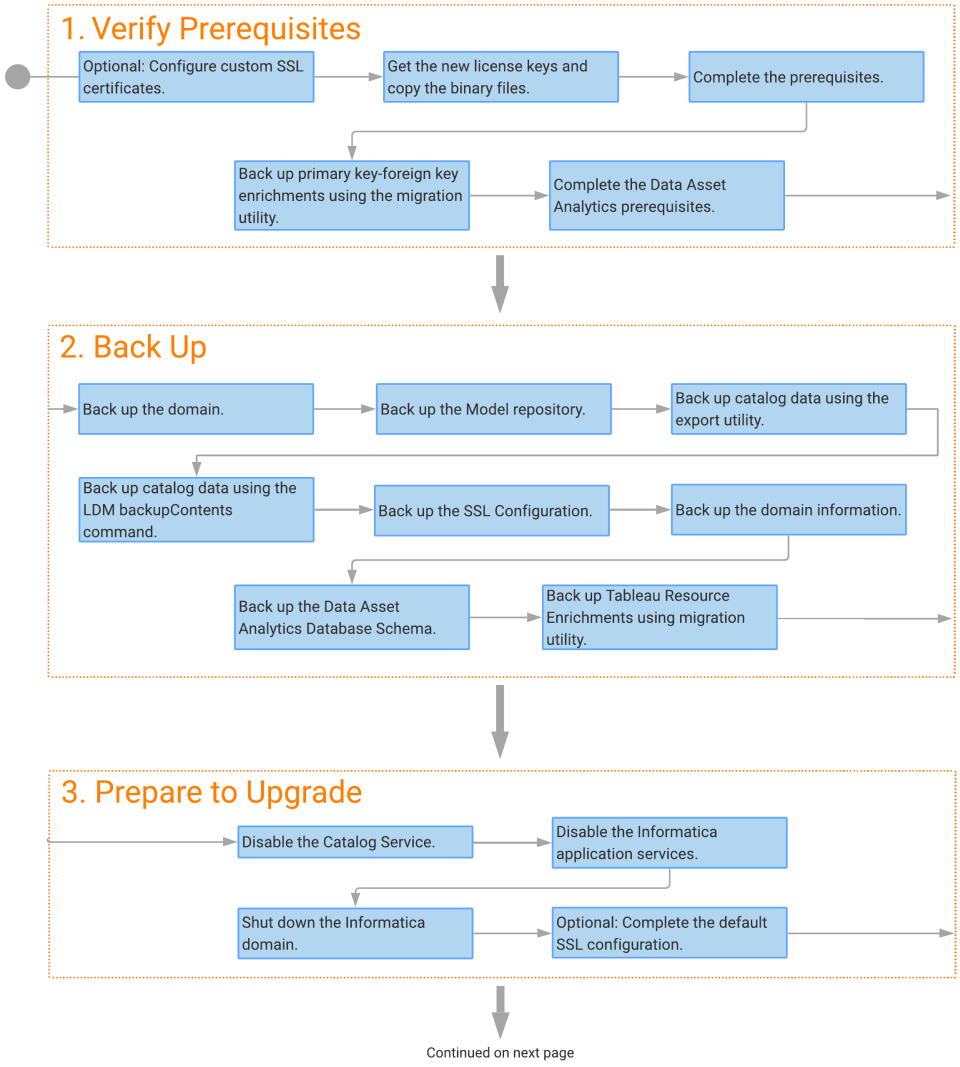
Upgrade Enterprise Data Catalog in an External Cluster

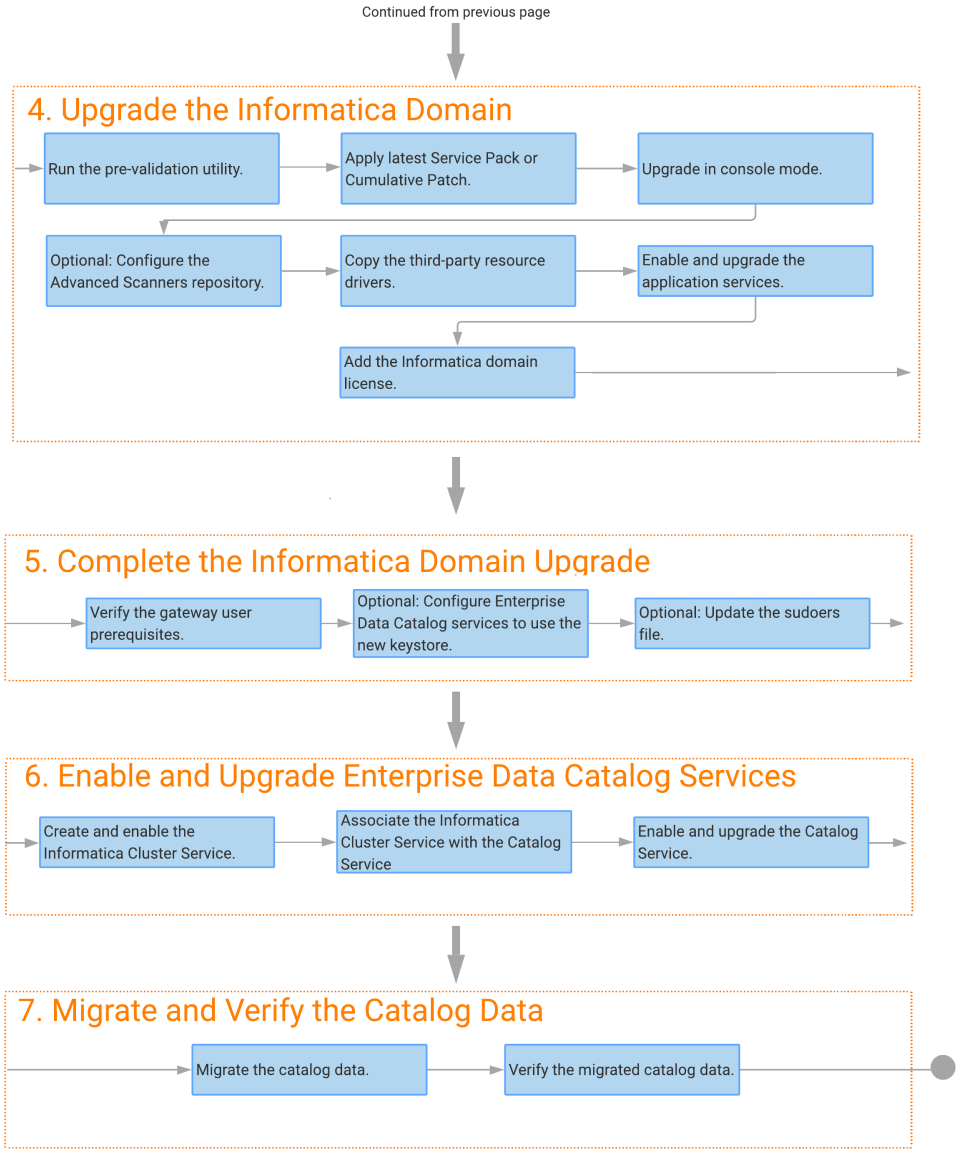
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Upgrade Enterprise Data Catalog in an External Cluster

The following image shows the steps that you need to perform to upgrade Enterprise Data Catalog:





Step 1. Verify Prerequisites to Upgrade

Before you start the upgrade as a non-root user with sudo privileges, validate the following prerequisites:

- Verify that at least 100 GB of free disk space is available on the machine where the Informatica domain runs.
- Verify that Java Development Kit (JDK) 17 is installed.
- If the cluster is enabled for Kerberos and SSL, back up the Informatica domain truststore and default keystore files.
- Verify that the connection to the domain configuration repository is available during the upgrade.
- Verify that ntpd is synchronized between the Informatica domain node and the cluster nodes.

- If you are upgrading from a cluster enabled for Kerberos and SSL, take a backup the Informatica domain truststore and default keystore files.
- Install the following applications and packages on all nodes before you upgrade Enterprise Data Catalog:
  - Bash shell
  - rsync
  - libcurl
  - xz-libs
  - systemctl
  - zip
  - unzip
  - tar
  - wget
  - scp
  - rpm
  - curl
  - nslookup
  - md5sum
  - netstat
  - ping
  - ifconfig
  - cksum
  - dnsdomainname
  - libncurses5
  - OpenSSL version 1.0.1e-30.el6_6.5.x86_64 or later.

  OpenSSL 3.0 is not supported.

  If the machine on which the PostgreSQL service runs is FIPS enabled, verify that compat-openssl10 is installed.

  Verify that the $PATH variable points to the `/usr/bin` directory to use the correct version of Linux OpenSSL.

  In 10.5.1.0.1 and later versions, rsync is used instead of scp for file transfers.

  In 10.5.1.0.2 and later versions, the MaxStartups parameter in the SSH server configuration file is validated. The maximum number of unauthenticated concurrent SSH connections must be set to a value greater than or equal to 30. To set the value of this parameter to 30, open the `/etc/ssh/sshd_config` file, change the value of the MaxStartups parameter to 30:30:100, and then restart the sshd daemon.

  Note: Unauthenticated SSH connections are socket connections to the SSH daemon that are established without user authentication.

- The following table lists the default ports that must be available for the Informatica Cluster Service:

Service	Default Port
HTTP/HTTPS	9075
Nomad Serf	4648
Nomad HTTP	4646
Nomad RPC	4647
ZooKeeper	2181
ZooKeeper peer	2888
ZooKeeper leader	3888
Solr	8983
MongoDB	27017
PostgreSQL	5432

- If you configured the repository for MetaDex, back up the MetaDex repository database.

  Note: If you upgrade from version 10.4.1.3.3 or earlier to version 10.5.11 and if you used the scanners.executions.max property in the `server.properties` file, rename the property to scanners.server.execution.slots.

  The default MetaDex server memory is 4 GB.
- Remove the values configured for the following environment variables using the `unset` command:

  - INFA_HOME

  - INFA_DOMAINS_FILE

  - DISPLAY

  - JRE_HOME

  - INFA_TRUSTSTORE

  - INFA_TRUSTSTORE_PASSWORD
- Verify that the LD_LIBRARY_PATH and the PATH environment variables do not contain earlier versions of Informatica.
- Delete the following JAR file from the nodes on which the Catalog Service runs: `<Informatica Installation Directory>/logs/<NodeName> /services/CatalogService/<CatalogServiceName>/migrate/restore.jar`
- Effective in version 10.5, Universal Connectivity Framework (UCF) is deprecated. To use UCF resources that you had configured in earlier versions, copy the UCF jar files from the existing installation location to the new upgrade location:`<$INFA HOME>/services/CatalogService/ScannerJars/bin`.

  Note: The UCF jar files have the `com.infa.ldm.extended. *.jar` file pattern.

  After you upgrade the Catalog Service, the existing UCF resources are listed in the **Select Resource Type** dialog box in Catalog Administrator.

Note: Keep a record of the asset count from the catalog, Catalog Administrator, and the Data Asset Analytics dashboard for reference. You can use the `/data/search` REST API to view the asset count in JSON format as shown in the following sample: `http://<CatalogAdmin>:<port>/access/2/catalog/data/search?defaultFacets=false&disableSemanticSearch=true&enableLegacySearch=false&facet=true&facetId=core.classType&facetId=core.resourceName&facetId=core.resourceType&highlight=false&includeRefObjects=false&offset=0&pageSize=1&q=*&tabId=all`.

For more information, see the *Enterprise Data Catalog REST API Reference*.

If you use Axon Data Governance with Enterprise Data Catalog, make sure that the Axon SuperAdmin has disabled the asset onboarding job and the reference link retrieval job before you upgrade. For more information, see the Informatica Knowledge Base article.

Configure Custom SSL Certificates Overview

During an install or an upgrade, you can use SSL certificates of your choice, referred to as custom SSL certificates, to secure components associated with the Informatica Cluster Service. You can choose to use CA-signed custom SSL certificates or self-signed custom SSL certificates as the client and cluster certificates.

If you choose to use CA-signed custom SSL certificates, use the generate_csr.sh and generate_certs.sh scripts to generate the required certificates.

If you choose to use self-signed custom SSL certificates, use the custom SSL utility bundled with the installer to generate the required certificates.

Requirements to Enable Custom SSL Certificates

Before you enable custom SSL for Enterprise Data Catalog, secure the Informatica domain with custom SSL certificates.

Verify the following prerequisites for the Informatica domain:

- The custom truststore and keystore files are placed in a custom directory. The truststore file name must be infa_truststore.jks.
- The directory that stores the keystore and truststore files are accessible to the command line programs.

Important: The installer places the default infa_truststore.jks and keystore files in the `<Informatica installation directory>/services/shared/security` directory on each node. Do not overwrite, delete, or move the default truststore and keystore files. Do not place the custom truststore and keystore files in the directory.

To know about how to convert a non-SSL Informatica domain to an SSL-enabled Informatica domain, see the Knowledge Article Convert a non-SSL Informatica domain to an SSL-enabled Informatica domain.

Generate CA-signed Custom SSL Certificates

You have a custom SSL certificate for the Informatica domain, and you want to use CA-signed SSL certificates as the client and cluster certificates for the Informatica Cluster Service.

Use the `generate_csr.sh` and `generate_certs.sh` scripts to generate the Certificate Signing Request (CSR) to send it to a CA and generate the required SSL certificates. You can download the scripts from the Informatica Electronic Software Download site.

Perform the following steps to use the scripts to generate the certificates:

1. Extract the generate_csr.sh and generate_certs.sh scripts from the following location: `<Location of installer files>/properties/utils/CustomSSLScriptsUtil_ExternalCA`.

2. Set the JAVA_HOME environment variable to point to JDK 17.
3. In the gen_csr.properties file, provide the values for the following parameters:

Parameter	Description
InfaDomainKeystorePassword	The Informatica domain keystore password in plain text.
ServerHosts	The Informatica Cluster Service hosts that include the data nodes, processing nodes, and gateway node. Enter a comma-separated list of FQDNs of cluster nodes.
ClientHosts	Comma-separated list of unique host names of domain nodes and cluster nodes.
InfaDomainName	The Informatica domain name.
ICSServiceName	The name of the Informatica Cluster Service.
KeysOutputDir	The directory to store the generated keys. Specify the $CUSTOM KEYSTORE LOC directory to avoid the additional steps to copy the generated keys. The $ICS SERVICENAME/client_certs and the $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM KEYSTORE LOC directory. $CUSTOM KEYSTORE LOC is the directory where the custom keystore for the Informatica domain (infa_keystore.jks) is located. $ICS SERVICENAME is the name of the Informatica Cluster Service.
CertsOutputDir	The $CUSTOM TRUSTSTORE LOC directory to store the generated truststore files. The $ICS SERVICENAME/client_certs and the $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM TRUSTSTORE LOC directory. The $CUSTOM TRUSTSTORE LOC is the directory where the custom truststore for the Informatica domain (infa_truststore.jks) is located.
DNSDomainName	The DNS domain name for the cluster nodes.
ClusterCert_OrganizationUnit	Optional. The value for the OrganizationUnit for the cluster nodes certificate.
ClusterCert_Organization	The value for the Organization for the cluster nodes certificate. **Note:** Verify that the combination of the Organization (O) and the Organizational Unit (OU) parameters in the certificate subject is distinct for the cluster and client certificates
ClusterCert_Location	The value for the Location for the cluster nodes certificate.
ClusterCert_State	The value for the State for the cluster nodes certificate.
ClusterCert_CountryCode	The value for the Country Code for the cluster nodes certificate.
DomainCert_OrganizationUnit	Optional. The value for the Organization Unit for the domain nodes certificate.

Parameter	Description
DomainCert_Organization	The value for the Organization for the domain nodes certificate. **Note:** Verify that the combination of the Organization (O) and the Organizational Unit (OU) parameters in the certificate subject is distinct for the cluster and client certificates
DomainCert_Location	Optional. The value for the Location for the domain nodes certificate. Default is the ClusterCert_Location parameter.
DomainCert_State	Optional. The value for the State for the domain nodes certificate. Default is the ClusterCert_State parameter.
DomainCert_CountryCode	Optional. The value for the Country Code for the domain nodes certificate. Default is the ClusterCert_CountryCode parameter.
Custom_Server_Certificate_CN	Optional. The value for the Common Name in the cluster nodes certificate that can be used instead of the default $InfaDomainName-$ICSServiceName.$DNSDomainName value. **Note:** You must enter RFC2253 compliant values. The following special characters are supported: `, + " \ < > ;`. Double quotes (") must be used in pairs. The characters \ and " must not be used together. The value cannot contain a space.
Custom_Client_Certificate_CN	Optional. The value for the Common Name in the domain nodes certificate that can be used instead of the default $InfaDomainName-$ICSServiceName.$DNSDomainName value. Default is the Custom_Server_Certificate_CN parameter. **Note:** You must enter RFC2253 compliant values. The following special characters are supported: `, + " \ < > ;`. Double quotes (") must be used in pairs. The characters \ and " must not be used together. The value cannot contain a space.

Note: If the values contain spaces or special characters, you must enclose the values within double-quotes.

4. Run the generate_csr.sh script using the following command to generate the .csr files to send to an external CA: `./generate_csr.sh gen_csr.properties`
The following files are generated for the cluster:

   infa_nodecert.csr

   infa_privkey.key

   infa_privkey.pem

   keystore.jks

   The following files are generated for the client:

   infa_nodecert.csr

   infa_privkey.key

   infa_privkey.pem

   keystore.jks

   browser_cert.csr

browser_keystore.jks

browser_privkey.key

browser_privkey.pem

5. Validate the contents of the .csr files. Run the following command to view the contents: `keytool -printcertreq -file $PATH TO CSR`
6. Send the following .csr files to an external CA for signing:
   - <CertsOutputDir>/<ICSServiceName>/client_certs/infa_nodecert.csr
   - <CertsOutputDir>/<ICSServiceName>/cluster_certs/infa_nodecert.csr
   - <CertsOutputDir>/<ICSServiceName>/client_certs/browser_cert.csr

   Note: The browser_cert.csr file is required if you want to create the browser certificates to view the scan job logs on Nomad.
7. After you receive the certificates or certificate chains from the CA in .pem format, "Validate the CA-signed Certificates" on page 44 and store the certificates to a location under the $INFA HOME directory on your machine.

   Note: If you receive the certificates or certificate chains from the CA in .cer format, run the following command to convert the files to pem format: `openssl x509 -inform der -in <certificate file name>.cer -outform pem -out <certificate file name>.pem`.

   If you receive a certificate chain from the CA, you must extract the root certificate, intermediate certificates, and the end user certificate.
8. In the gen_certs.properties file, provide the values for the following parameters:

Parameter	Description
InfaDomainKeystorePassword	The Informatica domain keystore password in plain text.
InfaDomainTruststorePassword	The Informatica domain truststore password in plain text.
ClusterCertificate	The path to the cluster certificate signed by the CA in .pem format. This is an end user certificate.
ClientCertificate	The path to the client certificate signed by the CA in .pem format. This is an end user certificate.
BrowserCertificate	The path to the browser certificate signed by the CA in .pem format.
ICSServiceName	The name of the Informatica Cluster Service.
IsCACertificateChainAvailable	Specify if the CA certificate chain is available as a single .pem file. Enter true or false. **Note:** The certificate chain must contain only the root and intermediate certificates.
SingleCACertificateChain	The path to the CA certificate chain in .pem format.

Parameter	Description
IndividualCertificatesFromCAChain	Optional. Only required if the IsCACertificateChainAvailable parameter is set to false. Comma-separated paths to the public certificates in the CA certificate chain in .pem format if the complete CA certificate chain is available as individual .pem files.
KeysOutputDir	The $CUSTOM KEYSTORE LOC directory store the generated keys. The $ICS SERVICENAME/client_certs and $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM KEYSTORE LOC directory. $CUSTOM KEYSTORE LOC is the directory where the custom keystore for the Informatica domain (infa_keystore.jks) is located. $ICS SERVICENAME is the name of the Informatica Cluster Service
CertsOutputDir	The $CUSTOM TRUSTSTORE LOC directory to store the generated truststore files. The $ICS SERVICENAME/client_certs and the $ICS SERVICENAME/cluster_certs directories are created under the $CUSTOM TRUSTSTORE LOC directory. The $CUSTOM TRUSTSTORE LOC is the directory where the custom truststore for the Informatica domain (infa_truststore.jks) is located.

9. Run the generate_certs.sh script using the following command to generate the certificates: `./generate_certs.sh gen_certs.properties`
   The keystore.jks keystore and the infa_privkey.pem private keys are stored at $CUSTOM KEYSTORE LOC/$ICSServiceName/client_certs and $CUSTOM KEYSTORE LOC/$ICSServiceName/cluster_certs directories.

   The truststore.jks truststore and the infa_nodecert.pem, infa_nodecertkey.pem, and infa_pubcert.pem public keys are stored at $CUSTOM TRUSTSTORE LOC/$ICSServiceName/client_certs and $CUSTOM TRUSTSTORE LOC/$ICSServiceName/cluster_certs directories.
10. Optional. The directories $CUSTOM KEYSTORE LOC and $CUSTOM TRUSTSTORE LOC are generally the same. If the <KeysOutputDir> location is not the same as $CUSTOM KEYSTORE LOC and <CertsOutputDir> location is not the same as $CUSTOM TRUSTSTORE LOC, move the keys and certificates to the respective directories.

    Note: Verify that the $CUSTOM KEYSTORE LOC and the $CUSTOM TRUSTSTORE LOC directories have the required user privileges. Also, validate that the user has minimum chmod 700 permissions configured for the directories and chmod 600 permissions configured for the files that are copied to the directories.

To access the Nomad Web UI and Solr Admin UI when the Informatica Cluster Service is SSL enabled, you must import the browser certificates. To know more about how to import the browser certificates, see the Knowledge Article
Access Nomad Web UI and Solr Admin UI when Informatica Cluster Service is SSL enabled in Enterprise Data Catalog.

Validate the CA-signed Certificates

You must use a single CA for the cluster and client certificates. After you receive the signed certificates from the CA, you must verify that each certificate is an X.509 certificate in .pem format.

Run the following command to view the contents of the signed cluster and client certificates: `keytool -printcert -file $PATH TO CERTIFICATE PEM FILE`.

Validate the following requirements for the cluster and client certificates:

Prerequisite	Certificate Requirement
Mandatory fields	For the cluster certificate, consider the following key usage requirements: keyUsage = digitalSignature,keyEncipherment extendedKeyUsage = serverAuth,clientAuth For the client certificate, consider the following key usage requirements: keyUsage = digitalSignature extendedKeyUsage = clientAuth
Subject Alternate Name (SAN)	For the cluster certificate, the SAN must include the list of cluster nodes in the following format: SAN=DNS:$CLUSTER HOST1 FQDN,DNS:$CLUSTER HOST2 FQDN,DNS:$CLUSTER HOST3 FQDN The client certificate must contain the FQDNs for the cluster nodes. For the client certificate, the SAN must include the list of all Informatica nodes in the following format: SAN=DNS:$INFA DOMAIN HOST1 FQDN,DNS:$INFA DOMAIN HOST2 FQDN The client certificate must contain the FQDNs for both the domain and cluster nodes.

CLUSTER HOST FQDN represents the fully qualified domain name for the cluster gateway host, processing nodes, and data nodes in the cluster.

INFA DOMAIN HOST FQDN represents the fully qualified domain name of the Informatica domain gateway host, domain nodes, cluster gateway host, processing nodes, and data nodes in the cluster.

Note: Verify that the custom certificate location for the domain nodes contains the infa_truststore.pem file. Also, verify that all the certificates in the CA certificate chain are present in the truststore.jks and the infa_pubcert.pem files.

Generate Self-signed Custom SSL Certificates

You have a custom SSL certificate for the Informatica domain, and you want to use self-signed custom SSL certificates as the client and cluster certificates for the Informatica Cluster Service.

Use the custom SSL utility bundled with the installer to generate the required SSL certificates.

Perform the following steps to use the custom SSL utility to generate the certificates:

1. Extract `GenerateCustomSslUtility.zip` from the following location: `<Location of installer files>/properties/utils/CustomSslCertsUtility/`.
2. Set the JAVA_HOME environment variable to point to JDK 17.
3. Configure the following parameters in the `input.properties` file that you extracted from the `GenerateCustomSslUtility.zip` file:

Parameter	Description
KeystoreFile	Path to a custom keystore file along with file name. The keystore type must be in JKS. X509 format. The file must contain a single private key entry with the complete certificate chain. Verify that the file has signing capability.
KeystorePassword	Password of the custom keystore file in plain text format.

Parameter	Description
TruststoreFile	Path to a custom truststore file along with file name. the truststore type must be in JKS. X509 format. The file must contain the public certificates corresponding to the private key entry in the keystore file.
TruststorePassword	Password of the custom truststore file in plain text format.
ISPDomainKeystorePassword	Password of Informatica domain keystore in plain text format. The utility uses the password for the cluster and client keystore.jks file.
ISPDomainTruststorePassword	Password of the Informatica domain truststore in plain text format. The utility uses the password for the cluster and client truststore.jks file
KeystoreOutputDir	Represents the location of the Informatica domain custom keystore that you provided when you installed Enterprise Data Catalog.
TruststoreOutputDir	Represents the location of the Informatica domain custom truststore that you provided when you installed Enterprise Data Catalog. **Note:** Verify that the KeystoreOutputDir and TruststoreOutputDir parameters point to the same directory.
ServerNodes	Comma-separated list of fully qualified domain names of nodes that you plan to configure as data nodes, processing nodes, service hosts, and gateway node when you configure the Informatica Cluster Service.
IcsServiceName	Name of the Informatica Cluster Service.
ClientNodes	Comma-separated list of fully qualified domain names of nodes that you plan to configure as Informatica domain hosts, data nodes, processing nodes, service hosts, and gateway node.
IspDomainName	The Informatica domain name.
ClusterNodeDNSDomain	Domain name of the gateway host that you plan to configure for the Informatica Cluster Service.

4. Run the utility using the following command: `java -jar GenerateCustomSslUtility.jar -in input.properties`. The utility generates the following keys and client and cluster certificates:
   - Keys:
     - keystore.jks
     - infa_privkey.pem

     Note: The cluster keys are generated in the following directory: `<Keystore output directory>/<IcsServiceName>/cluster_certs`. The client keys are generated in the following directory: `<Keystore output directory>/<IcsServiceName>/client_certs`
   - Certificates:
     - truststore.jks
     - infa_pubcert.pem
     - infa_nodecert.pem

- infa_nodecertkey.pem

  Note: The cluster certificates are generated in the following directory: `<Truststore output directory>/<IcsServiceName>/cluster_certs`. The client certificates are generated in the following directory: `<Truststore output directory>/<IcsServiceName>/client_certs`

5. Copy the generated certificates to the required folders.
6. Assign the ownership of the following directories to the Informatica domain user:
   - <Keystore output directory>/<IcsServiceName>/cluster_certs
   - <Truststore output directory>/<IcsServiceName>/cluster_certs
   - <Keystore output directory>/<IcsServiceName>/client_certs
   - <Truststore output directory>/<IcsServiceName/client_certs

Note: Verify that all the client certificates are included in a single directory. Similarly, verify that all the cluster certificates are included in a single directory. You must also verify that the directories have the `chmod 700` permission configured and the files under the directories have the `chmod 600` permission configured.

Download and Extract the Installer Files

The installer files are distributed as compressed files. You can get the Informatica installation file and distribution packages from the FTP link contained in your fulfillment email.

Download the Informatica installation tar file and the required distribution package ZIP files from the Informatica Electronic Software Download site. You can download them to a local directory or a shared network drive that is mapped on your machine.

To download other compressed files and utilities, raise a shipping request with Informatica Global Customer Support.

Extract the Informatica installer files to a directory on your machine. The user that runs the installer must have read and write permissions on the installer files directory and execute permissions on the executable file.

Note: Ensure that you extract the installer files to a local directory as you can't run the installer from a mapped file.

Copy the ZIP files of the distribution packages to the following location: `<Informatica installer files>/source`

Note: The installer fails if the ZIP files for distribution packages aren't available in the source directory.

Verify Installer Package Checksum on UNIX

Before you run the services installer, verify the install package integrity through the cksum command. The cksum command calculates the checksum value for the installers.

Verify the checksum for the specific installer files against the checksum of the installation files downloaded from the Informatica Electronic Software Download site.

The following table lists the checksum and file size for the Informatica services for UNIX:

File	Checksum Value	File Size
informatica_10510_server_linux-x64.tar	1346986189	17774100480 bytes

A checksum mismatch can occur when there are data errors during download due to network issues or when data corruption occurs in the file on disk. For more information about the checksum errors, see HOW TO: Identify file errors after downloading Informatica installation files.

Verify the License Keys and Copy the Binary Files

Before you start the upgrade, perform the following steps:

- Verify the new license keys for the Informatica domain and Informatica Cluster Service.
- Extract the installer binary files to a directory on the machine where you plan to upgrade Enterprise Data Catalog. You must have read and write permissions on the installer files directory and execute permissions on the executable file.
- Copy the scanner binaries from the Informatica Electronic Software Download site and place it in the following directory: `<Location where you extracted the installer binary files>/source`

Verify Distribution Package Requirements (Linux and UNIX)

You can use third-party distribution packages to integrate the Informatica domain with the Hadoop or Databricks environment.

The Informatica domain and client require the distribution packages to process complex files within the Informatica domain, or to connect to Hadoop or Databricks environment when you process within the Informatica domain, or to push processing to Hadoop or Databricks environment.

If you need a distribution package, you can install it through the installer or through Integration Package Manager (the package manager) at any time.

Process within the Informatica domain

You can use the same distribution package that you used in the previous Informatica version to process complex files within the Informatica domain or to connect to the Hadoop or Databricks environment when you process within the Informatica domain.

The following adapters require distribution packages for processing within the Informatica domain:

- PowerExchange for Amazon S3
- PowerExchange for Google Cloud Storage
- PowerExchange for Google Cloud Storage for PowerCenter
- PowerExchange for Hadoop for PowerCenter
- PowerExchange for HBase
- PowerExchange for HDFS
- PowerExchange for Hive
- PowerExchange for JDBC V2
- PowerExchange for Kafka for PowerCenter
- PowerExchange for MapR-DB
- PowerExchange for Microsoft Azure Blob Storage
- PowerExchange for Microsoft Azure Data Lake Storage Gen1
- PowerExchange for Microsoft Azure Data Lake Storage Gen2

Process with Hadoop or Databricks environment

When you push processing to the Hadoop or Databricks environment, the Informatica domain and client require distribution packages. For more information about the supported distribution packages, see the Product Availability Matrix.

Complete the Prerequisites

Before you upgrade the Informatica domain, complete the prerequisites such as downloading the installer files and copying the required binaries.

- Download the installation file from the FTP link contained in your fulfillment email. Download the Informatica installation tar file from the Informatica Electronic Software Download site to a directory on your machine and then extract the installer files. Extract the installer files to a directory on your machine. The user that runs the installer must have read and write permissions on the installer files directory and execute permissions on the executable file.
- Copy the new scanner binaries from the Informatica Electronic Software Download site to your installation directory.
- Clear the configured values for the INFA_TRUSTSTORE and INFA_TRUSTSTORE_PASSWORD environment variables if the domain is enabled for Secure Sockets Layer (SSL).
- Download and install the libidn.so.11 library if you use RHEL 8.3.

Back Up Key Enrichments Using the Migration Utility

To use the enhancements added to the primary key-foreign key enrichments in Enterprise Data Catalog version 10.5.11, back up the enrichments using the migration utility before you upgrade.

Perform the following steps to run the utility:

1. From the Informatica Electronic Software Download (ESD) site, download and extract the `Informatica_<version>_PkFkEnrichmentMigrationUtility.zip` file on the domain machine.
2. Run the following command:

```
java -jar PKFKEnrichmentsRestoreUtility.jar export <Catalog Service hostname>
<Catalog Service admin port number> <username> <password> <location to download
enrichments json> <comma separated resource type IDs> <SSL Enabled true false>
```

Note: Verify that the resource name length does not exceed the maximum file name length supported by the operating system.

Complete the Data Asset Analytics Prerequisites

If you configured Data Asset Analytics for Enterprise Data Catalog, you must configure the process termination timeout before you shut down the Informatica domain for an upgrade. You must run the Data Asset Analytics utility before you upgrade to Enterprise Data Catalog 10.5.11.

Configure the Process Termination Timeout

1. In the Administrator tool, go to the custom properties section for the domain.
2. Add the following custom property:
   ProcessTerminationTimeout=900
3. Stop the Informatica Services using the `infaservice.sh shutdown` command.
4. Start the Informatica Services using the `infaservice.sh startup` command.

Verify that the file size of the transaction or redo log files for the Data Asset Analytics repository database is large enough. For an Oracle database, ensure that the database has a minimum of three redo log files of 2 GB each.

Run the Data Asset Analytics Utility

Data Asset Analytics displays key metrics and key charts on the dashboard tabs based on pre-computed analytical data. Before you perform the upgrade, you must populate the pre-computed database tables with analytical data.

Perform the following steps to run the Data Asset Analytics utility:

1. Back up the Data Asset Analytics database schema.
2. Verify the daa.log log file and make sure that there are no unprocessed data asset events to avoid losing event data such as user logins, searches, and asset usage.
3. From the Informatica Electronic Software Download (ESD) site, download and extract the informatica_<version>_daa_utility.zip file on the Informatica domain.

   Note: If you applied a service pack, you must download the utility corresponding to the service pack version.
4. Go to the directory where you extracted the files and navigate to the `config` directory.
5. Open the input.properties file and configure the following connection parameters for the Data Asset Analytics repository.

Property	Description
databaseType	The Data Asset Analytics database type. Enter ORACLE, POSTGRESQL, or SQLSERVER.
databaseHostName	The fully qualified host name of the machine where the database is hosted. **Note:** If the value contains a special character, you must enclose the value within single quotes (').
databasePort	The port number for the database engine service.
databaseServiceName	The service name for the Oracle database.
databaseName	The database name for the PostgreSQL or SQL Server database.
databaseUserName	The name of the user account to connect to the database.
databaseAdvanceParameter	Optional. Enter the advanced JDBC connection parameters used to connect to the Data Asset Analytics repository. **Note:** You must enclose the parameters within single quotes (').

6. On a shell command line, enter the following command to run the utility from the directory where you extracted the files: `./daautil.sh`.
7. Press **1** to select the upgrade mode.
8. Enter the password to connect to the database.

Note: To know the execution status, see the daautil.log file from the directory from which you run the utility.

Step 2. Back Up the Repositories and Configuration Details

Back Up the Domain

Before you upgrade the domain, you must back up the configuration metadata for the domain.

Complete the following steps to back up the domain:

- Run the infasetup BackupDomain command to back up the domain configuration database tables to a file.
- Back up the metadata configuration files to any directory accessible by the machines where you install Informatica.

Informatica infasetup includes command line programs to back up and restore the domain. infasetup is located in the following directory:

```
<Informatica installation directory>/isp/bin
```

To back up the domain with infasetup, use the following syntax:

```
BackupDomain
<<-DatabaseAddress|-da> database_hostname:database_port|

<-DatabaseConnectionString|-cs> database_connection_string>

<-DatabaseUserName|-du> database_user_name

<-DatabasePassword|-dp> database_password

<-DatabaseType|-dt> database_type

[<-DatabaseServiceName|-ds> database_service_name]

<-BackupFile|-bf> backup_file_name

[<-Force|-f>]

<-DomainName|-dn> domain_name

[<-Tablespace|-ts> tablespace_name (used for IBM DB2 only)]

[<-SchemaName|-sc> schema_name (used for Microsoft SQL Server only)]

[<-DatabaseTlsEnabled|-dbtls> database_tls_enabled]

[<-DatabaseTruststorePassword|-dbtp> database_truststore_password]

[<-TrustedConnection|-tc> trusted_connection (used for Microsoft SQL Server only)]

[<-EncryptionKeyLocation|-kl> encryption_key_location]
```

Back up the metadata configuration files to any directory accessible by the machines where you install Informatica. The following table describes the metadata files and the locations where you can find them:

Metadata File	Description	Location
nodemeta.xml	Contains metadata for a node.	Stored in the isp/config directory on each node in the domain. If you use the same backup directory name on all the nodes, rename nodemeta.xml before copying it to the backup location. For example, you back up nodemeta.xml to the /nodebak directory on nodeA and nodeB. Rename the configuration files so that on nodeA the file is backed up to /nodebak/nodemeta_A.xml, and on nodeB the file is backed up to /nodebak/nodemeta_B.xml.
domains.infa	Contains connectivity information for the gateway nodes.	Stored in one of the following locations: - The Informatica installation directory on the client and server machines. - The location configured through the INFA_DOMAINS_FILE environment variable.

Back Up the Model Repository

Back up the contents of the Model repository using the Administrator tool.

To back up a Model repository, select the Model Repository Service in the Administrator tool. Then, on the Domain Actions menu, click **Repository Contents** > **Backup**.

Back Up Catalog Data Using the Export Utility

Back up the catalog data using the export.jar utility so that you can restore the catalog data after you upgrade.

Use the export utility if you upgraded from a Hadoop cluster version to a non-Hadoop cluster version, and you want to migrate the existing catalog data to version 10.5.11. Contact Global Customer Support to download the utility.
Before you run the utility, verify the following prerequisites:

- The Informatica Cluster Service is running.
- No jobs are running in Enterprise Data Catalog. To verify, see the **Monitoring** tab in Catalog Administrator.
- The Catalog Service is running.
- If backup contains large similarity profile data, make sure that infacmd process includes 4 GB heap size. You can set the JAVA heap size value with the following property in the Infacmd.sh file: `export ICMD_JAVA_OPTS=-Xmx4096m`
- Enable the read-only mode for the Catalog Service. See the following sample curl commands to enable the service in read-only mode:
  - If the Catalog Service is SSL-enabled, use the `curl -k -u <DomainUserName>:<DomainPassword> -X POST "https://<host name>:<Catalog Service port>/access/1/catalog/catalogMode/READONLYMODE" -H "accept: */*"` command.
  - If the Catalog Service is not SSL-enabled, use the `curl -k -u <DomainUserName>:<DomainPassword> -X POST "http://<host name>:<Catalog Service port>/access/1/catalog/catalogMode/READONLYMODE" -H "accept: */*"` command.
- From your existing installation, copy the `application.properties` file from `<$INFA_HOME>/logs/<NODE_NAME>/services/CatalogService/<SERVICE_NAME>/<SCN_NAME>/spark/` to the directory where you copied the `export.jar` file.

- If the Catalog Service is SSL enabled, set the following environment variables:
  - INFA_TRUSTSTORE. Set the variable to the location of the Informatica truststore file. Default location is `$INFA_HOME/services/shared/security`.
  - INFA_TRUSTSTORE_PASSWORD. Encrypt the password that you set. See the sample command to set the encrypted password: `export INFA_TRUSTSTORE_PASSWORD="84Ve/soUbpQ/Aae5uGKXQA=="`

    Note: See the following sample command to encrypt the password: `$INFA_HOME/server/bin/pmpasswd <password>`
- If catalog data is large, run the export.jar utility with 20 GB heap size. For example, `java -Xmx20960m -jar export.jar export` For more information about performance sizing, see the *Enterprise Data Catalog Performance Tuning Guide*.
- Set the commit interval option to 48 hours when you run the export utility. For example, `java -Dexport.commit.interval.ms=172800000 -jar export.jar export`

Run the utility using the following command: - `java -jar export.jar export`

You can verify the `export.log` file generated at the location from which you ran the utility.

Back Up the Catalog Data Using the ldm backupContents Command

Back up the catalog data before you upgrade Enterprise Data Catalog. Backing up the catalog ensures the availability of the existing metadata, attributes, associations, values, schedules, and tasks in the event of an upgrade failure. Back up the catalog data to retain the catalog data from the Hadoop environment in the event of a roll back or upgrade failure. You must have the Informatica Administrator role to perform the backup.

Before you run the infacmd ldm backupContents command, verify the following prerequisites:

- Set the following environment variables if the Catalog Service is enabled for SSL:
  - INFA_TRUSTSTORE. See the following sample command to set the variable: `export INFA_TRUSTSTORE=<Location of the Informatica truststore file>` . Default location is `$INFA_HOME/services/shared/security`.
  - INFA_KEYSTORE. See the following sample command to set the variable: `export INFA_KEYSTORE=<Location of the keystore file>`. Default location is `$INFA_HOME/services/shared/security`. This variable is required only if you used custom SSL configuration. For default SSL and non-SSL configurations, you must unset the variable.
  - INFA_TRUSTSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_TRUSTSTORE_PASSWORD="84Ve/soUbpQ/Aae5uGKXQA=="`.
  - INFA_KEYSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_KEYSTORE_PASSWORD="6cDE/ItyUL/Rtui9nhVRI=="`. This variable is required only if you used custom SSL configuration. For default SSL and non-SSL configurations, you must unset the variable.

  You can contact Informatica Global Customer Support for the default truststore and keystore passwords.

  Note: See the following sample command to encrypt the password: `$INFA_HOME/server/bin/pmpasswd <password>`
- Estimate the HDFS disk space consumption of the catalog data. Run the following command to view the directory size: `hdfs -du /Informatica/LDM/<SCN>`.
- Make sure that sufficient disk space is available in a domain host so that backup does not run out of disk space.

The infacmd ldm BackupContents command as shown in the following sample, creates a .zip file of the directory in the local machine: `./infacmd.sh LDM backupContents -dn <name of the Informatica domain> -un <username to connect to the Informatica domain> -pd <password for the username> -sn <name of the Catalog Service> -of <name of the backup ZIP file>`
For more information about the syntax, parameters, and description of the command, see Informatica Command Reference.

Note: If backup fails with an out of memory issue, you must increase the JAVA heap size for **ICMD_JAVA_OPTS** property in the following `infacmd.sh` file. For example, ICMD_JAVA_OPTS=-Xmx20g.

Back Up the SSL Configuration

Back up the following SSL configuration files from the default location: `<Informatica installation directory>/services/shared/security`.

- Back up the domain truststore and keystore files.
- Back up the default keystore files.
- Back up all the keystore and truststore files that the application services use.

Back Up the Domain Information

Back up the following files:

- `<Informatica installation directory>/isp/config/keys/siteKey`
- `<Informatica installation directory>/domains.infa`
- `<Informatica installation directory>/tomcat/conf/server.xml`
- `<Informatica installation directory>/isp/config/nodemeta.xml`

Note: You must back up the binaries from the Informatica installation directory.

Back Up the Data Asset Analytics Database Schema

If you have configured the repository for Data Asset Analytics, perform a backup of the Data Asset Analytics database schema.

Note: Use a database tool to take backup.

Back Up MetaDex

If you have configured the repository for MetaDex, back up the MetaDex repository.

Note: In version 10.4, the MetaDex job scheduler was disabled by default. Effective in version 10.5, by default, the job scheduler removes jobs every six hours, leaving only the last three jobs. You can change that behavior by editing parameters in the repository.properties file. For more information, see "Removing jobs" in the *MetaDex Administrator Guide*.

1. Use a database tool to perform a backup of the MetaDex database schema.

   Note: Ensure that you create a backup of the entire database, not just the data content.
2. Create a backup of the following configuration files:
   - `<workspace>/etc`
   - `<workspace>/setup.env`
3. Create a backup of the connection registry.

4. In the MetaDex tool, export your configurations and projects.
   See "Projects page" and "Configuration page" in the *MetaDex Administrator Guide*.

Back Up Tableau Resource Enrichments Using the Migration Utility

If the catalog contains Tableau resources, you can back up the Tableau resource enrichments using the Tableau migration utility before you upgrade. Or, back up the enrichments post upgrade before you run the resources again.

1. From the Informatica Electronic Software Download (ESD) site, download and extract the `Informatica_<version>_TableauEnrichmentMigrationUtility.zip` file on the domain machine.
2. Run the following command to back up the Tableau resource enrichments:

```
java -jar EnrichmentUtility.jar export <Catalog Service hostname> <Catalog Service admin port number> <username> <password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
```

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Step 3. Prepare to Upgrade

Disable the Catalog Service

To disable an application service, select the Catalog Service in Informatica Administrator and click **Actions** > **Disable Service**..

Disable the Application Services

You must disable the following Informatica application services before you back up the catalog:

1. Informatica Cluster Service
2. Content Management Service
3. Data Integration Service
4. Model Repository Service

To disable an application service, select the service in Informatica Administrator and click **Actions** > **Disable Service**.

Shut Down the Informatica Domain

From Informatica Administrator, shut down the Informatica domain.

Step 4. Upgrade the Informatica Domain

The upgrade wizard reads the domain information from files in the previous version and uses the same settings to configure the domain and server files for the upgrade. It upgrades the tables of the domain configuration repository in the same database as the previous version. You can upgrade in the console or silent mode.

Complete the pre-upgrade tasks before you start the upgrade. Run the installer on all machines that host previous versions of Informatica that you want to upgrade.

Note: In a multi-node domain, upgrade the gateway node before you upgrade other nodes and upgrade all worker nodes in parallel.

You can perform the upgrade from the directory where you downloaded the installation files.

After you upgrade the domain, upgrade the Informatica client tool to the same Informatica version.

Before you perform the upgrade, verify that the Informatica domain truststore file contains only the TrustedCertEntry. You must delete all the PrivateKeyEntry entries from the file.

Run the Cluster Validation Utility

Run the Cluster Validation utility to validate the prerequisites to upgrade Enterprise Data Catalog.

Perform the following steps to run the utility:

1. Verify that each cluster node has a minimum of four CPU cores, 16 GB of memory, and 60 GB of free disk space.

   Note: The hardware requirements listed here are the minimum recommended values to install the product and start the services. The requirements vary based on the size of deployment. See the Enterprise Data Catalog Performance Tuning Guide for additional information about hardware requirements based on deployment sizes.
2. Verify that the root directory (/) has a minimum of 10 GB of free disk space.
3. Download the Informatica Cluster Validation Utility for version 10.5.11 from the Informatica Electronic Software Download (ESD) site.

   Note: If you applied a service pack, you must download the utility corresponding to the service pack version.
4. Extract the ZIP file on the Informatica domain that you plan to upgrade.
5. Go to the directory where you extracted the files and open the input.properties file.

   Note: You must have read and write permissions on the directory and execute permissions on the executable files.
6. In the input.properties file, configure the required properties.
7. Disable the firewall on each host in the cluster.
8. Enable passwordless SSH between the following nodes:
   - The node that hosts the Informatica domain and the node that hosts the gateway.
   - The gateway host and the data nodes.
   - The gateway host and the processing nodes.
   - The backup nodes and the gateway node.
   - All nodes in the cluster and the data nodes.

Note: If you enabled advanced configuration for the Informatica Cluster Service, and the MongoDB, PostgreSQL, Solr, and Nomad services run on different hosts, enable passwordless access from the gateway host to the hosts that run these services.

9. On a shell command line, go to the directory where you extracted the utility files and run the following command: `java -jar InformaticaClusterValidationUtility.jar -in <Location of the input.properties file>`.
   The utility provides the following options:

Option	Description
-hp	Display the help contents.
-in	Specify the location of the input.properties file.
-lf	Optional. Specify the directory to store the log files generated by the utility.
-v	Optional. Enable the debug log.
-vf	Optional. If you moved the validation scripts to a different directory, specify the location where the validation scripts are stored.

   Note: If you see a permission validation error message for the cluster custom directory, you can ignore the error.

10. Verify the validation status from the following log files located in the directory where you ran the utility:
    - clusterPrevalidation.log
    - clusterPrevalidation.err

Upgrade in Console Mode

You can upgrade in console mode to upgrade the domain on the same machine and on the same domain configuration repository database. You can upgrade the domain in console mode.

When you run the installer in console mode, the words quit, back, and help are reserved words. Do not use them as input text.

1. Log in to the machine with the same user account that you used to install the previous version.
2. Stop all processes that access the directory and subdirectories of the Informatica product to upgrade, including command prompts and tail logs.
3. On a shell command line, run the install file.
   The installer displays the message to verify that the locale environment variables are set.
4. If the environment variables are not set, press **n** to exit the installer and set them as required.
   If the environment variables are set, press **y** to continue.
5. Press **2** to upgrade Informatica.
6. Press **2** to skip running the Informatica Upgrade Advisor.
   Informatica provides utilities to facilitate the Informatica services installation process. You can run the utility before you upgrade Informatica services. The Informatica Upgrade Advisor helps to validate the services and checks for obsolete services in the domain before you perform an upgrade.

The installer displays a warning to shut down the Informatica domain that you want to upgrade before you continue the upgrade.

7. Read the terms and conditions of Informatica product usage toolkit and press **2** to continue the upgrade.

   Subject to your opt-out rights described below, the software will automatically transmit to Informatica in the USA information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at https://www.informatica.com/in/privacy-policy.html. You may disable usage collection in Administrator tool.

8. The **Upgrade Prerequisites** page displays the upgrade system requirements.

   Verify the requirements before you continue the upgrade.

9. Review the upgrade information and press **Enter** to continue.

   The **Upgrade Directory** panel appears.

10. At the prompt, enter the directory of the Informatica version you want to upgrade and the directory in which you want to upgrade to Informatica 10.5.11.

    The following table describes the directories you must specify:

Directory	Description
Directory of the Informatica product to upgrade	Directory that contains the version of Informatica services that you want to upgrade.
Directory for Informatica 10.5.11	Directory in which to install Informatica 10.5.11. Enter the absolute path for the installation directory. The directory cannot be the same as the directory that contains the previous version of Informatica services.The directory names in the path must not contain spaces or the following special characters: @|* $ # ! % ( ) { } [ ] , ; ' **Note:** If you use a special character such as á or €, unexpected results might occur at run time.

11. From the following installation options, select the upgrade environment based on your requirement:
    - Sandbox. Used for proof of concepts or as a sandbox with minimal users.
    - Development. Used for a design environment.
    - Test. Used for high-volume processing environment that resembles a production environment.
    - Production. Used for production environments with high-volume processing and high levels of concurrency meant for end users.

12. Select if you want to change the node host name, port numbers, or domain configuration repository. Press **1** to use the same node configuration as the previous version. Press **2** to change the node configuration.

    The **Component Selection** panel appears.

13. Press **1** to upgrade Informatica services.

    The **Domain Security Encryption Key** panel appears.

14. If you created a Catalog Service on the node, you must validate the Catalog backup before you upgrade. Press **1** if the node does not have a Catalog Service or press **2** if the node has a Catalog Service and press **Enter** to continue.

15. Press **2** to confirm that you backed up the catalog using both the export utility and the infacmd LDM backupContents command.
16. Specify the location of the directory containing the catalog backup.

    The catalog backup is validated.
17. Press **1** if you want to configure the MetaDex repository and provide the following details:

Property	Description
Database type	Specify the database that you want to configure as the repository for MetaDex. You can select from the following options: - Oracle - SQLServer
Database user name	User name configured to access the database before upgrade.
Database password	Password configured for the user name.
Schema name	Not applicable if you selected Oracle as the database. Specify the schema that you want to use.
Is the database secure?	Specify if the database is enabled for SSL.
Path to the database truststore file	Applies if you specified that the database is enabled for SSL. Provide the fully qualified path to the database truststore file.
Truststore password	Applies if you specified that the database is enabled for SSL Password to access the truststore file.

Secure JDBC parameters	Applies if you specified that the database is enabled for SSL. Specify the secure database parameters as shown in the following sample:
	`EncryptionMethod=SSL;HostNameInCertificate=ORATLS.informatica.com;`
	`ValidateServerCertificate=true;CryptoProtocolVersion=TLSv1.2;`
	Enterprise Data Catalog appends the secure JDBC parameters to the JDBC connect string. If you include the secure JDBC parameters directly in the connection string, do not enter any parameter in the Secure JDBC Parameters field.

JDBC connect string	The JDBC connection string to connect to the database. Use the following formats to specify the connection strings: - Oracle. jdbc:Informatica:oracle://<host name> :<port number>;ServiceName= - SQLServer. jdbc:Informatica: sqlserver://<host name>: <portnumber>;SelectMethod=cursor;DatabaseName=
MetaDex repository server port	A unique port for the MetaDex repository server. Default is 48090.

Property	Description
Enable secure communication for the MetaDex repository ?	Specify if TLS is enabled for communication with the MetaDex repository.
SSL certificates to secure the MetaDex repository?	Specify if you want to use the default SSL certificates or custom SSL certificates to secure the MetaDex repository. Specify the following properties if you want to use custom SSL certificates: - Keystore type. Specify the type of keystore. - Keystore alias. Unique alias to access the keystore entry. - Keystore file. Specify the keystore file that contains the required keys and certificates. - Keystore password. Password to access the keystore file.

If you did not opt to configure the MetaDex repository during installation, you can configure the repository after you complete the installation. Perform the steps listed in "Configure the MetaDex Repository (Optional)" on page 65.

18. Choose whether you want to install distribution packages through the Informatica installer.
    - Press **1** if you don't need distribution packages or if you want to install them later.
    - Press **2** if you want to install distribution packages through the installer.

    Default is 1.
19. If you choose to install distribution packages, select one or more packages from the list that you want to install. Separate multiple packages with a comma.

    Default is 1.
20. Press **Enter**.

    The **Domain Security Encryption Key** panel appears.
21. Enter the user name and password for the Informatica domain.

    The **Pre-installation Summary** page appears.
22. Review the upgrade information and press **Enter** to continue.

    After the domain upgrade completes, the **Domain Configuration Repository** panel appears.
23. Press **Enter** to continue the upgrade.

    The installer copies the server files to the Informatica 10.5.11 installation directory.

    The installer displays the database and user account information for the domain configuration repository to upgrade. It displays the database connection string for the domain configuration repository based on how the connection string of the previous version was created at installation:
    - If the previous version used a JDBC URL at installation, the installer displays the JDBC connection properties, including the database address and service name.
    - If the previous version used a custom JDBC connection string at installation, the installer displays the custom connection string.
24. If you use a JDBC URL, you can specify additional parameters to include in the connection string.

    If you use a custom connection string, you cannot specify additional parameters.
25. Review the information and press **Enter**.

    The upgrade completes and the **Post-installation Summary** panel appears.
26. Press **Enter** to complete the installation procedure and exit the installer.

Review the upgrade.log file to get more information about the tasks performed by the upgrade wizard and to view the configuration of installed components.

Apply Latest Service Pack or Cumulative Patch

Apply the latest available service pack or cumulative patch after you upgrade the domain.

Configure the MetaDex Repository (Optional)

If you did not configure the MetaDex repository during upgrade, you can configure the repository after you upgrade Enterprise Data Catalog.

Perform the following steps to configure the repository:

1. Go to the directory where you extracted the installer files and navigate to the following sub directory: `\properties\MetaDex\`.
2. Open the `MetaDex.properties` file and configure the following properties:

Property	Description
USER_INSTALL_DIR	Directory in which Informatica is installed.
MDX_DB_UNAME	User name configured to access the database before upgrade.
MDX_DB_PASSWD	Password to access the database.
MDX_DB_TYPE	The type of database that you want to configure. Specify either `Oracle` or `MSSQLServer`.
MDX_SQLSERVER_SCHEMA_NAME	Applies if MDX_DB_TYPE=MSSQLServer. Specify the name of the schema that you want to use. If you do not specify a schema, Enterprise Data Catalog uses the default schema.
MDX_DB_CUSTOM_STRING	The JDBC connection string to connect to the database. Use the following formats to specify the connection strings: - Oracle. jdbc:informatica:oracle: //<host name>:<port number>;ServiceName= - SQLServer. jdbc:informatica:sqlserver: //<host name>:<portnumber>;SelectMethod=cursor;DatabaseName=
MDX_SERVER_PORT	A unique port for the MetaDex repository server. Default is 48090.
MDX_SSL_ENABLED	Specify `true` if the database is enabled for SSL, else specify `false`. Default is false.
MDX_DB_TRUSTSTORE_FILE	Applies if you specified that the database is enabled for SSL. Provide the fully qualified path to the database truststore file.
MDX_DB_TRUSTSTORE_PASSWD	Applies if you specified that the database is enabled for SSL Password to access the truststore file.

Property	Description

MDX_SSL_DEFAULT_STRING	Applies if you specified that the database is enabled for SSL. Specify the secure database parameters as shown in the following sample:
	`EncryptionMethod=SSL;HostNameInCertificate=`
	`ORATLS.informatica.com;`
	`ValidateServerCertificate=true;`
	`CryptoProtocolVersion=TLSv1.2;`
	Enterprise Data Catalog appends the secure JDBC parameters to the JDBC connect string. If you include the secure JDBC parameters directly in the connection string, do not enter any parameter in the Secure JDBC Parameters field.

MDX_CUSTOM_SELECTION	Applies if you specified that the database is enabled for SSL. Specify if you want to use the default SSL certificates or custom SSL certificates to secure the repository. Set the property to `true` if you want to use custom SSL certificates. Set the property to `false` if you want to use the default SSL certificate.
MDX_KEYSTORE_FILE	Applies if you set MDX_CUSTOM_SELECTION as `true`. Specify the keystore file that contains the required keys and certificates.
MDX_KEYSTORE_PASSWD	Applies if you set MDX_CUSTOM_SELECTION as `true`. Password to access the keystore file.
MDX_KEYSTORE_TYPE	Applies if you set MDX_CUSTOM_SELECTION as `true`. Specify the type of keystore.
MDX_KEYSTORE_ALIAS	Applies if you set MDX_CUSTOM_SELECTION as `true`. Unique alias to access the keystore entry.

3. Enter the following command: `./configureMetaDex.sh`

   Note: Install MetaDex only on the domain node.
4. If the Catalog Service is running, recycle the service.

Copy the Third-party Resource Drivers

Complete the following post requisite after you upgrade the Informatica domain:

- Copy the third-party JAR or ZIP files that you had configured for resources such as Teradata, JDBC, and IBM Netezza from the following location `<Informatica installation directory>services/CatalogService/ScannerBinaries` to the same location in the machine that hosts the upgraded Informatica domain.

Enable and Upgrade the Application Services

To enable an application service, select the service in Informatica Administrator and click **Actions** > **Enable Service**.

Enable the following Informatica application services:

1. Model Repository Service
2. Data Integration Service
3. Content Management Service

After you enable the Informatica application services, run the service upgrade wizard to upgrade the services.

Note: You must disable Data Asset Analytics before you upgrade the Informatica Cluster Service. To disable Data Asset Analytics, select the Catalog Service in Informatica Administrator. Edit the configuration properties for Data Asset Analytics and clear the **Enable Data Asset Analytics** check box.

Upgrade the Application Services Using the Service Upgrade Wizard

Use the service upgrade wizard to upgrade application services and the database contents of the databases associated with the services. The service upgrade wizard displays upgraded services in a list along with services and associated databases that require an upgrade. You can also save the current or previous upgrade report.

1. In the Informatica Administrator header area click **Manage** > **Upgrade**.
2. Select the application services and associated databases to upgrade.
3. Optionally, specify if you want to **Automatically recycle services after upgrade**.

   If you choose to automatically recycle application services after the upgrade, the upgrade wizard restarts the services after they have been upgraded.
4. Click **Next**.
5. If dependency errors exist, the **Dependency Errors** dialog box appears. Review the dependency errors and click **OK**. Then, resolve dependency errors and click **Next**.
6. Enter the repository login information.
7. Click **Next**.

   The service upgrade wizard upgrades each application service and associated database and displays the status and processing details.
8. When the upgrade completes, the **Summary** section displays the list of application services and their upgrade status. Click each service to view the upgrade details in the **Service Details** section.
9. Optionally, click **Save Report** to save the upgrade details to a file.

   If you choose not to save the report, you can click **Save Previous Report** the next time you launch the service upgrade wizard.
10. Click **Close**.
11. If you did not choose to automatically recycle application services after the upgrade, restart the upgraded services.

You can view the upgrade report and save the upgrade report. The second time you run the service upgrade wizard, the Save Previous Report option appears in the service upgrade wizard. If you did not save the upgrade report after upgrading services, you can select this option to view or save the previous upgrade report.

Add the Informatica Domain License

Perform the following steps to add the new license for the Informatica domain:

1. Log in to Informatica Administrator.
2. Select the Informatica domain.
3. Click **New** > **License**.
4. Provide a name for the new license.
5. Select the new license key from the **License** drop-down list.

Step 5. Complete the Informatica Domain Upgrade

Verify the Gateway User Prerequisites

The gateway user must be a non-root user with sudo access. You must enable a passwordless SSH connection between the Informatica domain and the gateway host for the gateway user. The gateway user must have the required privileges to run the ping command.

Configure Sudo Privileges for the Gateway User (Optional)

If you plan to change the properties for the Informatica Cluster Service or replace the SSL certificates configured for the Informatica Cluster Service, assign sudo privileges to the gateway user.

The following commands require sudo privileges for the gateway user when you enable the Informatica Cluster Service for the first time:

- mkdir
- chown
- chmod
- echo
- systemctl
- cp
- mv
- sysctl
- rm
- tee
- sed
- sshd

To configure sudo privileges for the commands, you must add the commands to the `/etc/sudoers` file as shown in the following sample: `/bin/mkdir, /bin/chown, /bin/chmod, /bin/echo, /bin/systemctl, /bin/cp, /bin/mv, /usr/sbin/sysctl, /bin/rm, /usr/bin/tee, /usr/bin/sed, /usr/sbin/sshd`.

Note: The `tee` command does not require sudo privileges in 10.5.1.1 and later versions.

You can determine the directory where each command is located using the `which <command name>` command.

After you enable the Informatica Cluster Service for the first time, you can choose to disable the validation for sudo permissions for the Informatica Cluster Service. To disable the validation, configure the following custom property for the Informatica Cluster Service: `IcsCustomOptions.IcsGatewayUserSudoEnabled` and set the value to false. After you disable the validation, sudo privileges are not required when you restart the Informatica Cluster Service. However, to shut down the Informatica Cluster Service, sudo permission is still required for the `systemctl` command.

To run the `infacmd ics cleanCluster` command to clean the Informatica Cluster Service, sudo permission must be configured for the `systemctl` and `rm` commands.

Note: The services associated with the Informatica Cluster Service such as Apache Solr, PostgreSQL, and Nomad require a non-root user with sudo privileges to run the services. Sudo allows a user to run programs

or commands with elevated privileges for a specific time frame. Enterprise Data Catalog requires a non-root user with sudo permissions to run certain commands when performing administrative tasks such as installation, upgrade, and service status monitoring. For more information on the commands used and the role of each command during the upgrade, see the
Enable sudo privileges for a non-root user in Enterprise Data Catalog KB article.

Complete the Default SSL Configuration (Optional)

If the Informatica domain is enabled for SSL using the default SSL certificates, you can configure the Informatica Cluster Service and Catalog Service to use the new keystore.
To configure the services to use the 10.5.11 keystore, perform the following steps:

1. Disable the Informatica Cluster Service and the Catalog Service.
2. Edit the process options for the Informatica Cluster Service and Catalog Service and provide the location of the 10.5.11 `Default.keystore` file. Default path to the file is `<Informatica installation directory>/tomcat/conf/Default.keystore`.
3. From the 10.5.11 `<Informatica installation directory>/services/shared/security/infa_truststore.jks` file, remove the existing value configured for the *infa_dflt* alias and import the `<Informatica installation directory>/tomcat/conf/Default.keystore` file.
4. Enable the Informatica Cluster Service and Catalog Service.

Step 6. Enable and Upgrade Enterprise Data Catalog Services

Create and Enable the Informatica Cluster Service

When you create the Informatica Cluster Service, you must use the license for the 10.5.11 version of Informatica Cluster Service.

Perform the following steps to create the Informatica Cluster Service:

1. In the Administrator tool, select a domain, and click the **Services and Nodes** tab.
2. On the **Actions** menu, click **New** > **Informatica Cluster Service**
The **New Informatica Cluster Service: Step 1 of 4** dialog box appears.

3. Configure the general properties in the dialog box:

Property	Description
Name	Name of the service. The name is not case-sensitive and must be unique within the domain. The name cannot exceed 128 characters or begin with @. The name cannot contain character spaces. The characters in the name must be compatible with the code page of the Model repository. The name cannot contain the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the service. The description cannot exceed 765 characters.
Location	Select the Informatica domain node.
License	License to assign to the Informatica Cluster Service. **Note:** Select the license for the 10.5.11 version of Informatica Cluster Service.
Node	Primary node on which the Informatica Cluster Service runs. If you change the node, you must recycle the Informatica Cluster Service.
Backup Nodes	Nodes on which the service can run if the primary node is unavailable.

4. Click **Next**.
The **New Informatica Cluster Service - Step 2 of 4** dialog box appears.
5. Configure the security properties in the dialog box:

Property	Description
HTTP Port	A unique HTTP port number for the service. Default is 9075.
Enable Transport Layer Security (TLS)	Select the option to enable TLS for the Informatica Cluster Service.
HTTPS Port	Port number for the HTTPS connection. Required if you select **Enable Transport layer Security**. Default is 9533.
Keystore File	Path and file name of the keystore file. The keystore file contains the keys and certificates required if you use the SSL security protocol with Catalog Administrator. Required if you select **Enable Transport layer Security**. **Note:** Verify that you specified the correct keystore file for the Informatica Cluster Service. The certificates in the keystore must be trusted by the domain truststore. The keystore file must contain CA-signed certificates for custom SSL configuration.
Keystore Password	Password for the keystore file. Required if you select **Enable Transport Layer Security**.
SSL Protocol	Secure Sockets Layer protocol to use.

6. Click **Next**.
The **New Informatica Cluster Service - Step 3 of 4** dialog box appears.

7. Configure the cluster properties in the dialog box:

Property	Description
Gateway Host	Fully qualified domain name (FQDN) of the node that you want to configure as the gateway host. You can use the `hostname -f` command to view the FQDN of the node. The node that you configure as the gateway host must be a data node or a processing node. **Note:** You must configure a new gateway host for the Informatica Cluster Service.
Data Nodes	Comma-separated list of fully qualified domain names of nodes that you want to configure as data nodes. You can use the `hostname -f` command to view the FQDN of the node.
Processing Nodes	Comma-separated list of fully qualified domain names of nodes that you want to configure as processing nodes. You can use the `hostname -f` command to view the FQDN of the node.
Gateway User	User name for the gateway host. The gateway user must be a non-root user with sudo access. You must enable a passwordless SSH connection between the following nodes for the gateway user: - Informatica domain and the gateway host. - Informatica domain to the gateway host and gateway host to the data nodes and processing nodes. - If you enabled Advanced Configuration, enable the connection between the gateway host and all service nodes configured.
Cluster Custom Directory	Directory for the service. Default is `/opt/informatica/ics`. **Note:** If you want to modify the cluster custom directory, you must click **Enable Advanced Configuration**. After you change the directory, you must modify the directories configured for the services associated with the Informatica Cluster Service.
Cluster Shared File System Path	Applies if you deploy the service in multiple nodes. The shared directory on all cluster nodes. The service uses this directory on all cluster nodes to back up Apache Solr data. Verify the following directory prerequisites: - The directory must be empty. - The directory must have the NFS file system mounted. - The user ID to access the directory must be the same in all cluster nodes. - The user configured to access the directory must be a non-root user.

8. Optional. Click **Enable Advanced Configuration** if you want to configure the properties of the applications and associated services. By default, the services use the values that you provided for the data nodes as the host names. The PostgreSQL database uses the value specified for the gateway host as the host name.
   See the Informatica Cluster Service Advanced Configurationon page 106 section for information about the parameters that you must configure for the associated services.
9. Select **Enable Service** to enable the service after you click **Finish**.

   Note: By default, the associated services use the values that you provided for the data nodes as the host names. The PostgreSQL database uses the value specified for the gateway host as the host name.
10. Click **Next**.
    The **New Informatica Cluster Service - Step 4 of 4** dialog box appears.
11. Click **Finish**.

Informatica Cluster Service Advanced Configuration

If you selected **Enable Advanced Configuration** while configuring the Informatica Cluster Service, you must configure all the properties for the associated services.

Configure the following properties for the services associated with the Informatica Cluster Service:

Nomad Service Properties

Property	Description
Nomad Server Hosts	Comma-separated list of fully qualified domain names of nodes that host the Nomad servers.
Nomad HTTP Port	HTTP port number configured for the Nomad server. Default is 4646.
Nomad Serf Port	Serf port configured as the gossip protocol for the Nomad servers. Default is 4648.
Nomad RPC Port	The Remote Procedure Call (RPC) port configured for communication. Default is 4647.
Nomad Server Working Directory	The directory that includes sub directories with tasks running on the Nomad server. Default is `<Cluster custom directory>/nomad/nomadserver`.
Nomad Client Working Directory	The directory configured for tasks in the Nomad client. Default is `<Cluster custom directory>/nomad/nomadclient`.
Nomad Custom Options	Specify any custom options for the service in the following format: [OptionGroupName.OptionName=OptionValue]. You can separate multiple options using a white space character. If the OptionValue includes a white space character, you must enclose the OptionValue within double quotes as shown in the following sample: “sample value”.

ZooKeeper Service Properties

Property	Description
ZooKeeper Hosts	Comma-separated list of fully qualified domain names of nodes that host the Apache ZooKeeper server.
ZooKeeper Port	Port number configured for the Apache ZooKeeper Server. Default is 2181.
ZooKeeper Peer Port	Port number configured for Apache ZooKeeper peer communication. Default is 2888.
ZooKeeper Leader Port	Port number configured for the ZooKeeper Sever identified as the Leader. Default is 3888.
ZooKeeper Installation Directory	Specify the path to the directory where you want to install Apache ZooKeeper. Default is `<Cluster custom directory>/zk/install`.

Property	Description
ZooKeeper Data Directory	Specify the path to the directory where you want to store data from Apache ZooKeeper. Default is `<Cluster custom directory>/zk/data`.
ZooKeeper Custom Options	Specify any custom options for the service in the following format: [OptionGroupName.OptionName=OptionValue]. You can separate multiple options using a white space character. If the OptionValue includes a white space character, you must enclose the OptionValue within double quotes as shown in the following sample: “sample value”.

Apache Solr Service Properties

Property	Description
Solr Hosts	Comma-separated list of fully qualified domain names of nodes that host the Apache Solr server.
Solr Port	Port number configured for Apache Solr Server. Default is 8983.
Solr Installation Directory	Specify the path to the directory where you want to install Apache Solr Server. Default is `<Cluster custom directory>/solr/install`.
Solr Data Directory	Specify the path to the directory where you want to store data from Apache Solr. Default is `<Cluster custom directory>/solr/data .`
Solr Custom Options	Specify any custom options for the service in the following format: [OptionGroupName.OptionName=OptionValue]. You can separate multiple options using a white space character. If the OptionValue includes a white space character, you must enclose the OptionValue within double quotes as shown in the following sample: “sample value”.

Mongo DB Service Properties

Property	Description
MongoDB Hosts	Comma-separated list of fully qualified domain names of nodes that host the MongoDB database.
MongoDB Port	Port number configured for MongoDB. Default is 27017. **Note:** You cannot change the MongoDB port after you enable Informatica Cluster Service.
MongoDB Log Directory	Specify the path to the directory where you want to store the log files. Default is `<Cluster custom directory>/mongo/log`.
MongoDB Data Directory	Specify the path to the directory where you want to store data from the MongoDB database. Default is `<Cluster custom directory>/mongo/data`.
MongoDB Custom Options	Specify any custom options for the service in the following format: [OptionGroupName.OptionName=OptionValue]. You can separate multiple options using a white space character. If the OptionValue includes a white space character, you must enclose the OptionValue within double quotes as shown in the following sample: “sample value”.

PostgreSQL Service Properties

Property	Description
PostgreSQL DB Host	Fully qualified domain name of the machine that hosts the PostgreSQL database. Default is the gateway host. **Note:** If you did not select the Enable Advanced Configuration option, the service uses the gateway host value specified as the host value
PostgreSQL DB Port	Port number configured for PostgreSQL. Default is 5432.
PostgreSQL DB Installation Directory	Specify the path to the directory where you want to install the PostgreSQL database. Default is `<Cluster custom directory>/postgres/install`.
PostgreSQL DB Log Directory	Specify the path to the directory where you want to store the log files from the PostgreSQL database. Default is `<Cluster custom directory>/postgres/log`.
PostgreSQL DB Data Directory	Specify the path to the directory where you want to store PostgreSQL data. Default is `<Cluster custom directory>/postgres/data`.
PostgreSQL DB Custom Options	Specify any custom options for the service in the following format: [OptionGroupName.OptionName=OptionValue]. You can separate multiple options using a white space character. If the OptionValue includes a white space character, you must enclose the OptionValue within double quotes as shown in the following sample: “sample value”.

Associate Informatica Cluster Service with the Catalog Service

Edit the Catalog Service in Informatica Administrator to associate the Informatica Cluster Service with the Catalog Service.

Enable and Upgrade the Catalog Service

To enable and upgrade the Catalog Service, select the service and then click **Enable Service** from the **Actions** menu in Informatica Administrator.
After you start the Catalog Service, Informatica Administrator prompts you to upgrade the Catalog Service.

Perform the following steps:

1. Upgrade the Catalog Service.
2. Enable the email service if you had configured the service for the Catalog Service.

Note:

- If you enabled Data Asset Analytics, it takes longer to start the Catalog Service. The service will not be available until the Data Asset Analytics repository upgrade is complete.
- To know the status of the Data Asset Analytics repository upgrade, review the daa_upgrade.log file from the installation directory.
- If you enabled Data Asset Analytics, run the “Complete the Data Asset Analytics Prerequisites” on page 49.

Step 7. Migrate and Verify the Catalog Data

Migrate the Catalog Data

Use the infacmd.sh ldm migrateContents command to migrate the data that you backed up using the export utility.

Perform the following steps to migrate catalog data:

1. Set the following environment variables if the Catalog Service is enabled for SSL:
   - INFA_TRUSTSTORE. See the following sample command to set the variable: `export INFA_TRUSTSTORE= <Location of the Informatica truststore file>` . Default location is `<Informatica installation directory>/services/shared/security`.
   - INFA_KEYSTORE. See the following sample command to set the variable: `export INFA_KEYSTORE=<Location of the keystore file>`. Default location is `<Informatica installation directory>/services/shared/security`. This variable is required only if you used custom SSL configuration. For default SSL and non-SSL configurations, you must unset the variable.
   - INFA_TRUSTSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_TRUSTSTORE_PASSWORD="84Ve/soUbpQ/Aae5uGKXQA=="`.
   - INFA_KEYSTORE_PASSWORD. Encrypt the password that you set. See the following sample command to set the encrypted password: `export INFA_KEYSTORE_PASSWORD="6cDE/ItyUL/Rtui9nhVRI=="`. This variable is required only if you used custom SSL configuration. For default SSL and non-SSL configurations, you must unset the variable.

   You can contact Informatica Global Customer Support for the default truststore and keystore passwords.

   Note: See the following sample command to encrypt the password: `<Informatica installation directory>/server/bin/pmpasswd <password>`.
2. If exported catalog data is large, update the `xdocs.content.systemProperties=-Xmx20G` property in the `<Informatica installation directory>/services/CatalogService/Binaries/MigrationModuleConfigurations.properties` file, and then reset `ICMD_JAVA_OPTS=-Xmx20g` property in the `<Informatica installation directory>/isp/bin/infacmd.sh` file. For more information about performance sizing, see *Enterprise Data Catalog Performance Tuning Guide*.
3. Run the infacmd.sh LDM migrateContents command as shown: ./infacmd.sh LDM migrateContents -un <Informatica domain user name> -pd <Informatica Domain Password> -dn <Informatica Domain Name> -sn <Name of the Catalog Service> -id <Export directory>

Note: The <Export Directory> signifies the directory where the backed up content is available.

If you want to tune the performance parameters for migrating data or skipping failed resources, modify the parameters specified in the `MigrationModuleConfigurations.properties` file available at the following location: `<Informatica installation directory>/services/CatalogService/Binaries`

See the following table for more information about the performance parameters:

Property	Description
enableModule=	Set the value to true if you want to enable a specific module, else, set the property to false. For example, similarity.content.enableModule=true enables the migration of similarity profiling data.
Dblacklist=	List the resources that you want to skip in the following format: -Dblacklist=<resource name1>,<resource name2>
DmaxParallelRestores=	Specify the number of threads that you want to run in parallel when you migrate data.
systemProperties=	Use the property to specify the system resources that you want to use. For example, xdocs.content.systemProperties=-Xmx6G -XX:+HeapDumpOnOutOfMemoryError

Verify the Migrated Catalog Data

You can use the `infacmd migrateContents -verify` command as shown in the following sample to verify the migrated content: `./infacmd.sh LDM migrateContents -un Administrator -pd Administrator -dn Domain -sn CS -id /data/Installer1050/properties/utils/upgrade/EDC/export -verify`.

After you run the `infacmd migrateContents -verify` command, the command produces a file named `results.txt`.

The following table describes the messages that appear in the `results.txt` file:

Error Message	Description
`missing from backup`	Indicates that some resources were scanned before verifying the migrated catalog data.
`missing from warehouse`	Indicates that assets were not included in version 10.5.11.

Note: Make sure that the resource is not purged or reloaded before you to verify the migrated catalog data.

CHAPTER 5

After You Upgrade Enterprise Data Catalog

This chapter includes the following topics:

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General Post Upgrade Tasks

After you upgrade the Catalog Service, perform the following steps:

- After you upgrade to version 10.5.11, a warning message appears in Informatica Administrator for the Catalog Service stating that you have not configured Data Asset Analytics. You can configure Data Asset Analytics for Enterprise Data Catalog to view analytical insights about the catalog in the form of reports and charts.
- Copy the odbc.ini files from the external directories to the corresponding new directories created after the upgrade. Verify that the odbc.ini files include the correct entries.
- After you upgrade, case sensitivity is enabled by default for all resources of type **File Management** and **Cloud**.
- After you upgrade, case sensitivity is disabled for all resources of type **Database Management**.

  Note: You can configure the resources for case sensitivity as required and run the resource again.
- Install the Enterprise Data Catalog agent. See the Informatica Catalog Administrator Guide for more information.
- Define the stop words again if you defined them in the previous version of Enterprise Data Catalog.
- If you configured reference resources, you must run the resources again.
- If you upgraded the Data Asset Analytics repository, remove the `ProcessTerminationTimeout` custom property that you set for the domain.

.

Perform the following steps to edit or run the UCF resources after you upgrade the Catalog Service:

1. Contact Informatica Global Customer Support for a UCF license.
2. Apply the new UCF license to the Catalog Service.
3. Recycle the Catalog Service.
4. Run the UCF resource.
5. If the UCF resource scan fails, contact Informatica Global Customer Support.

Enrichment Migration Utility for Resources

The Enrichment Migration Utility can migrate enrichments, such as business title, custom attributes, descriptions, people properties, questions, and subscriptions for the following resources:

- Erwin
- IBM Cognos
- IBM Netezza
- Oracle
- Oracle Business Intelligence
- Qlik Sense
- Qlik View
- Salesforce
- SAP BusinessObjects

- Tableau

Restore the Enrichments with the Enrichment Migration Utility

After you upgrade and run the resources again, the enrichments for the supported resources might disappear because of the object ID changes. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments with the import command for the Enrichment Migration Utility.

Note: If you run the resources without exporting enrichments after upgrade, the enrichments will be lost.

1. Extract the `enrichment-migration-utility.jar` file from the following location on the Informatica domain machine:

   `<Informatica installation directory>/services/CatalogService/ScannerBinaries/enrichment-migration-utility/lib`

2. You can export the enrichments either by specifying each command option and argument with the export command or by utilizing the enrichment values from the JSON file to export.

   - To export the enrichments specifying each command option and argument, run the following export command:

     `java -jar enrichment-migration-utility.jar export -e <Catalog Service hostname>: <Catalog Service admin port number> -u <user name> -p <password> -s <resource name> -c <cache location where you extracted the enrichments jar file>`

     For example:

     `java -jar enrichment-migration-utility.jar export -e http://InfaHostName001.informatica.com:9085 -u Administrator -p admin -s Qliksenseresource -c c:/newfolder/utility`

   - To export the enrichments values directly from the JSON file, use the following command:

     `java -jar enrichment-migation.jar export -f <file location with the resource information>`

     Where, the file can contain the parameters and sample values for the URL, user name, password, resource name, and cache location.

     For example, the JSON file contains the following entries:

     `{ "edcUrl": "http://InfaHostName001.informatica.com:9085", "userName": "Administrator", "password": "admin", "srcResName": "Qliksense_util", "cacheLocation": "C:/utility/Qliksense_util" }`

     To bulk export all the enrichments from multiple resources, see "Bulk Export Resource Enrichments with the Enrichment Migration Utility" on page 114.

     You can encrypt the password before you use it in the export command.

     To encrypt the password, run one of the following commands from the command line:

     `Java -jar enrichment-migration-utility.jar encryptPassword <password>` or

     `Java -jar enrichment-migration-utility.jar ep <password>`

     The command displays an encrypted password that you can use in the export command to export the enrichments.

3. Run the resources again.

4. To restore the enrichments, run the following import command:

   ```
   java -Xms512M -Xmx19524M -jar enrichment-migration-utility.jar import -e <Catalog Service host name>: <Catalog Service admin port number> -u <user name> -p <password> -s <resource name> -c <cache location where you extracted the enrichments jar file>
   ```

   For example:

   ```
   java -jar enrichment-migration-utility.jar import -e http://InfaHostName001.informatica.com:9085 -u Administrator -p admin -s Qliksenseresource -c c:/newfolder/utility
   ```

   Verify that the enrichments and data are valid.

Bulk Export Resource Enrichments with the Enrichment Migration Utility

You can bulk export the enrichments from the supported resources before you run a resource scan post-upgrade. Bulk export of resource enrichments allows to export all enrichements from the supported resources at once.You can then restore the resource enrichments with the import command for the Enrichment Migration Utility.

Complete the following steps to back up all enrichments from multiple resources in bulk:

Extract the enrichment-migration-utility.jar

Extract the `enrichment-migration-utility.jar` file from the following location on the Informatica domain machine:

<Informatica installation directory>/services/CatalogService/ScannerBinaries/enrichment-migration-utility/lib

Prepare the configuration file

1. Create a JSON configuration file and name it as `config_file.json`.
2. Add the parameters for the export process to the json file in the following format::

   ```
   { "edcUrl": "Catalog Service hostname:Catalog Service admin port number", "userName": "username", "password": "password", "srcResName": "", "cacheLocation": "file location for cache", "upgradeTypeList": "location of upgradeScannerTypes.txt" }
   ```

   For example:

   ```
   { "edcUrl": "https://invedcsatyam04.informatica.com:9085", "userName": "user name", "password": "password", "srcResName": "", "cacheLocation": "C:\\file location for cache", "upgradeTypeList": "C:\\file location\\upgradeScannerTypes.txt" }
   ```

3. Save the `config_file.json` file to your local system.

Note: You can encrypt the password before you update the `config_file.json` file.

To encrypt the password, run one of the following commands from the command line:

`Java -jar enrichment-migration-utility.jar encryptPassword <password>` or

`Java -jar enrichment-migration-utility.jar ep <password>`

The command displays an encrypted password that you can use in the `config_file.json` file.

Optionally Prepare the resource type list

Note: When you do not configure `upgradeScannerTypeList.txt` file, Enrichment Migration Utility bulk exports all enrichments from each supported resource type.

1. Create a text file and name it as `upgradeScannerTypeList.txt`.
2. List the resource types that require enrichment migration after the upgrade. Add each resource type on a separate line.
   The following list shows the supported resource types as they might appear in the file:

   ```
   Cognos
   Erwin
   Generic.QlikView
   Netezza
   OBIEE
   Oracle
   QlikSense
   Salesforce
   SAPBO
   Tableau
   ```
3. Save the `upgradeScannerTypeList.txt` file to your local system.

Run the bulk export:

Run the following command to bulk export enrichments from multiple resource types after you upgrade:

```
java -jar enrichment-migration-utility.jar pre_upgrade_enrichment_export -f "/file location/config_file.json"
```

Alternatively, you can run the following command to bulk export enrichments with **pue** instead of **pre_upgrade_enrichment_export**:

```
java -jar enrichment-migration-utility.jar pue -f "/file location/config_file.json"
```

Import the SAP Transports

Perform the following steps if you use SAP BW, SAP BW/4HANA, and SAP S/4HANA resources:

1. Download the compatible versions of the SAP transports.

   Note: The transports are located in the `SAP_Scanner_Binaries.zip` file that is downloaded from the Informatica installer location.
2. Import the SAP transports into the SAP server.

Update Binaries for Informatica Platform Resources

In Enterprise Data Catalog version 10.5.11, Informatica updated some binaries that you must download and install to run Informatica Platform resources.

The ScannerBinaries.zip file includes the Informatica Platform resource binary files for 10.5.11 and the ExtendedScannerBinaries.zip file includes the Informatica Platform resource binary files from version 10.2 through 10.5.2.
After you upgrade Enterprise Data Catalog, perform the following steps if you want to run Informatica Platform resources:

1. Download the compressed file from the Informatica Electronic Software Download (ESD) site.
2. Based on the resource version, extract the required version of the zip file to the following location where you installed Enterprise Data Catalog:

   `<Informatica installation directory>/services/CatalogService/ScannerBinaries`

   Important: Verify that the directory contains only the latest version of the binaries. The directory must have at least 50 GB of free disk space.

Restore Resource Enrichments for Tableau

After you upgrade and run the resources again, the enrichments for Tableau resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Export the enrichments using the following command: `java -jar EnrichmentUtility.jar export <Catalog Service hostname> <Catalog Service admin port number> <Username> <Password> <Location to download enrichments json> <Resource name> <SSL enabled true/false>`.

   Note: You must specify a new directory to store the JSON files.
2. Run the Tableau resource again.
3. Run the following command to restore the enrichments: `java -Xms512M -Xmx19524M -jar EnrichmentUtility.jar import <Catalog Service host name> <Catalog Service admin port number> <Username> <Password> <Location to the enrichment scanned in an earlier release> <Location to the enrichment scanned in the current release> <SSL Enabled true/false>`

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Restore Resource Enrichments for Erwin

After you upgrade and run the resources again, the enrichments for Erwin resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Export the enrichments using the following command: `java -jar ErwinEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service port number> <Catalog administrator user name> <Catalog administrator password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>`

   Note: You must specify a new directory to store the JSON files.
2. Run the Erwin resource again.
3. Run the following command to restore the enrichments: `java -Xms512M -Xmx19524M -jar ErwinEnrichmentsCLIMigrator.jar import <Catalog Service hostname> <Catalog Service port number> <Catalog administrator user name> <Catalog administrator password> <location to enrichment scanned in an earlier release> <location to enrichment scanned in the current release> <SSL Enabled true/false>`

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Backup and Restore Resource Enrichments for SAP Business Objects

After you upgrade and run the resources again, the enrichments for SAP Business Objects resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Copy and extract the `BOEnrichmentsMigrator.jar` file from the following location on the Informatica domain machine:
   `<Informatica installation directory>/services/CatalogService/ScannerBinaries/bo-utility/lib`
2. Export the enrichments using the following command:

   ```
   java -jar BOEnrichmentsMigrator.jar export <Catalog Service hostname> <Catalog Service port number> <Catalog administrator user name> <Catalog administrator password> <location to download enrichments json> <resourceName> <SSL Enabled true/false>
   ```

   Note: You must specify a new directory to store the JSON files.
3. Run the resources again.
4. Run the following command to restore the enrichments:

   ```
   java -Xms512M -Xmx19524M -jar BOEnrichmentsMigrator.jar import <Catalog Service hostname> <Catalog Service port number> <Catalog administrator user name> <Catalog administrator password> <location to enrichment scanned in an earlier release> <location to enrichment scanned in the current release> <SSL Enabled true/false>
   ```

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Post Upgrade Tasks for Qlik Sense Resource

After you upgrade and run the resources again, the enrichments for Qlik Sense resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Export the enrichments using the following command: `java -jar QliksenseEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service admin port number> <Username> <Password> <Location to download enrichments json> <Resource name> <SSL enabled true/false>`.

   Note: You must specify a new directory to store the JSON files.
2. Run the Qlik Sense resource again.
3. Run the following command to restore the enrichments: `java -Xms512M -Xmx19524M -jar QliksenseEnrichmentsCLIMigrator.jar import <Catalog Service host name> <Catalog Service admin port number> <Username> <Password> <Location to the enrichment scanned in an earlier release> <Location to the enrichment scanned in the current release> <SSL Enabled true/false>`

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Post Upgrade Tasks for Salesforce Resource

After you upgrade and run the resources again, the enrichments for Salesforce resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Export the enrichments using the following command: `java -jar EnrichmentUtility.jar export <Catalog Service hostname> <Catalog Service admin port number> <Username> <Password> <Location to download enrichments json> <Resource name> <SSL enabled true/false>`.

   Note: You must specify a new directory to store the JSON files.
2. Run the Salesforce resources again.
3. Run the following command to restore the enrichments: `java -Xms512M -Xmx19524M -jar EnrichmentUtility.jar import <Catalog Service host name> <Catalog Service admin port number> <Username> <Password> <Location to the enrichment scanned in an earlier release> <Location to the enrichment scanned in the current release> <SSL Enabled true/false>`

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Post Upgrade Tasks for Relational Resources

After you upgrade and run the resources again, the enrichments for IBM Netezza and Oracle resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Export the enrichments using the following command: `java -jar RelationalEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service admin port number> <Username> <Password> <Location to download enrichments json> <Resource name> <SSL enabled true/false>`.

   Note: You must specify a new directory to store the JSON files.
2. Run the IBM Netezza and Oracle resources again.
3. Run the following command to restore the enrichments: `java -Xms512M -Xmx19524M -jar RelationalEnrichmentsCLIMigrator.jar import <Catalog Service host name> <Catalog Service admin port number> <Username> <Password> <Location to the enrichment scanned in an earlier release> <Location to the enrichment scanned in the current release> <SSL Enabled true/false>`

Note: Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112

Post Upgrade Tasks for IBM Cognos Resource

After you upgrade and run the resources again, the enrichments for IBM Cognos resources might be lost. To resolve this issue after you upgrade, ensure that you export the enrichments before you run the resource scan. You can then restore the resource enrichments using the migration utility.

1. Export the enrichments using the following command: `java -jar CognosEnrichmentsCLIMigrator.jar export <Catalog Service hostname> <Catalog Service admin port number> <Username> <Password> <Location to download enrichments json> <Resource name> <SSL enabled true/false>`.

   Note: You must specify a new directory to store the JSON files.
2. Run the IBM Cognos resources again.
3. Run the following command to restore the enrichments: `java -Xms512M -Xmx19524M -jar CognosEnrichmentsCLIMigrator.jar import <Catalog Service host name> <Catalog Service admin port number> <Username> <Password> <Location to the enrichment scanned in an earlier release> <Location to the enrichment scanned in the current release> <SSL Enabled true/false>`

Informatica recommends to back up and restore the enrichments with the generic `enrichment-migration-utility` after you upgrade to Enterprise Data Catalog as explained here: "Enrichment Migration Utility for Resources" on page 112.

Note: The Enrichment Migration Utility might not successfully migrate IBM Cognos enrichments that have incomplete object hierarchies. To migrate such enrichments, complete the following steps:

1. After you upgrade, perform a bulk export of enrichments before you run the resource scan.
2. Run the resource scan.

3. Copy the object IDs that the Enrichment Migration Utility did not migrate.

View Column Similarity Results

After you upgrade Enterprise Data Catalog, perform the following steps to view column similarity results of the previous Informatica version in the current version:

1. In Catalog Administrator, rerun all the existing resources that contains similarity results.
2. Create an Informatica Similarity Discovery resource. For more information about the similarity discovery resource creation, see the Informatica Similarity Discovery Resource topic in the *Informatica Scanner Configuration* guide.
3. In the **Resources to be grouped** field, select the resources from which you want to view column similarity results in Enterprise Data Catalog.
4. Run the Informatica Similarity Discovery resource.

Synchronize Data Asset Events for Data Asset Analytics

Also, perform the following steps to synchronize data asset events if you have configured Data Asset Analytics for the first time in the current version:

1. In Informatica Administrator, add the `LdmCustomOptions.ingest.store.events.on.reindex.bool` custom property for the Catalog Service and set the value as `true`.
2. Recycle the Catalog Service.
3. Re-index the Catalog Service.

Restore the Key Enrichments

After you upgrade to Enterprise Data Catalog version 10.5.11, you must perform the following steps to restore the primary key-foreign key enrichments using the migration utility:

1. Rerun all the relational resources, and export the enrichments using the following command:

   ```
   java -jar PKFKEnrichmentsRestoreUtility.jar export <Catalog Service hostname>
   <Catalog Service admin port number> <username> <password> <location to download
   enrichments json> <comma separated resource type IDs> <SSL Enabled true false>
   ```

   Verify that you provide a different location to store the JSON files.
2. Run the following command to restore the enrichments:

   ```
   java -Xms512M -Xmx19524M -jar PKFKEnrichmentsRestoreUtility.jar import <Catalog
   Service hostname> <Catalog Service admin port number> <username> <password>
   <location to enrichment scanned in an earlier release> <comma separated resource
   type IDs> <SSL Enabled true false> <location to enrichment scanned in the current
   release>
   ```

Run or Create Resources

After you upgrade Enterprise Data Catalog, you need to run or create the resources again to use certain functionalities.

See the following table to decide if you need to run or create a resource again after upgrade:

Resource Name	Criteria to Run the Resource
PowerCenter	Run the resource to get the column-level lineage for a PowerCenter resource with sources and targets involving local files and lineage for WebService Java transformation.
SQL Server Integration Service	Run the resource to get the following functionalities: - The detailed lineage for SQL Server Integration Service - The column-level lineage for sources and targets involving files - SQL Server Integration Service database repository bridge and control lineage.
Informatica Intelligent Cloud Service	Run the resource to get the following functionalities: - Connect Informatica Intelligent Cloud Service with cloud-based Common Client Interface connectors, such as: - Amazon S3 - Microsoft Azure Blob - Microsoft Azure Data Lake Store - Ensure column-level lineage for sources and targets involving local files and cloud-based sources.
IBM InfoSphere DataStage	Run the resource to get column-level lineage for DataStage sources and targets involving the local files.
Informatica Platform	Run the resource to get column-level lineage for Data Engineering Integration with the following sources and targets: - Local files - Microsoft Azure Data Lake Store - Microsoft Azure Blob - Amazon S3
Hive	Run the resource to include reference objects and manual and auto-connection assignment for cross schema cases when you scan Hive as separate resources.
Microstrategy	Run the resource to get the changes required for lineage correction, clutter removal, and incremental scan. Enrichments are lost after the first re-scan, but they are retained after the next re-scans.
Tableau	Run the resource to get fixes and object ID changes. Enrichments are lost. To retain the enrichments, use the Tableau enrichment migration utility. See "Back Up Tableau Resource Enrichments Using the Migration Utility " on page 55.
Qlik Sense	Run the resource to get object ID changes. Enrichments are lost. To retain the enrichments, use the Qlik Sense enrichment migration utility. For more information, see "Back Up Qlik Sense Resource Enrichments Using the Migration Utility " on page 55.
IBM Cognos	Run the resource to get object ID changes. Enrichments are lost.
Oracle Business Intelligence	Run the resource to get object ID changes. Enrichments are lost.

Resource Name	Criteria to Run the Resource
Salesforce	Run the resource to get object ID changes. Enrichments are lost. To retain the enrichments, use the Salesforce enrichment migration utility. For more information, see "Back Up Salesforce Resource Enrichments Using the Migration Utility " on page 55.
Microsoft SQL Server	Run the resource to include reference objects and manual and auto-connection assignment for cross schema cases when you scan Microsoft SQL Server as separate resources.
Amazon S3	Run the resource to include Parquet files profiling on S3 filesystem. **Note:** Change the execution mode to Spark and run the resource again with profiling.
Microsoft Azure Data Lake Store	Run the resource to include metadata extraction and profiling of Parquet files. **Note:** Change the execution mode to Spark and run the resource with profiling.
SAP BW	Run the resource to include the following functionalities: - Lineage for the Function Module - Support for Data Profiling - Filter based on different asset types - Support for a secure network connection
SAP BW/4HANA	Run the resource to include the following functionalities: - Lineage for the Function Module - Support for Data Profiling - Filter based on different asset types - Support for a secure network connection
SAP S/4HANA	Run the resource to include the following functionalities: - Lineage for the Function Module - Support for Data Profiling - Filter based on different asset types - Support for a secure network connection
Apache Atlas	Create the resource to extract the entities filter parameter that allows the new Atlas search syntax.
Workday	Run the resource to include fixes related to a distributed entity asset.

Purge and Run the Reference Resources Again

After you upgrade Enterprise Data Catalog, to use the features added to the reference resources in Enterprise Data Catalog version 10.5.11, you must purge and rerun the resources if you had used the resources before upgrade.

Post Upgrade Tasks for MetaDex

After you upgrade Enterprise Data Catalog, to use the features added to MetaDex, perform the post-upgrade tasks.

1. On the MetaDex server machine, verify that the following environment variables are set:
   - SCANNERS_HOME
   - SCANNERS_WORKSPACE
2. Associate Enterprise Data Catalog with MetaDex by performing the following actions:
   a. In Informatica Administrator, start the Catalog Service.
   b. Log in to the MetaDex tool as an administrator.

      Note: You can access the web interface by providing the URL in the <host>:<port> format. Where:

      <host> represents the host name configured for the Informatica domain

      <port> represents the port number configured for the MetaDex repository.

      See "Logging in to the MetaDex tool" in the *MetaDex Administrator Guide*.

      For enhanced security, you must change the password after you log in for the first time.
   c. In the header, click **Administration**, and go to **EDC Instance**.
   d. Configure the following variables:

Variable	Value
EDC_USER	The user name of the catalog administrator.
EDC_PASSWORD	The password configured for the administrator. **Note:** When you add the password, in the **New global variable** window, click the **Encrypt password** icon to encrypt the password.
EDC_URL	The host name and port number configured for Enterprise Data Catalog in the following format: <host>:<port>

   e. Shut down the MetaDex server using the `server.sh stop` command available in the following directory: `<INFA_HOME>/services/CatalogService/AdvancedScannersApplication/app/`
   f. Start the MetaDex repository server using the following command: `<Informatica installation directory>/services/CatalogService/AdvancedScannersApplication/app/server.sh`
3. Load the model definition to the catalog by running the following script: `<Informatica installation directory>/services/CatalogService/AdvancedScannerApplication/app/utils/scanners/installScanners.sh`
4. Recycle the Catalog Service.

5. Depending on the enrichments for MetaDex, perform one of the following actions:

Option	Description
If there are no enrichments configured for MetaDex	Refresh the Enterprise Data Catalog MetaDex resources with the same name by performing the following actions: 1. Delete the existing MetaDex resource in Enterprise Data Catalog using the Catalog Administrator. 2. Run the resource configuration from the MetaDex tool with the same resource name.
If there are enrichments configured for the MetaDex	Perform the following actions: 1. Edit the existing configuration from the MetaDex tool and rename the Enterprise Data Catalog resource. 2. Run the updated configuration from the MetaDex tool. **Note:** An Enterprise Data Catalog resource is created in the Catalog Administrator tool with the name that you provided in the configuration. This resource does not include the enrichments. You must assign the enrichments manually. The existing MetaDex resource with custom metadata is retained with enrichments.

Post Upgrade Tasks for Microsoft SQL Server MetaDex scanner

After you upgrade to Enterprise Data Catalog 10.5.3 or later, you must enable the encrypt property when you configure the Microsoft JDBC Driver for Microsoft SQL Server MetaDex scanner.

By default, the Microsoft JDBC Driver for SQL Server uses encrypt=true. You can disable encryption if the database connection is not encrypted. Use encrypt=false in the JDBC URL to disable encryption.
Note: Informatica recommends that you always encrypt the connection to the database.

If you use encrypt=false in the JDBC URL, and the Microsoft SQL Server database mandates encryption, you must validate the certificate using the trustServerCertificate connection setting.

For more information, see the *MetaDex Scanner Administrator Guide*.

Update the Driver Files for Resources

After you upgrade Enterprise Data Catalog, update the driver JAR files from the previous installation.

Perform the following steps to copy the driver JAR files:

1. Disable the Catalog Service.
2. Copy the JAR files from the following directory: `<Backup directory>/services/CatalogService/ScannerJars/externalDependencies`
3. For each resource type available in the catalog, copy the JAR files to the respective locations as shown:
   Oracle

   Copy ojdbc7.jar to the following directory: `<Informatica installation directory>/services/CatalogService/ScannerJars/externalDependencies/OracleScanner`

Microsoft SQL Server

Copy mssql-jdbc-7.4.1.jre8.jar to the following directory: `<Informatica installation directory>/services/CatalogService/ScannerJars/externalDependencies/MSSQLServerScanner`

Azure Microsoft SQL Server

Copy mssql-jdbc-7.4.1.jre8.jar to the following directory: `<Informatica installation directory>/services/CatalogService/ScannerJars/externalDependencies/AzureMssqlScanner`

Azure Microsoft SQL Data Warehouse

Copy mssql-jdbc-7.4.1.jre8.jar to the following directory: `<Informatica installation directory>/services/CatalogService/ScannerJars/externalDependencies/AzureSqlDWScanner`

IBM DB2

Copy db2jcc4.jar to the following directory: `<Informatica installation directory>/services/CatalogService/ScannerJars/externalDependencies/IBMDB2Scanner`

4. Recycle the Catalog Service.

Remove Deleted Resources Migrated by the Export Utility

When you run the export.jar utility to migrate data as part of the upgrade from version 10.4 or 10.4.1, the utility exports xdocs of deleted resources, and the deleted resources might appear in the catalog. To retrieve the list of deleted resources and remove the deleted resources from the catalog, run the infacmd ldm removeDeletedMigratedResources command.

For information on how to run the command, see the *Informatica Command Reference*.

Post Upgrade Tasks for Cloudera Distributed Hadoop (CDH) 6.3 Version Binaries

Informatica dropped support for Cloudera Distributed Hadoop (CDH) 6.3 version in Enterprise Data Catalog. You cannot run scanners that are compatible with Cloudera CDH or Hortonworks CDH. To run these scanners post-upgrade in Enterprise Data Catalog, you need to back up and restore Cloudera Distributed Hadoop (CDH) version 6.3 binaries.

1. After you upgrade, restore Cloudera Distributed Hadoop (CDH) version 6.3 binaries to `<infa_home>/services/shared/hadoop/`. For more details about backing up Cloudera Distributed Hadoop (CDH) 6.3 version binaries, see "Back Up Cloudera Distributed Hadoop (CDH) 6.3 Version Binaries" on page 57.
2. Restart the Catalog Service.

Post Upgrade Observation

In Catalog Administrator, you might notice a difference in the number of objects discovered for resources pre-upgrade and post-upgrade.

The object mismatch occurs due to the following reasons:

- All objects that do not have class types associated with them are not stored in MongoDB.
- For objects marked as case insensitive in the catalog and marked as case sensitive in the data source, only one object is included in the total number of objects.
- For assets of type table that have the same name but with differences in casing, only one object is used to count the total number of objects.
- For reference objects, proxy objects are not created for each directory, but a maximum of two proxy objects are added to the count of objects.
- In the Catalog Administrator tool, in the **Security** tab, the resource names for reference resources appear in the following format: <Resource name>$$<Provider ID>$$<Connection>.
- In the Catalog Administrator tool, in the **Connection Assignment** tab, only the connection assignments that are contained in the XDOC file for the resource appear.
- If there are objects with the same ID, only one object is added to the count of objects.
- If an object has multiple values associated with the an attribute value, all values are included in the object count.
- For Teradata and Azure Microsoft SQL Server resources, the post upgrade count indicates the total count of all assets, which was not visible in version 10.4.1.
- After you migrate the catalog data as part of the upgrade, the ID and name of the reference objects are converted to uppercase if the resource to which you assigned the connection is case insensitive.
- For a reference resource, access filter ResourceName_ProviderId_Connection will be removed and access filter ResourceName$$ProviderId$$Connection will be retained

CHAPTER 6

Troubleshooting Upgrade

This chapter includes the following topics:

- Troubleshooting Upgrade Overview, 127
- Troubleshooting Upgrade Issues , 127

Troubleshooting Upgrade Overview

The following are the issues that you might encounter while upgrading Enterprise Data Catalog. You can refer to the solutions provided to resolve the issues.

Troubleshooting Upgrade Issues

Catalog Upgrade

ICMD_10033] Command [BackupData] failed with error [[BackupRestoreClient_10002] Following error occurred while taking backup of data : [BackupRestoreClient_00027] Snapshot with name [ldmCluster] in HDFS snapshot directory [/Informatica/LDM/ldmCluster] already exists.

The error can occur as a result of an external snapshot of the catalog for Enterprise Data Catalog deployed in an internal cluster.

Run the removesnapshot command to remove the external snapshot and then run the BackupContents command to create a snapshot.

org.apache.hadoop.ipc.RemoteException(java.io.IOException): File /tmp/cat_ihs_backup/scanner/MRS_INFACMD_10.zip could only be replicated to 0 nodes instead of minReplication (=1). There are 1 datanode(s) running and no node(s) are excluded in this operation.

You might encounter this error while restoring HDFS data if there is insufficient free disk space in the machine where data must be restored.

Verify that the machine where HDFS data is restored has sufficient free disk space.

java.util.concurrent.ExecutionException: java.lang.OutOfMemoryError: GC overhead limit exceeded

You might encounter this error when you restore the similarity store backup and the default Java heap size of 512 MB is configured.

Set the ICMD_JAVA_OPTS environment variable to -Xmx4096m to increase the heap size to 4096 MB.

If the backup and restore fails for the Catalog Service, set the following properties for the cluster and restore the catalog:

- `set -mode=HDFS`
- For an external cluster, verify that the YARN property `yarn.nodemanager.aux-services` includes `mapreduce_shuffle`.

The Informatica keystore password is incorrect or corrupted.

Verify that you set the INFA_KEYSTORE_PASSWORD environment variable correctly.

Command [restorecontents] failed with error [/data/Informatica/LDM/services/shared/security/infa_truststore.jks (No such file or directory)]

Verify that you set the INFA_TRUSTSTORE and the INFA_TRUSTSTORE_PASSWORD environment variables correctly.

If upgrade fails for a nomad job with the following error message: Upgrade monitoring found [{0}] failed jobs for resources: [{1}]. Where can I view the log file details?

You can find the log file at the following location: `$ServiceLogDirectory/UpgradeJobs/`. For example, `$ServiceLogDirectory/UpgradeJobs/Dummy_Upgrade_Module/DummyResource/0dafea11-adc0-4abe-8ca9-d48681b097d7/Publish/log.zip`

Where can I monitor the upgrade progress?

You can monitor the upgrade progress in the LDM_Upgrade.log file at the following location: `$ServiceLogDirectory`.

When you migrate the backed up catalog data, the following error message appears in the Migration.log file: Error from server at https://host.informatica.com:8983/solr/collection1: Error from server at null: Could not parse response with encoding ISO-8859-1org.apache.solr.client.solrj.impl.HttpSolrClient$RemoteSolrException: Error from server at https://host.informatica.com:8983/solr/collection1: Error from server at null: Could not parse response with encoding ISO-8859-1

Verify that ntpd is synchronized between the Informatica domain node and the cluster nodes

Domain Upgrade

If SAML login fails with the following error message: "AADSTS900235: SAML authentication request's RequestedAuthenticationContext Comparison value must be 'exact'. Received value: 'Minimum'."

Change the property value for the Enable Authorization Context Comparison=EXACT option in Informatica Administrator under the **SAML Configuration** section of the **Catalog Service** > **Processes** tab.

After you enable the Catalog Service, the Catalog Administrator tool is not accessible

Delete the following directories:

```
<Informatica installation directory>/tomcat/temp/<Catalog Service name>
<Informatica installation directory>/services/CatalogService/ldmadmin
<Informatica installation directory>/services/CatalogService/ldmcatalog
```

The Informatica Cluster Service fails to start

Open the sshd_config file from the following directory: `/etc/ssh`. Change the value of the MaxSessions parameter to 60 and the MaxStartups parameter to 1024, and then restart the SSH daemon service.

The Catalog Service fails with the following error message: "unable to create INITIAL extent for segment in tablespace"

Increase the size of the tablespace for the Model Repository Service database.

CHAPTER 7

Informatica Client Upgrade

This chapter includes the following topics:

- Informatica Client Upgrade Overview, 129
- Before You Upgrade the Informatica Client , 129
- Upgrading in Graphical Mode, 130
- Upgrading in Silent Mode, 131
- After you Upgrade , 133

Informatica Client Upgrade Overview

Use the client installer to upgrade a previous version of the Informatica client tools. Based on the product that you're upgrading, you can upgrade the Developer tool, the PowerCenter Client, or both.

Run the installer on all machines hosting a client that you want to upgrade. You can upgrade the Informatica client tools in one of the following ways:

- Upgrade in graphical mode. Upgrades the Informatica client tools in graphical mode. The installer guides you through the upgrade process.
- Upgrade in silent mode. Upgrades the Informatica client tools using a properties file that contains the upgrade options.

The upgrade process also upgrades the JRE libraries and the DataDirect ODBC drivers.

You can perform the upgrade from the root of the directory where you download the installation files.

Before You Upgrade the Informatica Client

Verify that the prerequisites and minimum requirements for installation are met. If the machine where you upgrade the Informatica clients tools is not configured correctly, the upgrade can fail.

Verify the installer package checksum

Before you run the client installer, verify the install package integrity through the cksum command. The cksum command calculates the checksum value for the installer.

Verify the checksum for the specific installer files against the checksum of the installation files downloaded from the Informatica Electronic Software Download site.

The following table lists the checksum and file size for Informatica client on Windows:

File	Checksum Value	File Size
informatica_10510_client_winem-64t.zip	1758402761	4216046089 bytes

A checksum mismatch can occur when there are data errors during download due to network issues or when data corruption occurs in the file on disk. For more information about the checksum errors, see HOW TO: Identify file errors after downloading Informatica installation files.

Verify requirements for Informatica Developer

The following table lists the minimum requirements to upgrade Informatica Developer:

Component	Requirement
Processor	1 CPU
RAM	1 GB
Disk spaace	8.505 GB

Upgrading in Graphical Mode

When you upgrade in graphical mode, you provide information about the version you're upgrading and the directories associated with the upgrade.

1. Close all applications.
2. Run install.bat from the root directory.

   If you encounter problems, run install.exe from the following location:

   ```
   <Informatica installation directory>\client
   ```
3. Read the terms and conditions for Informatica installation and the product usage toolkit and select **I agree to the terms and conditions**.

   Subject to your opt-out rights described below, the software will automatically transmit to Informatica in the USA information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at https://www.informatica.com/in/privacy-policy.html. You can disable usage collection in Administrator tool.

   a. Press **1** if you don't want to accept the terms and conditions.

   b. Press **2** to accept the terms and conditions.
4. On the **Upgrade Pre-requisites** page, verify the system requirements before you continue the installation and click **Next**.

5. On the **Select Directory** page, enter the directory of the Informatica version and the source and target directories.

   The following table describes the directories to specify:

Directory	Description
Directory of the Informatica client to upgrade	Directory that contains the version of the Informatica client tool that you want to upgrade.
Directory for the client tools	Directory where you want to install the client tools. Enter the absolute path for the installation directory. The installation directory must be on the current machine. The directory names in the path can't contain spaces or the following special characters: @|* $ # ! % ( ) { } [ ] , ; ' **Note:** Informatica recommends using alphanumeric characters in the installation directory path. If you use a special character such as á or €, unexpected results might occur at run time.

6. Click **Next**.
7. On the **Pre-installation Summary** page, review the installation information and click **Install**.

   The installer copies the Informatica client files to the installation directory.
8. On the **Post-installation Summary** page, verify that the upgrade completed successfully and click **Done** to close the installer.
9. After you complete an upgrade of client, log off the Windows machine and then log back on to complete the system configurations.

You can view the installation log files to get more information about the upgrade tasks performed by the installer.

Upgrading in Silent Mode

To upgrade the Informatica client tools without user interaction, upgrade in silent mode. Use a properties file to specify the upgrade options. The installer reads the file to determine the upgrade options. You can use silent mode upgrade to upgrade the Informatica client tools on multiple machines on the network or to standardize the upgrade process across machines.

Copy the Informatica installation files to the hard disk on the machine that hosts the Informatica client you plan to upgrade.

To upgrade in silent mode, complete the following tasks:

1. Create the upgrade properties file and specify the upgrade options.
2. Run the installer with the upgrade properties file.

Creating the Properties File

Informatica provides a sample properties file that includes the upgrade parameters that are required by the installer. You can customize the sample properties file to specify the options for your upgrade.

The sample properties file is named SilentInput.properties and is located in the root of the client installer directory.

1. Go to the root of the directory that contains the client installation files.
2. Locate the file named SilentInput.properties.

   Back up the file before you modify it.
3. Use a text editor to open the file and modify the values of the upgrade parameters.

   The following table describes the upgrade parameters you can modify:

Property Name	Description
INSTALL_TYPE	Indicates whether to install or upgrade the Informatica client tools. To upgrade from a previous version of Informatica, set the value to 1.
USER_INSTALL_DIR	Directory in which to install the new version of the Informatica client tools.
UPG_BACKUP_DIR	Directory of the previous version of the Informatica tools that you want to upgrade.
INSTALL_HADOOP_LIBRARIES	Determines whether to install distribution packages through the installer. Set the value to true if you want to install distribution packages through the installer. Set the value to false if you don't need distribution packages or if you want to install them later.
SELECTED_HADOOP_LIBRARIES	Determines the distribution packages that you want to install from the supported packages list. Enter the distribution packages that you want to install, separating multiple packages with a comma.

4. Save the properties file.

Running the Silent Installer

After you create the properties file, open a command prompt to start the silent upgrade.

1. Open a command prompt.
2. Go to root of the client installer directory.
3. Verify that the directory contains the file SilentInput.properties with the upgrade options.
4. To start the silent upgrade process, run silentInstall.bat.

   The silent upgrade runs in the background. The process can take a while. The silent upgrade process is complete when the Informatica_<Version>_Client_InstallLog.log is created in the installation directory.

   The silent upgrade fails if you incorrectly configure the properties file or if the installation directory is not accessible. If the upgrade fails, view the installation log files and correct the errors. Then run the silent installer again.
5. After you complete an upgrade of Informatica Developer, log off the Windows machine and then log back on to complete the system configurations.

After you Upgrade

After you upgrade, import the data domains for the profiles.

To add predefined data domain groups and related data domains to the data domain glossary, import the Informatica_IDE_DataDomain.xml file into the Developer tool using the **Windows > Preferences > Informatica > Data Domain Glossary > Import** menu option.

To view and make changes to rules associated with data domains, import the Informatica_IDE_DataDomainRule.xml file using the **File > Import** menu option in the Developer tool.

APPENDIX A

Upgrade in Silent Mode

To upgrade the Informatica domain and application services without user interaction, you can upgrade in silent mode. You can use the properties file bundled with the installer to specify the upgrade options. The installer reads the file to determine the upgrade options. You can use silent upgrade to upgrade the Informatica domain services on multiple machines on the network or to standardize the upgrade process across machines.

To upgrade in silent mode, complete the following tasks:

1. Copy the Informatica installation files to the machine that hosts the Informatica instance you plan to upgrade.
2. Configure the upgrade properties file and specify the upgrade options in the properties file.
3. Run the installer with the upgrade properties file.

Configuring the Properties file

Informatica provides two versions of the properties file. You can use any of the following files to specify options for that upgrade.

SilentInput_upgrade_NewConfig.properties file

Use the silent input upgrade new configuration properties file to upgrade Informatica domain and application services with changes to the node configuration.

The silent input upgrade new configuration properties file contains the configuration properties required to upgrade the Informatica domain and application services in silent mode. Use the file if you want to view the sample values specified for each property.

Default SilentInput_upgrade.properties file

Use the silent input upgrade properties file to upgrade Informatica domain and application services without user interaction.

The default silent input properties file contains default values for many configuration properties. You can Use the file if you plan to upgrade the Informatica domain services using the default property values.

Informatica provides a sample properties file that includes the upgrade parameters that are required by the installer. You can customize the sample properties file to specify the options for the upgrade.

The sample upgrade properties file is named SilentInput_upgrade.properties and is located in the location where you downloaded the installer files. After you customize the file, save it with the file name, SilentInput.properties.

1. Go to the directory where you downloaded the installer files.

2. Find the file named SilentInput_upgrade.properties.
3. Back up the SilentInput_upgrade.properties file before you modify it.
4. Use a text editor to open the file and modify the values of the upgrade parameters as required.

   The following table describes the upgrade parameters that you can modify:

Property Name	Description
ENABLE_USAGE_COLLECTION	Enables Informatica DiscoveryIQ, a product usage tool that sends routine reports on data usage and system statistics to Informatica. Informatica DiscoveryIQ uploads data to Informatica 15 minutes after you install and configure Informatica domain. Thereafter, the domain sends the data every 30 days. You can choose to not send any usage statistics to Informatica. For more information on how to disable sending usage statistics, see the *Informatica Administrator Guide*. You must set the value to 1 to upgrade.
INSTALL_TYPE	Indicates whether to install or upgrade Informatica. Default is 1. To perform a new installation of Informatica, use the SilentInput.properties file. To upgrade Informatica to a different node configuration, use the SilentInput_upgrade_NewConfig.properties file.
INSTALL_LDM	Indicates whether to upgrade Enterprise Data Catalog too along with the domain services. Default is 0. If the value is 1, the installer upgrades a previous version of Informatica. If the value is 0, the installer upgrades Informatica without Enterprise Data Catalog.
USER_INSTALL_DIR	Directory in which you install the new version of Informatica services. The directory cannot be the same as the directory that contains the previous version of Informatica services.
UPG_BACKUP_DIR	Directory that contains the previous version of Informatica services that you want to upgrade.
INSTALLATION_ENVIRONMENT	Represents the installation environment. You can set one of the following properties as the installation environment type: - Sandbox. A basic environment used as a proof of concept with minimal users. - Development. The design environment. - Test. High volume processing that is closest to a production environment. - Production. High volume processing with high levels of concurrency for end users. The advanced production environments are typically multi-node setups. The value is case-sensitive.
EDC_CATALOG_INST	Set the property to true if the Catalog Service is running on the node that you want to upgrade.
EDC_UPGR_BACKUP	Set the property to true if you ran the catalog backup utility and the infacmd ldm BackupContents command before upgrade.
EDC_BACKUP_DIR	Enter the location of the catalog backup directory.

Property Name	Description
INSTALL_HADOOP_LIBRARIES	Determines whether to install distribution packages through the installer. Set the value to true if you want to install distribution packages through the installer. Set the value to false if you don't need distribution packages or if you want to install them later.
SELECTED_HADOOP_LIBRARIES	Determines the distribution packages that you want to install from the supported packages list. Enter the distribution packages that you want to install, separating multiple packages with a comma.
DOMAIN_USER	User name for the Informatica domain.
DOMAIN_PSSWD	Password for the Informatica domain.
DOMAIN_CNFRM_PSSWD	Confirm the password for the Informatica domain.
SECURITY_DOMAIN_NAME	If you do not want to use the default security domain, specify the security domain name that you want to use.
KEY_DEST_LOCATION	Directory in which you want to store the encryption key for the node that was created during the installation.
CREATE_MDX_SERVICE	Set the property value to true if you want to create the MetaDex repository. Set the property value to false if you do not want to create the MetaDex repository.
MDX_DB_TYPE	The property represents the database type for the MetaDex repository. Set the property to one of the following database types: - Oracle - MSSQLServer The value is case-sensitive.
MDX_DB_UNAME	The property represents the database user account name for the MetaDex repository.
MDX_DB_PASSWD	The property represents the database password for the database user account.
MDX_SQLSERVER_SCHEMA_NAME	The property represents the name of the schema. You must configure the property if MDX_DB_TYPE=MSSQLServer
MDX_DB_SSL_ENABLED	Indicates whether the database for the MetaDex repository is secure. To create the domain configuration repository in a secure database, set the property value to true. To use the custom string option, if the property is set to true, you must set the MDX_DB_CUSTOM_STRING property value to 1. The MDX_DB_CUSTOM_STRING property must include the following secure database parameters: - EncryptionMethod=SSL - HostNameInCertificate= - ValidateServerCertificate=

Property Name	Description
MDX_DB_TRUSTSTORE_FILE	Indicates the path and file name of the truststore file for the MetaDex repository database. The property is required if you set the MDX_DB_SSL_ENABLED property value to true.
MDX_DB_TRUSTSTORE_PASSWD	The truststore file password for the MetaDex repository database. The property is required if you set the MDX_DB_SSL_ENABLED property value to true.
MDX_SSL_DEFAULT_STRING	The property is required only if the MDX_DB_SSL_ENABLED property value is set to true. The property must include the following secure database parameters: - EncryptionMethod=SSL - HostNameInCertificate= - ValidateServerCertificate=
MDX_DB_CUSTOM_STRING	The property represents a valid custom JDBC connection string.
MDX_SERVER_PORT	The property represents the MetaDex repository database server port.
MDX_SSL_ENABLED	Optional. Set the property to true if you want to enable secure communication to the MetaDex database.
MDX_CUSTOM_SELECTION	You must configure the property if the MDX_SSL_ENABLED property is set to true. To use the default SSL certificate files, set the MDX_CUSTOM_SELECTION property to false. To use the custom SSL certificate files, set the MDX_CUSTOM_SELECTION property to true. Default is false.
MDX_KEYSTORE_FILE	You must configure the following properties if the MDX_CUSTOM_SELECTION property is set to true: - MDX_KEYSTORE_PASSWD - MDX_KEYSTORE_TYPE - MDX_KEYSTORE_ALIAS You need not configure the MDX_KEYSTORE_FILE property if the MDX_CUSTOM_SELECTION property is set to false.
MDX_KEYSTORE_PASSWD	Password for the keystore.
MDX_KEYSTORE_TYPE	Indicates the type of the keystore certificate.
MDX_KEYSTORE_ALIAS	The property represents the keystore alias.

5. Save the properties file with the name SilentInput.properties.

Running the Silent Installer

Perform the following steps to run the silent installer:

1. Go to the directory that contains the installation files.

2. Verify that the directory contains the SilentInput.properties that you updated with the upgrade options.
3. On a shell command line, enter the following command to run the silent installer: ./silentInstall.sh.

The silent upgrade runs in the background. The process can take a while. The silent upgrade process is complete when the Informatica_<Version>_Services_InstallLog<timestamp>.log is created in the installation directory.

The silent upgrade fails if you incorrectly configure the properties file or if the installation directory is not accessible. If the upgrade fails, correct the errors in the silentErrorLog.log log file and run the command again.

APPENDIX B

Managing Distribution Packages

This appendix includes the following topics:

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- Before You Begin, 139
- Install or Remove Distribution Packages in Console Mode, 140
- Install or Remove Distribution Packages in Silent Mode, 141
- After You Install, 141

Managing Distribution Packages Overview

You can use Integration Package Manager (the package manager) to install and remove distribution packages from the Informatica service and client machines.

A distribution package is a set of distribution binaries that you install within the domain for the following processing requirements:

- To push processing to the Hadoop or Databricks environment.
- To process complex files within the Informatica domain.
- To connect to the Hadoop or Databricks environment when you process within the Informatica domain.

You can install distribution packages if you didn't do so during the upgrade or install process or if you want to add a distribution package. You can remove a distribution package if you want to use a different package or if you installed a package that you don't use.

When you install or remove distribution packages, verify that you perform the operation on all service and client machines.

Before You Begin

Before you run the package manager, perform tasks such as setting environment variables and downloading files.

1. Shut down the Informatica services.

2. Set one of the following environment variables:

Variable	Description
INFA_JDK_HOME	Location of the folder containing the supported Java Development Kit (JDK). Set the INFA_JDK_HOME environment variable in the following scenarios: - Informatica domain is on Windows or Linux - Informatica client
INFA_JRE_HOME	Location of the folder containing the supported Java Runtime Environment (JRE). If the Informatica domain is on AIX, set the INFA_JRE_HOME environment variable.

3. Verify that the user that runs the package manager has read and write permissions on the Informatica installation directory and execute permissions on the executable file.
4. Download the following files from the Informatica Electronic Software Download site:
   - Integration Package Manager
   - Distribution packages
5. Extract the Integration Package Manager ZIP files to a local drive.
6. Copy the ZIP files of distribution packages that you need to the following location: `<Integration Package Manager directory>/source`

   Note: The package manager fails if the ZIP files for distribution packages aren't available in the source directory.

Install or Remove Distribution Packages in Console Mode

You can run the package manager in console mode to install or remove distribution packages.

1. From the package manager directory, run one of the following commands:
   - `./Server.sh console` for Linux or UNIX
   - `Server.bat console` for Windows
   - `Client.bat console` for client

   Note: To run the command on Windows, use the administrator command prompt.
2. Enter the installation directory of the services or client and press **Enter**.
3. Choose the operation type and press **Enter**.
   - Select 1 to remove existing distribution packages.
   - Select 2 to install one or more distribution packages.

   The console lists the distribution packages that you can install or remove.
4. Enter the distribution packages that you want to install or remove, separating multiple packages with a comma, and press **Enter**.
5. Verify the installation or removal status in the package manager log file.

   You can find the log file in the following location: `<Integration Package Manager directory>/IntegrationPackageManager_<date and timestamp>.log`

Install or Remove Distribution Packages in Silent Mode

You can run the package manager in silent mode to install or remove distribution packages. The silent input properties file contains the properties for the package manager to run in silent mode for service and clients. Set the appropriate value for each property in the file.

1. Find the IntegrationPackageManager.properties file in the following location: `<Integration Package Manager directory>/`
2. Edit the properties file in a text editor.

   The following table describes the properties that you can modify:

Property Name	Description
USER_INSTALL_DIR	The installation directory of the service or client.
OPERATION_TYPE	The operation that you want to perform: - Set to DELETE to remove existing distribution packages. - Set to EXTRACT to install one or more distribution packages.
SELECTED_HADOOP_LIBRARIES	Lists the distribution packages and versions. Enter the distribution packages that you want to install or remove. Separate multiple packages with a comma.

3. Save the properties file.
4. From the package manager directory, run one of the following commands:
   - `./Server.sh silent` for Linux or UNIX
   - `Server.bat silent` for Windows
   - `Client.bat silent` for client

   Note: To run the command on Windows, use the administrator command prompt.
5. Verify the installation or removal status in the package manager log file.

   You can find the log file in the following location: `<Integration Package Manager directory>/IntegrationPackageManager_<date and timestamp>.log`

After You Install

To use the distribution packages that are installed using the package manager, configure the property or environment variable in service and client machines.

Configure the Developer Tool

After you install the distribution packages in the Developer tool, update the developerCore.ini file with the installed distribution package.

1. Find the developerCore.ini file in the following location: `<Informatica installation directory>\clients\DeveloperClient`

2. Edit the file to update the following property:

   ```
   -DINFA_HADOOP_DIST_DIR=hadoop\<Hadoop distribution name>_<version>
   ```

   For example,

   ```
   -DINFA_HADOOP_DIST_DIR=hadoop\CDH_7.1
   ```

3. Restart the Developer tool.

Configure Environment Variables

Some adapters require environment variables for the Data Integration Service and Metadata Access Service to access the distribution packages. For more information, see Configure environment variables to process complex files.

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